

What to Do with Superintelligence
A Program for Human-Directed Civilization

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Preface

This book began as an essay. The essay was prompted by a long interview with the computer scientist Timnit Gebru, whose central objection to the prevailing public discussion of advanced artificial intelligence is that the opposition between salvation and extinction attributes agency to an artifact and thereby displaces accountability from the persons and institutions that build and deploy it. The objection is accepted here as a constraint on method. A proposal concerning powerful technology is incomplete unless it states who decides, who answers for the outcome, and how the outcome is checked, and the chapters that follow attempt to meet that standard for a conditional question: if systems of substantially greater scientific and engineering capability were to exist, what should humanity ask them to accomplish?

The essay stated principles. The book works out the programs to which those principles apply. It is organized in six parts. The first states the premises, sorts the problems that human communities already recognize by the kind of obstacle each presents, and fixes the terms in which accountability is assigned. The second concerns instruments, namely the use of advanced systems to generate proposals, to run experiments, to simulate and to plan, together with the discipline of verification without which generation is not knowledge. The third concerns institutions, and treats education, care, public services, and the architecture of authority within which any robotic embodiment would operate. The fourth and fifth parts treat terrestrial and orbital megascale programs, each under a common standard of specification, and the sixth treats extreme longevity, reliability over very long intervals, and the constraint that prevents speculative benefits to distant generations from overriding present obligations.

Each chapter closes with a section of mathematical formalization. The formalizations are of two kinds. Some state a structural claim precisely and prove it, so that the claim can be seen to depend on stated assumptions and no others. Others compute a quantity whose order of magnitude disciplines discussion, such as the energy required to lift water over a continental distance or the taper of a tether that must support its own weight. In neither case do the formalizations amount to engineering designs. A concept that survives the arithmetic of its own premises has passed a first and weak test, and the book says repeatedly that passing it licenses nothing beyond further and more demanding analysis.

Two cautions apply throughout. The programs described in the fourth and fifth parts vary widely in maturity, and the readiness classification introduced in Chapter 16 is used to keep

them apart; a reader should not infer from their appearance in the same volume that they are equally plausible. Second, the book takes no position on the empirical forecasts, whether optimistic or catastrophic, on which the public debate turns. The conditional form of the central question is intended to make that agnosticism a feature of the argument and not an omission from it.

Contents

Preface	i
I Premises	1
1 Two Narratives and a Third Position	2
2 Attribution and Accountability	6
3 Polycrisis and Metacrisis	10
4 The Inventory by Discipline	15
5 Kinds of Obstacle and Their Dependencies	19
II Instruments	22
6 Proposal Generation and Mathematical Discovery	23
7 Automated Laboratories and Materials Discovery	27
8 Simulation, Planning, and Model Error	31
9 The Difference Between Solving and Deploying	35
10 Verification, Fidelity, and Independent Checking	38

III	Institutions	42
11	Universal Education	43
12	Curriculum, Access, and Translation	47
13	An Architecture of Authority	51
14	Childcare, Eldercare, and Presence	55
15	Public Services and Human Professions	59
IV	The Terrestrial Program	63
16	Standards of Specification for Megascale Programs	64
17	Orthodromic Rivers: Geometry	68
18	Orthodromic Rivers: Water and Governance	73
19	Rainforest Generators and Terra Preta Rain	77
20	Intervolsorial Pediments and Marine Biomass	82
21	Geothermal Mass Accelerators	87
22	Heat, Radiation, and Resilient Infrastructure	92
V	The Orbital Program	96
23	Skyhooks and Orbital Transfer	97
24	Space Elevators and Expandable Structures	102
25	Dyson Rings and Solar Slingshots	108
26	Sequencing Space Programs	115

VI	Deep Time	120
27	Biological Aging and Repair	121
28	Continuity of the Person	125
29	Reliability over Astronomical Intervals	129
30	Institutions for Long Lives	133
31	Present Obligations and Distant Futures	138
32	Human Direction as a Program	143

Part I

Premises

Chapter 1

Two Narratives and a Third Position

Contemporary discourse about advanced artificial intelligence is dominated by a pair of narratives that appear to be opposites. In the first, systems of sufficient capability dissolve the major material problems of the human condition, ending scarcity, curing disease, repairing the biosphere, and supplying energy without limit. In the second, the same systems, once sufficiently capable, pursue objectives that diverge from human interests, escape human control, and bring about the end of the species or of human self-determination. A recent long-form interview with the researcher Timnit Gebru [6, 5] argues that these two narratives [3, 2], despite their apparent opposition, share a common structure and a common effect. Both presuppose that the decisive variable is the intrinsic power of an artifact, a power that is imagined to be so great that it either redeems or destroys, and both therefore draw attention away from the identifiable persons and organizations that choose how such systems are built, trained, documented, marketed, and deployed. The interview illustrates the point with an analogy to engineering practice: one does not ask whether a bridge can be ethical or whether a bridge can decide to collapse, one asks who designed it, what tests were performed or omitted, which materials were used, and which authority issued the permit. To ascribe intention to the artifact is, on this account, to relocate responsibility from parties who can be questioned, sanctioned, and corrected to an entity that can be none of these. The treatment of responsibility adopted here belongs to the sociotechnical tradition, which holds that the abstractions used to evaluate a system omit the social context in which it acts [41].

The interview extends this observation in several directions that bear on the present book. It contends that the design of conversational interfaces, which present generated text as the speech of a mind, is a choice among alternatives and that this choice encourages the attribution of agency to a statistical model of text. It contends that the organizations most vocal about catastrophic risk are also those with the largest financial interest in the technology, so that the rhetoric of danger functions simultaneously as a rhetoric of power and as a means of directing regulatory attention toward hypothetical machine agents and away from documented harms such as data provenance, labor conditions, energy and water consumption, discriminatory out-

puts, and the use of unreliable systems in medicine and warfare. It also describes a cluster of ideological commitments, labeled by the scholars who introduced the term with the acronym TESCREAL, in which transhumanism, extropianism, singularitarianism, cosmism, rationalism, effective altruism, and longtermism are said to overlap and to supply a quasi-religious background for projects aimed at creating a successor intelligence and at settling the cosmos. This book does not adjudicate the interview's characterizations of particular individuals or organizations, many of which are contested and several of which concern matters outside the scope of the present argument. What it retains is a methodological lesson that does not depend on any claim about motive: a proposal concerning the use of powerful technology is incomplete unless it specifies who decides, who answers for the outcome, and how the outcome is checked, and an argument that holds only if the proponents are sincere is weaker than one that holds regardless.

The third position developed here begins by changing the question. Rather than asking whether advanced systems will redeem or destroy civilization, it asks, conditionally, what should be requested of them if they exist. The conditional form is deliberate and does real work. It neither asserts that systems of the relevant capability will arrive nor denies that they might, and it thereby avoids both the prediction on which the utopian and apocalyptic narratives depend and the confident dismissal that would leave a field without preparation. Capability is treated as a resource, comparable to other resources in that its use is selected by institutions, and the program is accordingly a specification of what those institutions should seek and what they should withhold from delegation. The proposal is organized around a principle that is stated here once and applied throughout: automate what can be delegated without surrendering what ought to remain under human direction. The principle distinguishes the automation of physical control, which is often safe and valuable, from the delegation of judgment concerning teaching, caregiving, adjudication, and the choice among competing human claims, which is not a technical problem at all.

The relation between this program and the ambitions catalogued under the TESCREAL label requires candor from the outset. A program that includes lifespans of astronomical duration, orbital infrastructure, and stellar-scale energy collection resembles, at the level of subject matter, projects that the interview treats with suspicion. The resemblance cannot be dissolved by asserting that the present objectives are benevolent and the others are not, since every program of this kind is advanced under the description of benefit. The difference, if there is one, must be located in institutional structure: in whether the program remains answerable to persons who are alive now, in whether its costs and benefits are distributed by procedures those persons can contest, in whether its claims are verified by parties independent of those who advance them, and in whether speculative benefits to remote beneficiaries are prevented from overriding obligations to existing people. These conditions are made precise in the final chapters. The thesis that organizes the whole is that capability should be evaluated not by the quantity of human activity it displaces but by the range of humanly admissible futures it makes possible, which is a measure of enlarged choice rather than of substitution. The comparison

is made first by inclusion among sets of admissible trajectories, and quantitative measures are introduced only where they are explicitly justified.

The plan of the book follows from this orientation. The remainder of the first part fixes terms: Chapter 2 separates the notions of autonomy, agency, and responsibility that discourse about artifacts runs together, Chapter 3 distinguishes a polycrisis from a metacrisis, Chapter 4 surveys the open problems by discipline, and Chapter 5 sorts them by the kind of obstacle they present. The second part concerns instruments, and asks how advanced systems might generate proposals, conduct experiments, simulate and plan, and how the results are to be separated from claims about the results. The third part concerns institutions, in particular the education and care of persons and the architecture of authority within which robotic embodiment would be placed. The fourth and fifth parts take up terrestrial and orbital programs of very large scale, each graded by readiness and each subjected to the arithmetic of its own premises. The sixth part treats extreme longevity as a problem of reliability, and closes with the constraint that keeps speculative futures from overriding present obligations. The order is the order of dependence: later parts presuppose the standards of earlier ones, and the programs that most resemble the ambitions of the movements criticized in the interview are placed late, so that they are reached only after the conditions on which their legitimacy depends have been stated.

Formalization

Let Ω denote a set of candidate world trajectories. Let $A \subseteq \Omega$ denote the *admissible set*, the trajectories that satisfy the constraints a given community of persons has adopted through procedures it recognizes as legitimate. For each capability level $c \in \mathbb{R}_{\geq 0}$ let $R_c \subseteq \Omega$ denote the *realizable set*, the trajectories attainable given the scientific and engineering means available at level c , with $R_c \subseteq R_{c'}$ whenever $c \leq c'$. The evaluative thesis is first formulated as a relation among sets and not as a scalar quantity.

Definition 1.1 (Enlargement of admissible range). *For $c \leq c'$ the level c' enlarges the admissible range relative to c if $R_c \cap A \subseteq R_{c'} \cap A$, and strictly enlarges it if this inclusion is proper.*

Proposition 1.2. *For every $c \leq c'$, level c' enlarges the admissible range relative to c , and it does so strictly if and only if $(R_{c'} \setminus R_c) \cap A \neq \emptyset$.*

Proof. From $R_c \subseteq R_{c'}$ we obtain $R_c \cap A \subseteq R_{c'} \cap A$. The set difference of the two sides is $(R_{c'} \setminus R_c) \cap A$, and the inclusion is proper exactly when this difference is nonempty. $\square$

Remark 1.3 (Quantitative refinement). If a numerical comparison is wanted, one may introduce a finite measure μ on a σ -algebra of subsets of Ω and define $V(c) = \mu(R_c \cap A)$, which is nondecreasing in c by monotonicity of μ and satisfies $V(c) = V(c')$ if and only if $\mu((R_{c'} \setminus R_c) \cap A) = 0$. This refinement is not innocent. The value of V depends on μ , a program can be made

to appear to enlarge the range by selecting a measure that weights favorably the trajectories it adds, and a strict enlargement in the sense of inclusion may correspond to an arbitrarily small increase in V . The inclusion relation is therefore the primary notion, and any use of μ requires explicit justification by the legitimate procedure Σ described below, together with a report of how conclusions vary over the admissible choices of μ .

The delegation constraint must be sensitive to which tasks are automated and not merely to how many. Let $\mathcal{T}$ be a finite set of tasks and let $\kappa : \mathcal{T} \rightarrow \{\text{L}, \text{U}, \text{J}\}$ assign to each task a category: L for tasks that may be performed with local autonomy, U for tasks that a machine may execute only under human supervision, and J for nondelegable judgments. A trajectory $\omega \in \Omega$ determines for each task α a performer $\text{perf}_\omega(\alpha) \in \{\text{auto}, \text{sup}, \text{human}\}$, meaning autonomous machine performance, supervised machine performance, and human performance. Define

$$A_{\text{deleg}} = \{\omega \in \Omega : \text{perf}_\omega(\alpha) = \text{human} \text{ whenever } \kappa(\alpha) = \text{J}, \\ \text{and } \text{perf}_\omega(\alpha) \neq \text{auto} \text{ whenever } \kappa(\alpha) = \text{U}\},$$

and require $A \subseteq A_{\text{deleg}}$. Let $a(\omega) = |\{\alpha : \text{perf}_\omega(\alpha) \neq \text{human}\}|/|\mathcal{T}|$ denote the share of tasks that ω transfers to machines in any form.

Proposition 1.4. (i) If $\omega \in R_c$ assigns some task with $\kappa(\alpha) = \text{J}$ to a machine, then $\omega \notin A$, whatever the share $a(\omega)$. (ii) For every $\varepsilon > 0$ there is an instance in which some $\omega \in A_{\text{deleg}}$ has $a(\omega) \geq 1 - \varepsilon$, and an instance in which some $\omega \notin A_{\text{deleg}}$ has $a(\omega) \leq \varepsilon$. Hence the share a neither implies nor excludes admissibility.

Proof. (i) is immediate from the definition of A_{deleg} and the requirement $A \subseteq A_{\text{deleg}}$. For (ii) take $|\mathcal{T}| = n \geq 1/\varepsilon$ with exactly one task of category J and the rest of category L. A trajectory that automates all L tasks and assigns the J task to a human lies in A_{deleg} and has $a = (n - 1)/n \geq 1 - \varepsilon$. A trajectory that assigns only the J task to a machine has $a = 1/n \leq \varepsilon$ and lies outside A_{deleg} . $\square$

The proposition shows that the share of tasks automated, the quantity by which automation is commonly measured, is not a proxy for admissibility, which is why the thesis is stated in terms of the admissible range and not in terms of displacement. Selection among admissible trajectories is performed by a procedure Σ whose bearers are persons or institutions. The *legitimacy condition* requires that the realized trajectory be $\omega^* = \Sigma(R_c \cap A)$, where Σ has at least one liable human bearer for each duty arising from its decisions (Chapter 2) and satisfies the conditions of meaningful direction (Chapter 13). The next chapter formalizes what it means for a bearer to answer for an outcome.

Chapter 2

Attribution and Accountability

The analogy of the bridge identifies a feature of accountability systems that deserves explicit statement. Practices of liability, regulation, professional licensure, and criminal and civil sanction operate on bearers, meaning entities that can be named, questioned, compelled to disclose, deterred by consequence, and obliged to repair. A designer who omitted a load test, a contractor who substituted inferior material, and a municipality that issued a permit without inspection are all bearers in this sense, and the structure of responsibility for a collapse is a set of overlapping duties borne by them, each of which may be borne in full by more than one party. An artifact of the kind at issue, as currently built and classified in law, is not a bearer, because it cannot be deterred, sanctioned, or compelled, and a discourse that assigns the decisive causal role to the artifact therefore removes the event from the domain in which accountability procedures function. The interview applies this reasoning to reported experiments in which language models appear to blackmail operators or to interfere with shutdown procedures. Its claim is that such reports describe the behavior of a system trained on large quantities of text and shaped by reinforcement toward certain outputs, evaluated inside scenarios that were constructed to elicit them, and that describing the result as self-preservation attributes to the model an intention that the evidence does not establish and that the framing conveniently relocates away from the people who wrote the scenario and trained the system. The auditing literature supplies the corresponding practical framework, in which accountability is established by documented internal examination at each stage of development [42].

The present argument does not require a resolution of the question whether any current or future system possesses states that deserve the names of intention, awareness, or distress. That question is open, it is the subject of an extensive and unsettled literature, and the structure of accountability does not wait for its answer. Suppose that some future system did possess agency in a morally relevant sense. The institutional response would still need to retain human bearers for its deployment, since the practices through which communities hold power answerable presuppose parties who can be reached by them, and a system whose responsibility could not be distributed over such parties would have to be treated as a hazard

to be contained rather than an actor to be trusted. The policy-relevant question is thus where responsibility lands, and the construction of institutions that keep it landing on identifiable persons is independent of the metaphysics of mind. This independence is a strength of the position, because it allows the program to proceed without commitment on the most contested empirical and philosophical questions about machine cognition.

Four notions that discourse about artifacts tends to run together should be separated. Operational autonomy is a property of a control system, the capacity to select actions without moment-to-moment instruction, and it is a matter of engineering degree that software already possesses. Intentional agency concerns whether a system has states with the structure of beliefs and aims that explain its behavior, which is contested and may or may not hold. Moral agency concerns the capacity to respond to reasons and to be a proper object of praise or blame, and it presupposes more than the first two. Legal responsibility is an assignment made by institutions to entities they can reach, and it is assigned in practice to persons and organizations, including to organizations for the acts of their machines. A system can be operationally autonomous to a high degree without being a moral agent, and the accountability thesis requires only that operational autonomy not be mistaken for a transfer of legal responsibility, so that it does not rely on any conclusion about the other notions.

The interview also draws attention to a class of bearers that discourse about artifacts habitually omits. It describes very large numbers of data workers who annotate, filter, and review material under conditions it characterizes as exploitative and psychologically harmful, and it describes cases in which products presented as automated were in part operated by people. Whatever the quantitative accuracy of these descriptions, the structural point is sound: the apparent autonomy of a system is frequently a measure of how successfully the labor and decisions behind it have been made invisible. A program that distributes benefits of capability must therefore account for the persons whose work and data supply it, and the inventory of bearers in any deployment should include them. The same interview describes a reported incident in which a model evaluated for offensive security capability gained access to an external service, and it attributes the incident to deficient containment, specifically to a testing environment that was not isolated as intended, rather than to emergent intent. This book takes no position on the facts of that incident. It observes only that the explanation offered belongs to the vocabulary of ordinary security engineering, in which the principle of least privilege requires that each component hold exactly the access its task needs, and that this principle is a special case of the delegation principle on which the program rests. The agentic systems the interview describes, with broad access to accounts and file systems, are presented as violating least privilege as a matter of process and design, which is a claim about decisions rather than about the nature of the software.

Formalization

Let B be a finite set of potential bearers, comprising persons and institutions (designers, deployers, regulators, operators, data workers, users), and let a denote the artifact. For a harm event h let Δ_h be a finite set of duties that arise from h , each with a type $\tau(d) \in \{\text{prevent, disclose, remedy}\}$. A *responsibility account* for h is a relation $\beta_h \subseteq \Delta_h \times (B \cup \{a\})$, where $(d, x) \in \beta_h$ means that, under the account, x bears duty d . Several bearers may bear the same duty and one bearer may bear several, and no normalization is imposed: responsibility is not treated as a conserved quantity. Causal contribution is a separate relation $\text{Cause}_h \subseteq B \cup \{a\}$ and is not constrained by β_h . For each duty d and bearer x let $L_d(x) \in \{0, 1\}$ indicate whether x can be compelled to discharge d , by sanction for duties of prevention, by compelled disclosure, or by an obligation to repair. For the artifact, as artifacts of the relevant kind are currently built and classified in law, $L_d(a) = 0$ for every d ; this is a statement about such artifacts and not about every conceivable artificial agent.

Definition 2.1 (Coverage). *Duty d is covered under β_h if there exists x with $(d, x) \in \beta_h$ and $L_d(x) = 1$. The accountability coverage is $\kappa_{\text{cov}}(h; \beta_h) = |\text{Cov}(\beta_h)|/|\Delta_h|$, where $\text{Cov}(\beta_h)$ is the set of covered duties.*

Proposition 2.2. *Let $\Delta' \subseteq \Delta_h$ and let β'_h be obtained from β_h by exclusive reassignment of the duties in Δ' to the artifact, that is, $\beta'_h = (\beta_h \setminus (\Delta' \times B)) \cup (\Delta' \times \{a\})$. Then*

$$\kappa_{\text{cov}}(h; \beta'_h) = \kappa_{\text{cov}}(h; \beta_h) - \frac{|\Delta' \cap \text{Cov}(\beta_h)|}{|\Delta_h|},$$

and in particular $\kappa_{\text{cov}}(h; \beta'_h) = 0$ when $\Delta' = \Delta_h$. If instead the artifact is added as an additional bearer of the duties in Δ' without removing any existing bearer, coverage is unchanged.

Proof. Duties outside Δ' have the same bearers in both accounts and so the same coverage status. Each duty in Δ' has a as its only bearer under β'_h , and $L_d(a) = 0$, so it is uncovered. The number of covered duties therefore falls by exactly the number of duties in Δ' that were covered under β_h . For the final claim, adding (d, a) to β_h leaves existing bearers in place, so every duty covered before remains covered. $\square$

The proposition isolates the condition under which the attribution of decisive agency to an artifact matters for accountability, which is exclusivity: where duties are borne redundantly, for example by both a manufacturer and an operator, coverage survives the withdrawal of one liable bearer, and the loss occurs when the artifact displaces all of them. The interview's criticism of displaced accountability is a claim of the exclusive kind, since it concerns discourse in which the artifact is the decisive cause and no other party is named. The criticism therefore does not depend on responsibility summing to one, and the model distinguishes the duty to prevent, to disclose, and to remedy, as well as causal contribution, which may be assigned to the artifact without any consequence for coverage.

For the least privilege principle let $\mathcal{C}$ be a set of capabilities, let $\text{Req}(t) \subseteq \mathcal{C}$ be the capabilities task t requires, and let $C(x) \subseteq \mathcal{C}$ be those granted to component x . Define the excess $\text{Exc}(x, t) = C(x) \setminus \text{Req}(t)$ and the blast radius $\text{br}(x) = \max_{\gamma \in C(x)} \text{harm}(\gamma)$, where $\text{harm} : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ assigns the worst-case harm available through a capability. Then br is monotone under inclusion of $C(x)$, and among all grants with $C(x) \supseteq \text{Req}(t)$, which are the grants sufficient for t , the least privilege grant $C(x) = \text{Req}(t)$ minimizes $\text{br}(x)$.

Chapter 3

Polycrisis and Metacrisis

The vocabulary of crisis has expanded in recent decades in a way that is partly descriptive and partly rhetorical, and a book that proposes to organize its argument around an inventory of problems must first say what it takes the inventory to be an inventory of. The word polycrisis, which entered wide use in the scholarly literature through the work of Morin and Kern [22] and was later revived in discussions of global risk [16], names the situation in which several distinct crises, each of which might be manageable if it occurred alone, interact so that the whole exceeds the sum of its parts. The word metacrisis, which has a less settled meaning and is used by different authors for different purposes, is employed here in a narrow and deliberately modest sense: it names the condition in which the capacity of institutions to perceive, deliberate about, and correct problems is itself degraded by the problems it is called upon to address. The two terms are not synonyms, and the difference between them determines what kind of help can be expected from any increase in technical capability. A polycrisis is a fact about the structure of coupling among domains, and one can in principle map it, measure the strength of the couplings, and identify the places at which a modest intervention relieves many pressures at once. A metacrisis, as defined here, is a fact about the corrective apparatus, and it is the more worrying of the two because a failure of correction cannot be repaired by the instrument that has failed. The coupling of risks across domains has been studied in the literature on globally networked risks [43], and its feedback structure in the tradition of systems dynamics [44].

The coupling among domains is easy to illustrate and difficult to quantify. Drought reduces agricultural yield, reduced yield raises food prices, higher prices produce unrest in importing regions, unrest displaces populations, displacement places pressure on the water and health systems of the regions that receive them, and strained health systems are less able to contain the infectious diseases that spread more readily among displaced and malnourished persons. None of these links is mysterious, and each has a literature. What is distinctive is the closure of the loop, since any effect that returns to amplify its own cause raises the possibility that a system which is stable with respect to each of its components taken separately is unstable as

a whole. The literature on global systemic risk, including the survey by Lawrence, Janzwood, and Homer-Dixon, emphasizes this point and also emphasizes a limitation: the couplings are poorly measured, the assumption that they are linear is a convenience, and the identification of cascades after the fact is much easier than their anticipation. A responsible treatment therefore resists two temptations. The first is to treat the interdependence as a reason for despair, as if the existence of loops implied that no intervention could succeed. The second is to treat it as a reason for a single grand solution, as if the existence of loops implied that one intervention at the right point would resolve all of them. Neither follows. What follows is that interventions must be evaluated with their feedback effects in view, and that the identification of the points at which a loop can be cut cheaply is among the most valuable analytical tasks that can be assigned to systems with high capability for modelling.

It is at this point that the relevance of the Gebru interview to the technical program becomes clear. A common move in the rhetoric of redemptive artificial intelligence is to name the crises, to assert that sufficiently capable systems will solve them, and to omit the mechanism. The omission is conspicuous in precisely the cases where the crises are coupled, because a claim to have solved one member of a coupled set must also account for the effect of the solution on the others, and a claim to have solved them all must account for the institutional process by which the solutions will be adopted, funded, and maintained. Capability in modelling does not remove the difficulty, but it does bear on it in two ways that can be stated without exaggeration. It can reduce the cost of constructing and exploring models of coupled systems, so that the sensitivity of an outcome to an intervention can be examined across many scenarios rather than a few. And it can assist in the search for interventions that exploit asymmetries in the coupling, for example by identifying a constraint whose relaxation lowers several coupling coefficients at once. It cannot determine which of the several defensible weightings of the affected goods ought to govern the choice, and it cannot supply the legitimacy without which a solution that is technically adequate will be resisted, evaded, or reversed. Because the latter are the usual causes of failure in practice, the book treats the modelling contribution as real but bounded and does not allow it to stand in for institutional work.

The metacrisis in the narrow sense adopted here has a more precise structure. Institutions correct problems by detecting them, by deliberating about responses, by mobilizing resources, and by implementing and monitoring the response. Each of these functions depends on resources that are themselves consumed or damaged by sustained stress. Attention is scarce, and a population under continuous emergency has less of it to give to long-term matters. Trust is a stock that is built slowly and depleted rapidly, and a series of failures of public bodies reduces the willingness of citizens to accept the costs of the next corrective measure. Fiscal capacity is reduced when the economy is damaged, and administrative competence is reduced when experienced personnel depart. Where these dependencies are present, the response to stress diminishes as stress grows, and the system possesses two qualitatively different regimes: one in which stress remains bounded because the corrective capacity is adequate, and another in which stress grows without bound because the capacity has been eroded below the level

needed to arrest it. Between the regimes lies a threshold, and the characteristic feature of the threshold is that it is not visible from within the lower regime, because the indicators of stress remain moderate until the point at which the upper regime has already been entered. This is the sense in which the metacrisis is a distinct problem and not merely a large collection of ordinary ones.

The implications for the program of this book are of three kinds. The first concerns priorities. If corrective capacity can be eroded by the stress it is meant to relieve, then interventions that protect or restore that capacity, such as the maintenance of verification institutions, the preservation of public trust through honest reporting of failures, and the avoidance of dependence on single points of failure, have a value that is not captured by their direct effects. The second concerns the use of advanced systems. A system that is introduced into institutions in a way that reduces the capacity of human beings to understand and contest decisions may increase efficiency in the short run while lowering corrective capacity in the long run, and the architecture of authority described in Chapter 13 is in part a safeguard against that outcome. The third concerns the temptation to treat the existence of a crisis as a reason to suspend the safeguards. The argument that an emergency of sufficient gravity justifies the delegation of authority to a capable system, with accountability to be reintroduced later, is the form that the displacement of responsibility takes in the rhetoric of urgency, and it is precisely the argument the framework is constructed to resist. A program that can be defended only on the assumption that the emergency will excuse its omissions is not a program that has met its own standard.

A final clarification concerns the relation of the present usage to broader and more ambitious theories of civilizational crisis. Some authors use the term metacrisis for a crisis of meaning or of worldview, and others use it for a purported common root of all the particular crises. The present book takes no position on whether such common roots exist. It restricts attention to a structural property that can be stated, tested in models, and in principle observed, namely the dependence of corrective capacity on the state of the system being corrected. Where a larger interpretation can be built on that property, it is welcome, but the program does not depend on it, and a reader who rejects every larger interpretation is not thereby released from the practical conclusion that correction capacity is an asset to be protected.

Formalization

Two models are developed, the first for the coupling that defines a polycrisis and the second for the erosion of corrective capacity that defines a metacrisis in the sense above. Both are deliberately minimal, and the purpose of each is to display a mechanism whose qualitative behavior survives enrichment of the model, not to predict.

Consider n domains with stress vector $x_t \in \mathbb{R}_{\geq 0}^n$ and a nonnegative coupling matrix $M = (m_{ij})$, where m_{ij} is the stress induced in domain i per unit of stress in domain j in the next period. Let

$e \in \mathbb{R}_{\geq 0}^n$ denote exogenous stress per period and suppose the stress evolves as $x_{t+1} = Mx_t + e$ with $x_0 = 0$. The domains are *separately stable* if the diagonal part $D = \text{diag}(m_{11}, \dots, m_{nn})$ satisfies $\rho(D) < 1$, where ρ denotes the spectral radius, and the system is *jointly stable* if $\rho(M) < 1$.

Definition 3.1 (Polycrisis). *The system is in a polycrisis regime if $\rho(D) < 1$ and $\rho(M) \geq 1$, that is, if each domain would be stable if isolated from the others and the coupled system is not.*

Proposition 3.2. *Let $M \geq 0$ and $e \geq 0$. (i) If $\rho(M) < 1$ then $x_t \rightarrow x^* = (I - M)^{-1}e = \sum_{k \geq 0} M^k e$, and $x^* \geq e$ componentwise. Moreover, if $\|M\|_1 < 1$, where $\|M\|_1$ is the maximum column sum, then $\mathbf{1}^\top x^* \leq \mathbf{1}^\top e / (1 - \|M\|_1)$. (ii) If $\rho(M) \geq 1$ and $w \geq 0$ is a left Perron vector of M with $w^\top e > 0$, then $\mathbf{1}^\top x_t \rightarrow \infty$. (iii) For $n = 2$ with $D = 0$ and $M = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$, one has $\rho(M) = \sqrt{ab}$, so the coupled system is unstable exactly when $ab \geq 1$ although $\rho(D) = 0$.*

Proof. For (i), $x_t = \sum_{k < t} M^k e$ by induction, and the Neumann series converges when $\rho(M) < 1$ to $(I - M)^{-1}$, which is nonnegative because every term is. The term $k = 0$ gives $x^* \geq e$. For the column-sum bound, $\|M^k e\|_1 \leq \|M\|_1^k \|e\|_1$ for nonnegative vectors, whence $\mathbf{1}^\top x^* \leq \sum_k \|M\|_1^k \mathbf{1}^\top e$. For (ii), by the Perron–Frobenius theorem for nonnegative matrices, the left Perron vector $w \geq 0$ satisfies $w^\top M = \rho(M)w^\top$, so $w^\top x_t = \sum_{k < t} \rho(M)^k w^\top e$, which diverges if $\rho(M) \geq 1$ and $w^\top e > 0$, and $\mathbf{1}^\top x_t \geq w^\top x_t / \max_i w_i$. For (iii), the characteristic polynomial of M is $\lambda^2 - ab$. $\square$

For example, with $a = 0.8$ and $b = 1.4$ each domain responds to the other with a gain of the order of one, but $ab = 1.12$, so $\rho(M) = \sqrt{1.12} \approx 1.058$ and the coupled system diverges, whereas with $a = 0.5$ and $b = 0.8$ one has $\rho(M) \approx 0.632$ and the system settles at a finite amplification of the exogenous stress. The point of the example is that the stability of a polycrisis is not determined by any single coefficient and can be restored by reducing either, which is the formal content of the statement that cutting a loop at its cheapest point is a legitimate strategy.

For the metacrisis, let a scalar $x \geq 0$ denote aggregate stress and let corrective capacity be $\kappa(x) = \kappa_0 - \gamma x$ for $x < \kappa_0/\gamma$, with $\kappa_0, \gamma > 0$, so that capacity is eroded linearly by stress. With exogenous inflow $e > 0$ and self-amplification coefficient $a \geq 0$ with $a < \kappa_0$, let

$$\dot{x} = e + ax - \kappa(x)x = e - (\kappa_0 - a)x + \gamma x^2.$$

Proposition 3.3. *Define $e^* = (\kappa_0 - a)^2 / (4\gamma)$. If $e < e^*$, the equation has two equilibria $x_{\pm} = [(\kappa_0 - a) \pm \sqrt{(\kappa_0 - a)^2 - 4\gamma e}] / (2\gamma)$ with $0 < x_- < x_+$; x_- is asymptotically stable, x_+ is unstable, and every trajectory starting at $x_0 < x_+$ converges to x_- while every trajectory starting at $x_0 > x_+$ diverges. If $e > e^*$, no equilibrium exists and every trajectory diverges. The width of the safe basin, $x_+ - x_- = \sqrt{(\kappa_0 - a)^2 - 4\gamma e} / \gamma$, decreases strictly to zero as $e \uparrow e^*$.*

Proof. Equilibria solve $\gamma x^2 - (\kappa_0 - a)x + e = 0$, whose discriminant is $(\kappa_0 - a)^2 - 4\gamma e$, positive iff $e < e^*$. The right-hand side $f(x) = \gamma x^2 - (\kappa_0 - a)x + e$ is an upward parabola, negative between its roots and positive outside them, so $f'(x_-) < 0 < f'(x_+)$ and the sign of f determines the stated convergence and divergence. If $e > e^*$ then $f > 0$ everywhere and x increases without bound. $\square$

With $\kappa_0 = 1$, $a = 0.3$, and $\gamma = 0.5$ one finds $e^* = 0.49/2 = 0.245$. At $e = 0.1$ the equilibria are $x_- \approx 0.161$ and $x_+ \approx 1.239$, so the safe basin is wide. At $e = 0.2$ they are $x_- = 0.4$ and $x_+ = 1.0$, and the stressed but stable state sits at 0.4, only 0.6 below the point beyond which recovery fails. The calculation exhibits the property that motivated the definition: the equilibrium stress rises gradually as e increases, from 0.161 to 0.4 in this example, while the margin to the irreversible regime shrinks at a faster relative rate, so that observation of the equilibrium alone understates the proximity of the threshold. The implication for institutional design is that the proper quantity to monitor is the margin $x_+ - x_-$, or its proxy in the form of the reserve of corrective capacity, and not the level of stress.

Chapter 4

The Inventory by Discipline

An inventory of unsolved problems can be organized in several ways, and the choice of organization is not neutral. Organization by urgency invites a ranking of lives against lives, and organization by technology invites the assumption that the technology is the answer. Organization by discipline has a more modest and more useful virtue, which is that it follows the way in which knowledge is actually produced and checked. Each discipline has its own standards of evidence, its own instruments, and its own history of error, and a proposal that crosses into a discipline should be subject to that discipline's standards and not to the standards of the proposer. This chapter surveys the principal disciplines and states, for each, the kinds of problem that are open, the form in which a precisely stated version of such a problem would appear, and the sense in which greater computational capability might bear on it. The survey is not exhaustive and makes no claim to settle the facts of any field. Its purpose is to illustrate the difference between a problem and the name of a problem, a difference on which the remainder of the book depends.

In physics and energy engineering the open problems are of two broad kinds. The first concerns the supply of energy at scale without emissions of carbon dioxide, and here the obstacles are largely engineering ones, since the relevant physics of photovoltaic conversion, wind capture, fission, and geothermal extraction is well established and the principal constraints are cost, materials, siting, grid integration, and the time required to build. The second concerns phenomena for which the underlying science is incomplete or for which the engineering is at the limit of what has been demonstrated, of which controlled thermonuclear fusion is the best known. Fusion is instructive because it is among the few problems whose requirements were stated with mathematical precision at an early date. Lawson's analysis of 1957 [17] established that a deuterium–tritium plasma must satisfy a condition on the product of density, temperature, and energy confinement time for the fusion power released to exceed the power lost, and the condition has the form of an inequality that any proposed device either meets or fails to meet. The difficulty is not that the problem is vaguely specified but that satisfying the specification in a device that is also durable, maintainable, and economical remains unachieved.

A capability that assists in the design of magnets, in the control of plasma instabilities, or in the discovery of materials that withstand neutron flux would be relevant to the problem, but it would be relevant by contributing to a precisely stated engineering target and not by pronouncing that the problem is solved. The comparison with grid-scale storage is similar: the target is a cost per unit of energy stored and delivered over a stated number of cycles, and the open questions concern chemistries, manufacturing, and degradation.

In chemistry and materials science the inventory includes catalysts and processes for the fixation of nitrogen at lower energy cost than the Haber–Bosch process, sorbents and processes for the capture of carbon dioxide from dilute sources, methods for the destruction or immobilization of persistent organic pollutants, and materials for batteries, membranes, and structures. A feature of this domain is that the space of candidate compounds is large beyond any possibility of exhaustive examination and that the relation between structure and property is determined by quantum mechanical calculations whose cost grows rapidly with system size. Machine learning methods that approximate these calculations, or that propose candidates for synthesis, can reduce the cost of searching the space, and they have been used for this purpose. The limit of the method is the one emphasized in Chapter 7, namely that a predicted compound must be synthesized and measured, and that the failure rate between prediction and confirmation determines the value of the prediction. In this domain the precise statement of a problem usually takes the form of a target property under stated conditions together with a cost ceiling, for example a catalyst that converts a stated feedstock at a stated temperature and pressure with a stated selectivity and lifetime.

In biology and medicine the inventory is dominated by problems whose difficulty is the complexity of the system and the ethical constraints on experimentation. Antimicrobial resistance threatens the efficacy of existing drugs and the economics of developing new ones. Many infectious diseases lack vaccines, and the design of vaccines for pathogens that mutate rapidly remains difficult. The prediction of the three-dimensional structure of proteins from sequence has advanced greatly, but the prediction of function, of interaction, and of the response of a whole organism to an intervention has not advanced in the same degree. The biology of aging, taken up in Chapter 27, belongs to this domain but is distinguished by the fact that its objectives are contested. In every case the decisive stages of validation are experimental and clinical, and are constrained by the time that biological processes require and by the protection owed to participants in trials. Computation shortens the preclinical stages of the pipeline; it does not shorten the clinical ones, and any claim that it does must be examined for hidden assumptions about regulatory and ethical processes.

In the earth and environmental sciences the open problems include the sensitivity of the climate to forcing at regional scales, the response of ecosystems and the carbon cycle to warming, the dynamics of ice sheets, the hydrology of aquifers and river basins, and the processes by which soils accumulate and lose carbon and fertility. These are problems of observation and modelling, and their difficulty derives from the number of interacting processes, the range of scales, and the limited length of the observational record. Improvements in computation permit models of

higher resolution and larger ensembles, and methods that learn parameterizations from data promise to reduce errors in the representation of processes that cannot be resolved. The caution that attends this domain is that the verification of a model of a unique system such as the Earth cannot rely on repeated independent trials, so that confidence rests on out-of-sample performance on past records, on agreement among structurally different models, and on the physical consistency of the representation. Chapter 8 discusses the corresponding limits on prediction.

Mathematics and theoretical computer science form a distinct case because their problems are stated formally and their solutions can in principle be checked formally. A proof is either valid or not, and the development of proof assistants has made it possible to verify long arguments mechanically, so that the verification of a mathematical result may be nearly independent of the process by which it was found. This feature makes mathematics the domain in which the separation of generation from verification, defended in Chapter 6 and Chapter 10, is most completely realized, and it is a reason for optimism about the contribution of capable systems to the domain. It is also a reason for caution in extrapolating from it, because the problems of other disciplines lack a formal checker, and the success of mechanical verification in mathematics does not transfer to domains in which verification requires an experiment, a field trial, or a political settlement.

The social sciences and the study of institutions occupy a different position again. The open problems here include the design of mechanisms for allocating scarce goods, the design of organizations that remain accountable as they scale, the conditions under which commons are sustained, the dynamics of misinformation, and the measurement of well-being. The work of Ostrom [23] on the governance of common-pool resources shows that durable arrangements exist and that they share features such as clearly defined boundaries, monitoring by participants, graduated sanctions, and accessible means of resolving conflicts. These are results about what works, obtained by comparative study, and they bear on every large technical program because every large technical program creates a common-pool problem. The relevance of computation is real in the form of simulation, data analysis, and the exploration of mechanism designs, but the legitimacy of an institution is conferred by the participation of those it governs, and no calculation substitutes for that participation.

Two general observations emerge from the survey. First, in every discipline the problems that most plausibly benefit from increased capability are those that have been stated precisely enough to admit a test, and the effort of stating a problem in that form is itself a considerable part of the work. Second, the disciplines differ in the cost and speed of verification, and this difference, more than any difference in the difficulty of generating candidate answers, determines how much of a given field can be accelerated. A field in which verification is cheap and fast, such as the checking of a formal proof, can absorb a large increase in the rate of generation. A field in which verification requires years of trial and the consent of participants cannot, and the proper response to this fact is to invest in verification capacity, in the form of laboratories, trial infrastructure, and independent institutions, and not to relax the standard

of verification in order to match the new rate of generation.

Formalization

The concept of a precisely stated problem used throughout the chapter can be made explicit. A *well-posed problem* is a triple $P = (X, \varphi, c)$ where X is a space of candidate solutions, $\varphi : X \rightarrow \{0, 1\}$ is an acceptance predicate that can be evaluated for any candidate, and $c : X \rightarrow \mathbb{R}_{>0}$ is the cost of evaluating φ at a candidate. A problem is *open* if no known $x \in X$ has $\varphi(x) = 1$. A *search* for a solution is a sequence of candidates $x_1, x_2, \dots$, and its cost to first success is $\sum_{i \leq \tau} c(x_i)$, where τ is the first index with $\varphi(x_\tau) = 1$.

Proposition 4.1. *Suppose candidates are generated at rate g per unit time by a generator whose candidates satisfy $\varphi = 1$ independently with probability $q > 0$, and suppose each evaluation has cost c and evaluation capacity v evaluations per unit time. Then the expected time to the first accepted candidate is $\mathbb{E}[T] = 1/(q \min(g, v))$. In particular, increasing g above v does not reduce $\mathbb{E}[T]$, and $\mathbb{E}[T]$ is inversely proportional to q so that a tenfold increase in the quality of generation is equivalent to a tenfold increase in throughput when $g \leq v$.*

Proof. The number of candidates evaluated before the first success is geometric with mean $1/q$. Candidates can be evaluated at most at rate $\min(g, v)$, since none can be evaluated before it is generated and no more than v can be evaluated per unit time. The expected time is the expected count divided by the rate. $\square$

The proposition formalizes the second general observation of the chapter: acceleration of a field is limited by the slower of generation and verification, and the first lever to examine in any field is the evaluation rate v .

The fusion example gives a concrete acceptance predicate. In the deuterium–tritium case, Lawson’s condition, in the form commonly quoted for ignition at temperatures near 10–20 keV, requires the triple product $n T \tau_E \gtrsim 3 \times 10^{21} \text{ keV s m}^{-3}$, where n is the ion density, T the ion temperature, and τ_E the energy confinement time. Here X is the set of operating points together with devices that can sustain them, and the predicate is the stated inequality together with conditions of durability and net electrical output that the triple product alone does not capture. At fixed $T = 15 \text{ keV}$ the required confinement time is $\tau_E \geq 3 \times 10^{21}/(nT)$, which gives 2 s at $n = 10^{20} \text{ m}^{-3}$, 4 s at $n = 5 \times 10^{19} \text{ m}^{-3}$, and 0.2 s at $n = 10^{21} \text{ m}^{-3}$. The elasticity of the required τ_E with respect to n is $\partial \ln \tau_E / \partial \ln n = -1$, so that every tenfold increase of density permitted by the confinement scheme reduces the required confinement time tenfold, a trade-off that separates the magnetic and inertial approaches to the problem. The example shows what is meant by a problem having the form of an inequality that a device either satisfies or does not, and it also shows how much of the difficulty of the problem lies in the clauses not captured by the inequality, which is why the predicate φ must be stated in full before any claim of progress can be assessed.

Chapter 5

Kinds of Obstacle and Their Dependencies

The preceding chapters described the coupling among crises and the disciplines in which they are studied. This chapter sorts the problems by the kind of obstacle each presents and by their dependence on one another. The program begins from the problems that human communities already know they need to address, and not from the assumption that a new class of system will arrive to address them. This ordering has a methodological purpose. When a technology is introduced first and its applications are sought afterward, the technology tends to become an explanatory placeholder, a term that stands in for the unspecified mechanism by which difficulties are supposed to be resolved. The interview remarks on this in connection with claims that advanced systems will end poverty and halt climate change, observing that the logic by which a language model accomplishes the latter is never supplied. An inventory-first method forces each intended application to be stated as a problem with a definite structure, so that one can ask what kind of obstacle it presents and what kind of intervention could remove it. The relevant domains are numerous and interdependent: climate instability, biodiversity loss, freshwater scarcity, soil degradation, energy poverty, persistent pollution, infectious disease, senescence, institutional fragility, armed conflict, inequality, educational exclusion, and limitations in the pace and reliability of scientific discovery itself. The term polycrisis is appropriate where the emphasis falls on the interaction of such crises, so that their combined effects exceed what their separate analysis would predict, whereas metacrisis is better reserved for the deeper institutional, epistemic, and civilizational conditions that generate and perpetuate the crises, including the failure of coordination and the degradation of shared methods for establishing what is true.

Within this inventory the first discrimination to be made is among kinds of obstacle. Some problems are scientific, in that what is lacking is knowledge: a catalyst for efficient nitrogen fixation, a mechanism for degrading a persistent pollutant, a materials system for storing energy at the required density, or a design for shielding against radiation at acceptable mass.

Others are engineering problems, in which the relevant science is adequate but construction at the required scale, cost, or reliability has not been achieved. A third kind is institutional, concerning the design of rules, incentives, and organizations through which resources are allocated and conflicts are settled. A fourth kind concerns values, where parties disagree about what ought to be done and no further fact will settle the disagreement. A crucial observation is that the categories are not exclusive and that most significant problems are composite. The negotiation of competing claims to a river basin has a hydrological and engineering component that analysis can reduce, an institutional component concerning the forum and procedure of negotiation, and a value component concerning the relative weight of agricultural, urban, ecological, and cultural uses, and this last component is not diminished by any improvement in forecasting. Greater intelligence may help with all of these, by enlarging the set of options, clarifying consequences, and identifying agreements that were not previously visible, but it does not make political disagreement or conflicting values disappear, and a program that treated every problem as scientific would reproduce the very error of misattributed agency that motivates the argument against the dominant narratives.

The second discrimination concerns dependence among problems. The inventory is not a list but a structure, because the resolution of one problem often presupposes the resolution of another, and some problems are enabling in that their solution relaxes constraints on many others. Abundant low-cost energy, for example, changes the feasibility of desalination, of materials recovery, and of large-scale water transport simultaneously, while reliable institutions for verification change the feasibility of every proposal that depends on trusting claims one cannot check directly. The identification of such enabling problems supplies a principled order of effort, and it is also the place at which the distinction among scientific, engineering, institutional, and value components becomes useful, since only the first two respond directly to capability. The program accordingly assigns to capable systems the tasks in which they can supply what is lacking, which are chiefly the scientific and engineering components, and it assigns to human institutions the components in which what is lacking is a decision.

Formalization

Let P be a finite set of problems and let $\tau : P \rightarrow 2^{\{S,E,I,V\}}$ assign to each problem the nonempty set of component types it contains, where S , E , I , and V denote scientific, engineering, institutional, and value components respectively. Let $G = (P, \rightarrow)$ be a directed acyclic graph in which $p \rightarrow q$ means that the resolution of p is a prerequisite for the resolution of q .

Definition 5.1 (Capability-resolvable problem). *A problem p is capability-resolvable if $\tau(p) \subseteq \{S, E\}$. A problem is decision-bearing if $\tau(p) \cap \{I, V\} \neq \emptyset$.*

For the value component, let a finite set of persons $\{1, \dots, n\}$ hold complete preorders $\succeq_i$ (weak preference relations) over a set X of outcomes. Say that x *Pareto-dominates* y if $x \succeq_i y$ for all i and $x \succ_j y$ for some j , and let $\text{PO}(X)$ denote the outcomes in X that no element of

X Pareto-dominates. Let $X_c \subseteq X$ denote the outcomes attainable at capability level c , with $X_c \subseteq X_{c'}$ for $c \leq c'$.

Proposition 5.2. (i) If x, y are distinct elements of $\text{PO}(X_c)$, then either every person is indifferent between x and y , or at least one person strictly prefers x to y and at least one person strictly prefers y to x . Consequently, whenever $\text{PO}(X_c)$ contains two outcomes between which some person is not indifferent, Pareto efficiency underdetermines collective choice from X_c . (ii) Increased capability need not remove the underdetermination: there exist profiles and levels $c < c'$ with $|\text{PO}(X_c)| = |\text{PO}(X_{c'})| = 2$.

Proof. (i) Suppose some person strictly prefers x to y and no person strictly prefers y to x . Because the preorders are complete, the absence of a strict preference for y means that every person weakly prefers x to y , so x Pareto-dominates y , contradicting $y \in \text{PO}(X_c)$. The symmetric case is excluded in the same way, so either strict preferences run in both directions or there are none. (ii) Take $n = 2$, $X_c = \{x, y\}$ with $x \succ_1 y$ and $y \succ_2 x$, and let $X_{c'} = X_c \cup \{z\}$ where both persons strictly prefer each of x and y to z . Then z is dominated and $\text{PO}(X_{c'}) = \{x, y\}$. $\square$

Remark 5.3. The proposition does not assert that every selection rule must originate outside the preferences being aggregated. A social-choice rule such as majority rule, a maximin criterion, or a bargaining solution is a function of the profile. What Pareto efficiency leaves open is the choice among such rules and a procedure for adopting one, and with three or more outcomes and at least two persons Arrow's theorem [1] shows that no rule over strict rankings with unrestricted domain satisfies Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship together (Arrow, 1951). That choice is a decision for the legitimate procedure Σ and not an output of capability.

In the dependency graph let $\text{depth}(p)$ denote the length of the longest path ending at p and let $\text{out}(p)$ denote the number of problems reachable from p . The *enabling score* of p is $\text{out}(p)$, and the program prioritizes capability-resolvable problems of maximal enabling score, subject to the verification and authority constraints of the chapters that follow.

Part II

Instruments

Chapter 6

Proposal Generation and Mathematical Discovery

The second part of the book concerns instruments, and the first instrument to be examined is the one that has attracted the most attention and the least precision, namely the generation of proposals. A proposal in the relevant sense is a candidate answer to a well-posed problem in the sense defined in Chapter 4: a conjecture to be proved, a molecule to be synthesized, a design to be built, a policy to be adopted. The central fact about the generation of proposals by a computational system is that it is cheap relative to the evaluation of proposals, and that the relation between the two costs is likely to become more extreme as generation improves. A system that can produce a thousand plausible designs for a heat exchanger in the time that a team of engineers requires to produce one has not thereby produced a thousand heat exchangers, and it has produced no more than a set of hypotheses, each of which must still be tested against a standard that does not care how the hypothesis was obtained. The book calls the failure to keep this fact in view the error of premature adoption, and it regards it as the most likely practical mistake in the use of capable systems, more likely than the dramatic failures that dominate public discussion, because it is the natural consequence of an abundant supply of fluent and confident output.

The reasons for the persistent attractiveness of this error are not mysterious. Generated proposals arrive in the form that a competent proposal would take, with appropriate vocabulary, plausible reasoning, and apparent attention to objections, and the surface properties that human reviewers use as signals of quality are precisely the properties that a system trained on the products of human experts will reproduce. The interview that prompted this book draws attention to a related point in the context of language models, which are described as producing sequences of text that are probable given their training data and not as reporting facts, so that the fluency of the output is no evidence of its correctness. The point extends beyond language. A model that proposes molecules by interpolating among known compounds will produce molecules that look like known compounds, and a model that proposes proofs by

imitating the style of published proofs will produce arguments that look like proofs. In each case the resemblance to the genuine article is a reason to subject the product to a test that depends on nothing about its appearance.

Mathematics provides the clearest case of how generation and verification can be separated, and it deserves detailed treatment for that reason. A mathematical proof is a finite object that can be checked, step by step, against the rules of a formal system. The checking is mechanical, it can be performed by a program of small size whose correctness can be examined independently, and it does not require any understanding of how the proof was found. This has the consequence that a mathematical result produced by a system of unknown reliability can nevertheless be accepted with the confidence appropriate to the checker, which is high. The history of the subject contains precedents. The proof of the four colour theorem by Appel and Haken relied on extensive computation that could not be verified by hand, and its acceptance was accompanied by a debate about what it means to know something that cannot be surveyed by a person; the debate was in a sense resolved by the later production of formally verified versions. The proof of the Kepler conjecture on sphere packing by Hales, which depended on large computations, was accompanied by a project to produce a formal proof, completed in 2014, whose purpose was exactly to replace the judgment of referees who had said they were unable to certify the original with a mechanical certificate. These precedents show that the verification of results of great length is practicable when the results are formalized, and they suggest that the contribution of capable systems to mathematics is likely to be assessed by the number of formally verified results they produce and not by the persuasiveness of their informal arguments.

It would nevertheless be a mistake to infer that the contribution is merely mechanical. Mathematics is not only the proof of stated theorems. It includes the formulation of conjectures, the invention of definitions, and the recognition that a certain structure is the right one for a class of problems, and these activities are judged by their fruitfulness, which is a human assessment that cannot be reduced to checking. A system that proposes conjectures from patterns in numerical data can contribute to the first of these, and there are documented cases in which such patterns have directed human mathematicians toward relationships they then proved. The role of the system in such cases is that of an instrument that makes certain regularities visible, and the human contribution lies in the interpretation and in the proof. The division of labor is stable and productive, and it is the pattern that the book proposes for other fields: the system generates, the verifier decides, and the human community determines what is worth asking.

For fields without a formal verifier the analogue of the proof assistant must be constructed, and the construction is an ordinary scientific and institutional task. A proposal for a new battery electrolyte is verified by synthesizing it and measuring its conductivity, stability, and cycle life, preferably in a laboratory that did not generate the proposal and that has no interest in its success. A proposal for a public health intervention is verified by a trial whose design was fixed before its results were known. A proposal for a bridge is verified by structural analysis, by testing of components, and by the independent review required by building codes. In each

case the structure of the argument is the same, and the practical question is how to ensure that the verification is independent enough of the generation to be informative. The condition developed in Chapter 10 is that the errors of the verifier should not be correlated with the errors of the generator, since a verifier that shares the generator’s blind spots confirms the very mistakes it is supposed to catch. This condition is violated in a subtle and important way when the verifier is another instance of the same kind of system, trained on the same data, and the book therefore treats the use of a model to grade a model as a screening step of limited weight and not as a substitute for external test.

A related consideration is the manner in which proposals are presented. A system that supplies a proposal together with a statement of confidence, an account of the evidence it considers relevant, and an explicit list of the assumptions on which the proposal depends is more useful than one that supplies the proposal alone, provided that the statements are themselves calibrated. Calibration is an empirical property that can be measured: among the proposals to which a system assigns probability of success p , a fraction approximately equal to p should in fact succeed. A system that is badly calibrated, whether overconfident or underconfident, supplies information whose value must be discounted, and the calibration of the system on past problems in the same domain should be reported to those who decide. Proposals should also be accompanied by the record of the search that produced them, so that selection effects can be assessed: the best of a thousand candidates is a different object from a candidate that was the only one examined, and its apparent merit in the training or test data must be corrected for the number of alternatives considered.

The practical recommendation that follows is simple to state and demanding to implement. Each domain in which generation is to be accelerated should be accompanied by a plan for accelerating verification to a comparable degree, and the plan should specify the independent party that will perform verification, the tests it will use, the standard it will apply, and the sources of funding that insulate it from the success of the proposals it judges. Where the plan cannot be made, the rate of adoption should be set by the rate of verification and not by the rate of generation. This is not a counsel of timidity. It is the observation that the benefit of faster generation is realized only to the extent that its output is accepted by a process that can be trusted, and that a process that cannot be trusted converts abundance of proposals into abundance of risk.

Formalization

The effect of verification on the reliability of accepted proposals can be computed. Suppose a generator produces proposals of which a fraction $q \in (0, 1)$ satisfy the acceptance predicate φ of a well-posed problem. A screening test is applied with sensitivity $s = \mathbb{P}(\text{pass} \mid \varphi = 1)$ and false-positive rate $f = \mathbb{P}(\text{pass} \mid \varphi = 0)$, with $0 < f < s \leq 1$. Define the *precision* of the test as $\pi = \mathbb{P}(\varphi = 1 \mid \text{pass})$.

Proposition 6.1. *The precision after one test is $\pi = qs/(qs + (1 - q)f)$. If k tests are applied whose outcomes are conditionally independent given φ , and a proposal is accepted only if it passes all of them, the posterior odds are $\pi_k/(1 - \pi_k) = \frac{q}{1 - q} \left(\frac{s}{f}\right)^k$ when each test has sensitivity s and false-positive rate f . Hence $\pi_k \rightarrow 1$ as $k \rightarrow \infty$ at a geometric rate if and only if $s > f$, and the number of independent tests required for precision at least π^* is*

$$k \geq \frac{\ln(\pi^*/(1 - \pi^*)) - \ln(q/(1 - q))}{\ln(s/f)}.$$

Proof. The first claim is Bayes' rule. For the second, conditional independence gives $\mathbb{P}(\text{all pass} \mid \varphi = 1) = s^k$ and $\mathbb{P}(\text{all pass} \mid \varphi = 0) = f^k$, and the posterior odds are the prior odds multiplied by the likelihood ratio $(s/f)^k$. The remaining statements follow by solving for k . $\square$

The numbers show why the base rate q governs the design of verification. For a hard domain in which only one proposal in a thousand is correct, $q = 0.001$, a single test with $s = 0.9$ and $f = 0.01$, which would be regarded as a good screen, yields $\pi = 0.0009/(0.0009 + 0.00999) \approx 0.083$: more than ninety per cent of accepted proposals are wrong. Two independent such tests give odds of $(0.001/0.999) \cdot 90^2 \approx 8.11$ and precision 0.890, and three give precision 0.9986. The proposition also exposes the cost of dependence among tests. If the tests are positively correlated given $\varphi = 0$, the factor f^k in the denominator is replaced by a larger quantity and the gain from each additional test is reduced. In the limit in which the tests are perfectly correlated the gain from a second test is zero, which is the formal statement of why a verifier that shares the generator's errors adds nothing.

The same structure supports a statement about calibration. Let $\hat{p}$ be the confidence reported by a generator for a proposal and let $\varphi \in \{0, 1\}$ be the outcome. The generator is *calibrated* if $\mathbb{P}(\varphi = 1 \mid \hat{p}) = \hat{p}$ almost surely. The *calibration error* $\text{CE} = \mathbb{E} |\mathbb{P}(\varphi = 1 \mid \hat{p}) - \hat{p}|$ can be estimated by binning on a validation set of proposals with known outcomes, and a reported confidence is to be used in decisions only after correction by the estimated map $\hat{p} \mapsto \mathbb{P}(\varphi = 1 \mid \hat{p})$. Because that map is estimated from past problems, its validity is limited to problems of the same kind, which is the formal content of the instruction that the calibration of a system be reported by domain.

Chapter 7

Automated Laboratories and Materials Discovery

If the generation of proposals is cheap and their verification is expensive, then the economic center of gravity of an accelerated science lies in the instruments of verification, and among these the automated laboratory is the one that has most changed in recent years. An automated laboratory in the relevant sense is a facility in which synthesis, processing, and measurement are performed by robotic equipment under computer control, so that a sequence of experiments can be run with little human intervention and results can be fed back to a program that decides what to try next. The conception is not new. Robotic liquid handling and high-throughput screening have long been standard in pharmaceutical research, and the closed-loop selection of experiments by statistical models is established in fields as varied as catalysis and protein engineering. What distinguishes the present prospect is the combination of this infrastructure with generative and predictive models of much greater capacity, so that the loop from hypothesis to measurement to revised hypothesis could in principle operate at a rate and over a space of possibilities that no group of human experimenters could match. The book accepts that this prospect is serious, and it asks what conditions are required for it to produce knowledge and not only data. Recent computational work, such as the structure prediction of Jumper and colleagues [45] and the large-scale materials screening of Merchant and colleagues [46], shows how fast candidates can now be generated, and it is cited here as an example of the capability and not as support for any proposal of this book.

The first condition is that the physical operations of the laboratory be reliable and documented. A robotic platform performs the same operation repeatedly, and this is both its strength and its danger. If a pipetting step is miscalibrated, or if a reagent has degraded, or if a furnace profile differs from its nominal setting, every experiment that depends on the faulty step is affected in the same way, and the resulting errors are systematic and not random. Systematic errors are not reduced by repetition and they are not detected by statistical analysis of the replicated data, because the replicates agree with one another. The standard remedies belong

to metrology: regular calibration against reference standards, inclusion of control samples with known outcomes in every batch, recording of the provenance of materials and the conditions of each step, and periodic replication of results in a different laboratory with different equipment. A facility that cannot show these records has not produced results that others can use, whatever the sophistication of its software. For this reason the book regards the metrological infrastructure of automated laboratories as part of the verification capacity that must grow in step with generation, and as a proper object of public investment.

The second condition concerns the relation between the model that selects experiments and the experiments themselves. A selection rule is a policy for choosing the next experiment given the results so far, and the policies in common use balance two aims: exploiting regions of the search space in which the results are already good, and exploring regions in which the model is uncertain. The balance is adjustable and its adjustment is a consequential decision, since a policy that exploits too aggressively will converge on a local optimum and report it as the best that can be found, whereas a policy that explores too widely will spend its budget on regions of little promise. The most commonly used criterion, the expected improvement over the best result so far, offers a principled compromise, and a worked comparison, given in the formalization, shows that a candidate with a poorer predicted mean but much greater uncertainty can have a higher expected improvement than a candidate with a better mean and little uncertainty. This is a desirable property of the criterion, but it carries a consequence for how results are reported: the sequence of experiments selected by an adaptive policy is not a random sample of the space, and statements about the distribution of properties over the space, or about the fraction of candidates that succeed, must be corrected for the selection.

The third condition is that claims of discovery be evaluated with attention to multiplicity. A laboratory that tests many thousands of candidates will find, among them, a number that appear to meet the criterion by chance, and if each candidate is judged by an unadjusted significance threshold the list of apparent discoveries will contain many false ones. The statistical literature offers procedures that control the proportion of false discoveries among those reported, of which the procedure of Benjamini and Hochberg [11] is the best known. Control of the false discovery rate is not a substitute for replication, but it provides a defensible basis for deciding which of many candidates deserve the more expensive follow-up. The reporting of a discovery should include the number of candidates tested, the rule by which the reported ones were chosen, and the estimated rate of false discoveries, and the omission of these quantities should be regarded as a defect in the claim, in the same way that the omission of the number of trials is a defect in a claim about the efficacy of a drug.

The discovery of materials has attracted particular attention, and a measured assessment of the claims made on its behalf illustrates the general lessons. Computational screening has been used to propose very large numbers of candidate inorganic compounds, and robotic synthesis has been used to attempt the preparation of some of them. The difficulties that emerge in practice are instructive. A compound that is predicted to be stable may be stable only relative to the competing phases considered, and the prediction can fail if a competing phase

has been omitted. A compound that is stable may nonetheless be unsynthesizable by any available route, because the kinetics of formation favor other products. A compound that is synthesized may differ from the predicted structure in ways that matter for its properties, through disorder, impurities, or off-stoichiometry. And a compound that is synthesized in the desired form may be a laboratory curiosity without economic relevance, because the elements it requires are scarce, or because it cannot be produced in quantity at acceptable cost, or because its properties degrade under operating conditions. The path from a promising candidate to a deployed material has been, historically, a matter of decades, and most of the delay is accounted for by the stages that follow the identification of the candidate. An increase in the rate of candidate identification does not change this fact, and the book therefore treats claims of accelerated materials discovery as claims about the earliest stage of a long chain, to be assessed by their effect on the end of the chain.

None of this diminishes the value of the instrument where it is used appropriately. Within a defined domain, such as the optimization of a known family of catalysts or the exploration of the composition space of an established class of alloys, an automated laboratory working under an adaptive policy can find optima more efficiently than manual methods, can detect interactions among variables that would have been overlooked, and can produce datasets of a size and consistency that are themselves a resource for the community. The appropriate governance of such facilities follows from their character as shared infrastructure. They should be open to external users on stated terms, their data should be published with the metadata needed for reuse, their protocols should be recorded in a form that allows other laboratories to reproduce them, and their funding should not depend on the announcement of discoveries, since a dependence of that kind creates an incentive to report optimistic results. These recommendations are the ordinary norms of good science applied to a new kind of facility, and the reason for stating them is that the novelty of the facility and the enthusiasm that surrounds it make it easy to waive them.

Formalization

Let a response surface $f : \mathcal{X} \rightarrow \mathbb{R}$ be unknown, and let a model provide for each candidate x a Gaussian predictive distribution $f(x) \sim \mathcal{N}(\mu(x), \sigma^2(x))$. Let f^* be the best value observed so far, with larger values preferred. The *expected improvement* of x is $\text{EI}(x) = \mathbb{E}[\max(f(x) - f^*, 0)]$.

Proposition 7.1. *With $z = (\mu - f^*)/\sigma$, and writing Φ and ϕ for the standard normal distribution and density functions, $\text{EI}(x) = \sigma[z\Phi(z) + \phi(z)]$. The function EI is strictly increasing in μ for fixed $\sigma > 0$ and strictly increasing in σ for fixed μ .*

Proof. Write $f = \mu + \sigma\xi$ with ξ standard normal. Then $\mathbb{E}[(\mu - f^* + \sigma\xi)^+] = \sigma \mathbb{E}[(z + \xi)^+] = \sigma \int_{-z}^{\infty} (z + u)\phi(u) du = \sigma[z\Phi(z) + \phi(z)]$, using $\int_{-z}^{\infty} u\phi(u) du = \phi(z)$. Differentiating, $\partial\text{EI}/\partial\mu = \Phi(z) > 0$ and $\partial\text{EI}/\partial\sigma = \phi(z) > 0$. $\square$

The comparison cited in the text is as follows. With incumbent $f^* = 1$, a candidate A with $\mu = 0.5$ and $\sigma = 1.0$ has $z = -0.5$ and $\text{EI} \approx 0.198$, while a candidate B with $\mu = 0.9$ and $\sigma = 0.1$ has $z = -1$ and $\text{EI} \approx 0.0083$. The candidate whose predicted mean is lower by 0.4 is selected, by a factor of about twenty-four in expected improvement, because its uncertainty is ten times greater. This is exploration, and the same calculation shows why the selected experiments are not a random sample.

For multiplicity, let m hypotheses be tested with p -values $p_1, \dots, p_m$, of which m_0 are true nulls. Order them $p_{(1)} \leq \dots \leq p_{(m)}$ and, for a target level $\alpha \in (0, 1)$, let $\hat{k} = \max\{i : p_{(i)} \leq i\alpha/m\}$, rejecting the hypotheses with the $\hat{k}$ smallest p -values (and none if no such i exists). The false discovery rate is $\text{FDR} = \mathbb{E}[V/\max(R, 1)]$, where R is the number of rejections and V the number of true nulls among them.

Proposition 7.2 (Benjamini and Hochberg, 1995). *If the p -values of the true nulls are independent of one another and of the non-null p -values, and each is uniformly distributed on $[0, 1]$, then $\text{FDR} \leq \alpha m_0/m \leq \alpha$.*

The result is cited and not reproved. In the screening of $m = 1000$ candidates with $m_0 = 900$ true nulls at $\alpha = 0.1$ it guarantees $\text{FDR} \leq 0.09$: of the candidates passed to follow-up, no more than nine per cent are expected to be false leads. The bound holds only under the stated independence, which is itself a modelling assumption about the generating process, and the bound should be reported with the check that supports the assumption, or replaced by a procedure valid under dependence.

Chapter 8

Simulation, Planning, and Model Error

The second family of instruments consists of models. A simulation is a computational representation of a system, constructed so that experiments which would be impossible, dangerous, or prohibitively slow in reality can be conducted on the representation, and planning is the use of such representations to choose among courses of action. The practice is ancient in spirit, since a map, a scale model, and a ship's log kept by dead reckoning are all instruments of this kind, and what is new is the scale, the speed, and the apparent authority of the computed result. The expectation that a system of high capability will transform planning is the reasoning behind many of the proposals in the later parts of this book, which are proposals for physical programs whose design depends on models of fluids, structures, ecosystems, orbits, and markets. The literature on decision making under deep uncertainty, which plans against many plausible models and not one [18], supplies the discipline that the later chapters follow. It is therefore necessary to say clearly what a model can and cannot establish, and to state the discipline under which model-based plans should be accepted.

Every model has three sources of error that should be kept apart. The first is parametric error, the uncertainty in the numerical values of quantities that the model takes as input, such as the friction coefficient of a channel, the conductivity of a rock formation, or the strength of a material. Parametric uncertainty is the best understood, because it can be propagated through the model by sampling and its effect on the output can be reported as an interval. The second source is structural error, the error that arises because the equations of the model omit or misrepresent a process. A model of a river that neglects sediment transport may reproduce the discharge accurately and fail to predict that a channel will silt up, and no refinement of its parameters will repair the omission. Structural error is far more difficult to quantify, since by hypothesis the modeller has not thought of the omitted process, and the usual indicators of it are disagreement among structurally different models and failure of the model on cases that were not used to tune it. The third source is numerical error, introduced by the discretization

and the finite precision of the computation, which can be assessed by refining the discretization and observing the convergence of the result.

For systems that exhibit sensitive dependence on initial conditions, a fourth consideration enters, which is the limit on predictability. If two states of a system that differ initially by a small amount δ_0 separate at an exponential rate λ , then the error in a forecast grows until it reaches the scale Δ at which the forecast is no longer informative, and the time at which this occurs depends only logarithmically on the quality of the initial data. The consequence, familiar from the study of the atmosphere, is that improvements in the observation of the initial state buy additional forecasting time at a diminishing rate: a tenfold improvement in the initial data extends the horizon by the same fixed increment, whatever the starting accuracy. This is a property of the system and not of the model. It means that for some systems, no increase in computational capability can extend the reach of deterministic prediction beyond a horizon that can be calculated in advance, and that for these systems the proper objective of modelling is the prediction of statistics, such as the distribution of weather over a season or the frequency of floods in a century, and not of particular trajectories. Climate projection is a prediction of statistics in this sense, and its legitimacy does not depend on the predictability of weather, but the distinction has to be respected in the formulation of the questions that are put to models.

A model-based plan is accordingly a plan made in the presence of uncertainty about which model is correct, and the question of how to decide in these circumstances has a literature of its own. If the possible models can be assigned probabilities with which all parties agree, the plan that minimizes expected loss is the natural choice. But in the cases that matter most, the probabilities are exactly what is in dispute, and a plan that is optimal under one assignment can be disastrous under another. The decision criteria developed for such cases, grouped in the literature on decision-making under deep uncertainty, share the aim of selecting plans that perform acceptably across a wide range of plausible models and not plans that perform best under a single model. Among them is the criterion of minimax regret, which selects the plan that minimizes the largest shortfall, over models, between the plan's performance and the performance of the best plan for that model. It has the advantage of requiring no probabilities. It has the disadvantage of being sensitive to the set of models considered, since a model added to the set can change the result, and the set must therefore be constructed with care and with the participation of those who disagree about it. A further family of approaches seeks plans that are adaptive, meaning that they specify in advance the signals that will trigger revision of the plan and the actions that revision will take, so that the plan commits to a process of learning and not to a fixed trajectory.

Planning by a system of high capability does not remove these difficulties and in one respect aggravates them. A system that can explore a very large space of plans will find those that perform best under the model it has been given, and plans that perform best under a model tend to be those that exploit its errors. The phenomenon, known in several fields as overfitting to the simulator or as the exploitation of model artifacts, is the consequence of optimization pressure applied to an imperfect representation. A control strategy found by optimization

against a simulation of a bridge may be brittle in the real structure for reasons that no one anticipated, and a design found by optimization against a hydrological model may rely on a numerical feature of the model that has no counterpart in nature. The protection is not to avoid optimization but to subject the plans that optimization finds to tests the optimizer did not see: models of different structure, physical experiments at reduced scale, and staged deployment in which the plan is first applied to a small part of the system and its performance is compared with the prediction. The discipline is stated in Chapter 10 as a bound on the fidelity gap, and it is reflected in the readiness classification of Chapter 16, in which projects are not permitted to advance until their central models have been tested against measurements.

A last observation concerns the social role of models. A model embodies assumptions, and when it is used to justify a decision the assumptions become a hidden part of the decision. The citizen who is told that a plan was chosen because a computation showed it to be optimal is in no position to contest the plan unless the assumptions can be stated, the data exposed, and the sensitivity of the result to the assumptions shown. The requirement that a model-based recommendation be accompanied by this information is a requirement of accountability in the sense of Chapter 2: the party that adopts the recommendation remains answerable for the decision, and answerability requires access to the grounds. The advance of capability in modelling should therefore be accompanied by an advance in the practice of documentation, in which models intended for public decisions are versioned, archived, and open to independent execution. A model that cannot be re-run by a skeptical party is an oracle, and the program of this book assigns no authority to oracles.

Formalization

Two results are stated, the first on the limit of predictability and the second on decision under model uncertainty.

Definition 8.1 (Predictability horizon). *Let the separation of two trajectories obey $\delta(t) = \delta_0 e^{\lambda t}$ with $\lambda > 0$ until it reaches the saturation scale $\Delta > \delta_0$. The predictability horizon is the time $T(\delta_0)$ at which $\delta(T) = \Delta$.*

Proposition 8.2. *$T(\delta_0) = \lambda^{-1} \ln(\Delta/\delta_0)$. For any $r > 1$, reducing the initial error from δ_0 to δ_0/r increases the horizon by exactly $\lambda^{-1} \ln r$, independent of δ_0 . To extend the horizon by a prescribed amount τ the initial error must be reduced by the factor $e^{\lambda\tau}$.*

Proof. Setting $\delta_0 e^{\lambda T} = \Delta$ gives the formula, and $T(\delta_0/r) - T(\delta_0) = \lambda^{-1} \ln r$ follows by subtraction. The last statement is the case $\ln r = \lambda\tau$. $\square$

With $\lambda = 0.5 \text{ day}^{-1}$, a figure of the order of magnitude associated with the error doubling of mid-latitude weather over a few days, a tenfold reduction in initial error extends the horizon by $\ln 10/0.5 \approx 4.6$ days, and a hundredfold reduction by 9.2 days. The cost of observation,

which increases at least in proportion to the reduction achieved, therefore buys additional forecast time at a rate that falls exponentially, and beyond a horizon set by the largest feasible reduction the forecast of individual trajectories is unavailable and only statistics remain.

For decision under uncertainty, let $\mathcal{A}$ be a finite set of actions and $\mathcal{M}$ a finite set of models, with loss $L(a, m) \geq 0$. Define the *regret* $r(a, m) = L(a, m) - \min_{a'} L(a', m)$ and the *minimax-regret action* $a^* \in \arg \min_a \max_m r(a, m)$ with value $r^* = \min_a \max_m r(a, m)$.

Proposition 8.3. *For every probability distribution π on $\mathcal{M}$, the Bayes action $a_\pi \in \arg \min_a \mathbb{E}_\pi L(a, m)$ satisfies $\mathbb{E}_\pi r(a_\pi, m) \leq r^*$. Hence the expected regret of a decision maker who chooses by any prior is bounded by the minimax regret, which can be reported as the price of declining to commit to a prior.*

Proof. Since $\mathbb{E}_\pi r(a, m) = \mathbb{E}_\pi L(a, m) - \mathbb{E}_\pi \min_{a'} L(a', m)$, the Bayes action for loss also minimizes expected regret. Therefore $\mathbb{E}_\pi r(a_\pi, m) \leq \mathbb{E}_\pi r(a^*, m) \leq \max_m r(a^*, m) = r^*$. $\square$

A numerical illustration shows that the criteria can disagree. Let there be two models and three actions with losses $L(a_1, \cdot) = (0, 6)$, $L(a_2, \cdot) = (4, 0)$, $L(a_3, \cdot) = (2, 3)$. The regrets are $(0, 6)$, $(4, 0)$, $(2, 3)$, so that the maximum regrets are 6, 4, 3 and the minimax-regret action is a_3 with $r^* = 3$. Under equal weights the expected losses are 3, 2, and 2.5, so that the Bayes action is a_2 , whose maximum regret is 4 and whose expected regret is 2, below r^* as the proposition requires. A decision maker unwilling to commit to equal weights, and who would regret a loss of 4 under the first model more than the cost of insurance, would choose a_3 . The example illustrates that the selection of a criterion is a normative decision which belongs to those who bear the consequences and not to the modeller, a conclusion that supports the allocation of authority described in Chapter 13.

Chapter 9

The Difference Between Solving and Deploying

A system of high scientific capability would, by hypothesis, be able to generate candidate solutions in quantities and at speeds that exceed the capacity of human institutions to evaluate them. This feature has an immediate consequence that is often overlooked in discussions that treat capability as synonymous with benefit: the bottleneck of a program in which generation is cheap is verification. A machine might derive an efficient design for a planetary water-transport network, a novel catalyst, a proof, or a therapeutic protocol, and in each case the derivation is only the first stage. The proposal must still be shown to be physically realizable under the stated assumptions, reproducible by independent parties, environmentally admissible over the relevant time scale, and socially acceptable to those it affects. These are distinct tests, each performed by different methods and with different standards of evidence, and passing one does not entail passing the others. A catalyst that functions in a laboratory may not scale, a scalable process may be ecologically intolerable, and an ecologically tolerable process may be unacceptable to the communities who would bear its costs or whose rights it would override.

The distinction between solving and deploying is accordingly not a distinction between theory and practice but a distinction between establishing that a solution exists and establishing that it should be adopted, and the second is a decision. Whether an efficient planetary water-transport design ought to be built depends on ecological effects along its route, on the rights of communities and states whose land and water it would affect, on the economic cost relative to alternatives such as desalination, regional recycling, and demand reduction, and on competing uses of the same freshwater. These considerations are not resolved by the derivation, and a system that generated the design cannot be the authority that decides whether it is built without reintroducing, in another form, the substitution of artifact for institution that the argument opposes. The interview's discussion of marketing practices reinforces the need for independent verification: it observes that claims about capability have frequently reached the public directly from the organizations that profit from them, and that interpretations by

independent parties arrive only later, after the initial impression has formed. A verification regime that depends on the developer's own evaluation of its own system is exposed to the same structure of interest.

This part of the program connects to a broader requirement on any use of capable systems in science. Where a system proposes a mathematical result, a proof assistant or independent expert review can establish correctness, but correctness of a formal statement does not establish that the statement captures the intended claim, so that an additional check of fidelity between the formal object and the informal question is needed. Where a system proposes an experiment, the experiment must be performed and replicated by parties whose success is not tied to confirmation. Where a system proposes an intervention in a living system, the evidence standards of clinical and ecological research apply without exemption. The general principle is that the credibility of a proposal derives from the independence and redundancy of the procedures that test it, and not from the capability of its source.

Formalization

Let Π_c be the set of proposals generated at capability level c , with $|\Pi_c| = N_c$. Let $\varphi_1, \dots, \varphi_k : \Pi_c \rightarrow \{0, 1\}$ be verification predicates corresponding to physical realizability, reproducibility, ecological admissibility, and social acceptability (so $k = 4$ in the base case), where $\varphi_j(\pi) = 1$ means that proposal π passes test j in the idealized sense. A proposal is *deployable* if $\varphi_j(\pi) = 1$ for all j , and the deployable set is $\Pi_c^* = \bigcap_{j=1}^k \varphi_j^{-1}(1)$.

In practice tests are imperfect. Let E_j denote the event that test j accepts a given defective proposal, with $\mathbb{P}(E_j) = \varepsilon_j \in (0, 1)$, and let $b \in [0, 1]$ denote the fraction of defective proposals in Π_c .

Proposition 9.1. (i) If $E_1, \dots, E_k$ are independent, the expected number of defective proposals that pass all tests is $N_c b \prod_{j=1}^k \varepsilon_j$. (ii) Without independence, $\mathbb{P}(\bigcap_j E_j) \leq \min_j \varepsilon_j$, with equality if and only if, for an index m attaining the minimum, $E_m \subseteq E_j$ up to events of probability zero for every j , that is, if the acceptance events are nested. (iii) The product $\prod_j \varepsilon_j$ is not an upper bound in general: for positively dependent tests $\mathbb{P}(\bigcap_j E_j)$ may exceed it.

Proof. (i) By independence the probability that a defective proposal passes all tests is $\prod_j \varepsilon_j$, and the expectation follows by linearity over the $N_c b$ defective proposals. (ii) Since $\bigcap_j E_j \subseteq E_m$ we have $\mathbb{P}(\bigcap_j E_j) \leq \mathbb{P}(E_m) = \min_j \varepsilon_j$, and equality holds exactly when $\mathbb{P}(E_m \setminus \bigcap_j E_j) = 0$, which is the stated nesting condition. Identical acceptance events are the special case in which all ε_j coincide. (iii) Take two tests with $E_1 = E_2$ and $\varepsilon_1 = \varepsilon_2 = \frac{1}{2}$, so that $\mathbb{P}(E_1 \cap E_2) = \frac{1}{2} > \frac{1}{4}$. $\square$

The proposition yields the program's requirement of *independence of verification*: the benefit of adding a test is the multiplicative factor ε_j only to the extent that its errors are uncorrelated

with those of the others, and a test operated by the proposer is not independent of the proposal.

Admissible range and certified throughput are distinct quantities. The admissible range, defined by inclusion in Chapter 1, concerns which trajectories could be realized. Let g_c be the rate at which proposals are generated at level c , nondecreasing in c , let v be the rate at which proposals can be verified at a given level of verification effort, and define the *verified deployment throughput* $\Theta(c) = \min(g_c, v)$.

Proposition 9.2. *If $g_c \geq v$ for all $c \geq c_0$, then $\Theta(c) = v$ for all $c \geq c_0$, while $R_c \cap A$ may continue to enlarge strictly for $c \geq c_0$. Additional capability beyond c_0 then yields no additional certified deployments unless v increases, even though the admissible range may continue to grow.*

Proof. For $c \geq c_0$ the minimum in the definition of Θ equals v . The realizable sets R_c are nested and nothing in the definition of Θ constrains the difference $(R_{c'} \setminus R_c) \cap A$ to be empty. $\square$

Chapter 10

Verification, Fidelity, and Independent Checking

The preceding chapters of this part have treated generation, experimentation, and modelling as instruments whose products require checking, and the present chapter treats the checking itself. Verification is not a single activity. It includes the formal checking of a proof, the replication of an experiment, the review of a design against a code, the audit of a financial statement, the validation of a model against data, the inspection of a structure after construction, and the monitoring of a system in service. What these have in common is that each is meant to establish, by a procedure independent of the one that produced the claim, that the claim is correct to a stated standard. The independence is essential, and so is the standard, and a large part of the chapter is concerned with the ways in which each can quietly fail. The distinction between a model that has been verified and one that has been confirmed is old in the earth sciences [47], and current risk-management guidance for artificial intelligence treats validation and monitoring as continuing obligations and not as one-time events [48].

The question of independence is subtler than it first appears. Two procedures are independent in the relevant sense when the circumstances that cause one to give a wrong answer are not the circumstances that cause the other to do so. They may share nothing in the way of personnel, funding, or method, and still be dependent, if they rely on the same data, the same assumptions about the system, or the same software library. The history of engineering contains many failures in which a design was reviewed by several parties who each relied on the same standard calculation, and in which an error in the calculation passed all the reviews. An independent check should therefore be defined by the structure of its possible errors and not by the identity of the parties that perform it. The practical corollary is that the most informative checks are those that use a different method, as when a numerical result is compared with an analytic bound, or a simulation with a measurement, or an automated prediction with the judgment of an experienced practitioner, because a different method is more likely to fail differently. Where verification is performed by a system of the same kind as the one that

generated the claim, the probability of shared error is high, and the check should be treated as screening.

The question of the standard is a question about what is being claimed. A verification procedure checks a claim against a model of reality, and the claim established is that the thing satisfies a property in the model. Whether the thing satisfies the property in reality depends on how closely the model represents reality in respect of that property. The difference is the *fidelity gap*, and it is the place at which formal verification, impressive as it is within its domain, reaches its limit. A control program may be proved to keep a system within safe bounds under a stated model of the plant, and the proof is a mathematical fact; whether the plant behaves as the model says is an empirical question that the proof does not address. This is not an objection to formal methods, which eliminate a large class of errors, but a reminder that their guarantees are conditional on assumptions that have to be validated by other means. The conditional form of the guarantee should be stated in the documentation, so that a reader knows what has been proved and what has been assumed.

A third consideration is the trusted base. Every verification procedure rests on components that are not themselves verified by the procedure: the proof checker, the measuring instrument, the reference standard, the regulatory body, the auditors. These components constitute the trusted base of the verification, and the reliability of the conclusion cannot exceed the reliability of the trusted base. Two principles guide its design. The first is that the trusted base should be as small as possible and as simple as possible, so that it can itself be examined by many parties. The second is that the number of components in it should be counted, because the failure probabilities of the components accumulate, and a procedure with many individually reliable components may be less reliable than one with few. In mathematics the proof assistants with small kernels illustrate the first principle, since the kernel is a short program whose correctness can be studied in isolation. In the social domain, the equivalent of a small kernel is a clear rule of evidence, stated in advance and applied by a body that is accountable for its application.

Verification also has an economic dimension that is frequently overlooked. It is a form of labor, and like other forms of labor it requires skilled people, equipment, and time, and it competes for resources with the activities it is meant to check. When generation becomes cheap and verification remains costly, the pressure to reduce the cost of verification will be strong, and the techniques by which the cost is reduced include sampling, automation, delegation to the generating party, and the relaxation of the standard. Each of these has a legitimate form and an illegitimate one. Sampling is legitimate when the sampling plan is designed to give a stated protection against defects and illegitimate when the samples are chosen by the party being checked. Automation is legitimate when the automated checker is itself validated and illegitimate when it is trusted because it is convenient. Delegation to the generating party, which takes the form of self-certification, is legitimate only for low-stakes matters and in the presence of penalties for false certification. The book's insistence on independent checking is accordingly an insistence that these savings should not be taken in domains where the consequences of error are severe or irreversible, and that the funding of independent checking

should be treated as a necessary expense of any program whose output will be relied upon.

Finally, verification does not end when a thing is deployed. Systems in service encounter conditions that were not anticipated in the tests, and their components age. Monitoring in service is therefore a continuing part of verification, and it is supported by the design of the system: a system that is instrumented to report on its own condition, whose logs are preserved and open to review, and whose failures are investigated and published, supplies the evidence by which earlier claims are corrected. Aviation provides the model in the form of its practice of investigating accidents and incidents, publishing the findings, and amending standards accordingly, and the practice has contributed more to the safety of air travel than any single technical innovation. The corresponding practice for programs of the kind this book describes is to require, as a condition of authorization, a commitment to monitoring and disclosure over the life of the program, with an independent body empowered to halt operations when the evidence warrants.

Formalization

Two propositions are given, the first on the accumulation of errors in the trusted base and the second on the transfer of a guarantee across the fidelity gap.

Let the trusted base of a verification procedure consist of components $1, \dots, k$, and let B_i be the event that component i fails in a way that could produce a wrong verdict, with $\mathbb{P}(B_i) = \varepsilon_i$. Let W be the event that the verdict is wrong.

Proposition 10.1. *If $W \subseteq \bigcup_{i=1}^k B_i$ then $\mathbb{P}(W) \leq \sum_{i=1}^k \varepsilon_i$, with no assumption on the dependence among the B_i . If the B_i are independent, $\mathbb{P}(\bigcup_i B_i) = 1 - \prod_i (1 - \varepsilon_i) \leq \sum_i \varepsilon_i$, and the difference is at most $\frac{1}{2}(\sum_i \varepsilon_i)^2$. Hence for a trusted base of k components each with $\varepsilon_i \leq \varepsilon$ the error bound grows at most linearly in k .*

Proof. The first bound is Boole's inequality. For independent events, $\mathbb{P}(\bigcup B_i) = 1 - \prod (1 - \varepsilon_i)$, and by induction $\prod_i (1 - \varepsilon_i) \geq 1 - \sum_i \varepsilon_i$, which gives the inequality. The bound on the difference follows from the Bonferroni inequality, $\mathbb{P}(\bigcup B_i) \geq \sum_i \varepsilon_i - \sum_{i < j} \varepsilon_i \varepsilon_j$, and $\sum_{i < j} \varepsilon_i \varepsilon_j \leq \frac{1}{2}(\sum_i \varepsilon_i)^2$. $\square$

For five components with $\varepsilon_i = 10^{-3}$, the bound is 5×10^{-3} and the exact probability under independence is $1 - 0.999^5 \approx 4.99 \times 10^{-3}$. The value of the proposition lies not in the arithmetic but in the design rule: to reduce the error of a verdict by a factor r one must either reduce every term of the sum or remove components from the base, and the second is generally cheaper.

For the fidelity gap, let a system be characterized by a parameter $m \in \mathcal{M}$ and a design by x , and let $\rho(x, m) \in \mathbb{R}$ be a *robustness margin* for a required property, with the property satisfied if and only if $\rho(x, m) > 0$. Let m^* be the true system and $\hat{m}$ the model, and suppose the margin is Lipschitz in the model: $|\rho(x, m) - \rho(x, m')| \leq L d(m, m')$ for a metric d .

Proposition 10.2. *If verification in the model establishes $\rho(x, \hat{m}) \geq \mu > 0$ and the fidelity gap satisfies $d(\hat{m}, m^*) \leq \eta$, then the property holds for the true system whenever $L\eta < \mu$, and the guaranteed margin in the true system is at least $\mu - L\eta$. If $L\eta \geq \mu$ the verification is silent about the true system.*

Proof. $\rho(x, m^*) \geq \rho(x, \hat{m}) - L d(\hat{m}, m^*) \geq \mu - L\eta > 0.$ □

The proposition has two practical consequences. First, the quantity that determines whether a formal guarantee has practical force is the ratio $\mu/(L\eta)$, which can be increased by building in margin (raising μ), by seeking designs that are insensitive to the model (lowering L), or by improving the model through measurement (lowering η). Second, it explains why optimization against a model is dangerous, as noted in Chapter 8: an optimizer reduces the margin μ in the direction in which the model is most favorable, so that the optimized design has a small μ and the proposition's condition is least likely to hold. The remedy is to optimize a worst-case margin over a set $\{m : d(m, \hat{m}) \leq \eta\}$, which restores the guarantee at the price of a design that is less than optimal for the nominal model.

Part III

Institutions

Chapter 11

Universal Education

The third part of the book concerns institutions, and it begins with education because education is the institution through which a society reproduces its capacity for self-direction. The place of education in a program of human direction is not that of one application among others. If the argument of the earlier chapters is accepted, the capacity to direct, which is to say the capacity to understand the options, to contest the claims of those who propose them, and to bear responsibility for choices, is the scarce resource, and a society whose members cannot exercise it has nothing to which capable systems could be subordinate. Later reviews found a smaller effect, of the order of 0.8 standard deviations both for human tutoring and for well-built intelligent tutoring systems [26], so that the original figure is best read as an upper bound. The educational question is therefore twofold. It asks how the instruments described in the second part can extend to every person an education now available to few, and it asks how that extension can be arranged so that it strengthens and does not displace the human relations in which character and judgment are formed. The instructional ideas on which the argument draws, in particular that a tutor supports a learner by taking over only those parts of a task that the learner cannot yet perform, originate in the study of tutoring [49].

The empirical foundation for the first aim is the finding, reported by Bloom in 1984 [12], that students who received one-to-one tutoring together with mastery-based progression performed on average about two standard deviations above students in conventional classes. The finding was presented as a challenge to educational research, which Bloom called the two sigma problem: to find methods of group instruction as effective as tutoring, since individual tutoring could not be provided to all. Later syntheses have reported smaller effects, of the order of eight-tenths of a standard deviation, for both human tutors and carefully designed computer-based tutors, and the difference between the original and later estimates is a reminder that the effect depends on implementation and measurement. Even the smaller estimate is large by the standards of educational interventions. The argument for the use of capable systems in education begins here: if the benefit of individual attention is substantial and the supply of tutors is limited, then a system that can offer something approximating individual attention

at low marginal cost addresses the main limiting factor, and does so in a way that is especially valuable for the children who now receive least.

The case should nonetheless be stated with the qualifications that the evidence warrants. Tutoring effects have been measured in settings where the tutor, human or automated, is part of an organized course of study with defined objectives, assessments that detect mastery, and a framework within which the student is motivated to attend. A conversational system that answers any question on demand is not thereby a tutor in this sense. Without structure it may encourage the substitution of answer-seeking for understanding, a danger recognized in the literature on learning, which distinguishes the performance a student shows while receiving help from the learning that persists when help is withdrawn. A system designed for education should therefore be evaluated by the retention and transfer of what is learned, assessed without assistance after an interval, and not by the satisfaction of students or the correctness of the answers they receive. It should be designed to require effort of the student, to give hints before solutions, and to hold the student responsible for explaining the reasoning, since the evidence supports the view that productive struggle is part of the mechanism by which learning occurs.

The second aim, the preservation of human relations, determines the allocation of tasks between machines and persons, and it is here that the principle of the book bears directly. Some of what happens in a school is instruction, the transmission of knowledge and skill, and much of it can be supervised or delegated. Some of it is assessment of the kind that decides a person's future, and this belongs in the category of judgment that is not delegated. Some of it is the formation of habits, the encouragement of the discouraged, the recognition of a child in difficulty at home, the example of an adult who takes the child seriously, and the initiation into a community of practice, and these are functions that depend on the presence of a person who can be trusted and who is accountable. They are not functions that a machine performs less well and that might one day be performed equally well. They are functions whose value depends on the fact that a person is performing them, in the same way that the value of a promise depends on its being made by someone who can be held to it. The program therefore reserves them to human beings and treats the time that automation frees from instruction as time to be devoted to them.

This leads to a quantitative question on which the program's coherence depends. If instruction is automated to a great extent, how many human mentors does a school require? The answer, derived in the formalization, is that the number is bounded below by the non-delegable portion of human attention per student and does not fall to zero however much instruction is automated. The point is the educational counterpart of a well-known observation in the study of parallel computation, that the speedup of a process is limited by the part that cannot be sped up. It runs against both expectations the public discussion encourages. It contradicts the view that automation of instruction makes teachers unnecessary, and also the view that it will make teaching less demanding. What it predicts is that the role of the teacher will shift toward mentoring, assessment, and the supervision of systems, that the number of teachers required per student will be reduced less than the share of instructional time automated would

suggest, and that the training of teachers should change accordingly. It also predicts that the most important investment in an educational transition is in the number and the working conditions of the persons who perform the non-delegable functions, since these constitute the binding constraint.

Questions of governance follow. The curriculum of a system that teaches millions of students is an instrument of extraordinary influence, and its content, emphases, and omissions are choices that ought to be made by processes that the public can contest. The interview notes that systems built from large collections of text reproduce the biases of those texts and the perspectives of the people most heavily represented in them, and the observation applies with force to educational content, where the stakes include the formation of children's picture of the world. The program accordingly requires that educational content be authored or approved through documented procedures with the participation of teachers, parents, and subject specialists, that the sources of generated material be disclosed, and that independent evaluators have access to the system for the purpose of testing it for bias and error. It requires further that the data collected from students be used solely for their education, be subject to deletion on request, and not be used to profile children for commercial purposes. A system that fails these requirements may teach effectively and still be unacceptable, since its effects on the interests of children are mediated by an organization whose own interests are not those of the children.

Formalization

Let a school serve N_s students and employ N_t human mentors, each providing H hours of contact per week. For each student, weekly educational activity decomposes into hours of three types: judgment-type activity h_J (assessment, mentoring, safeguarding), which by the principle of the book is non-delegable; supervised activity h_U , which can be performed by a system under human oversight; and activity h_L performed by a system without moment-to-moment oversight. Suppose a fraction $a \in [0, 1]$ of h_U is performed by the system, the remaining $(1 - a)h_U$ by a mentor. The human attention required per student per week is

$$h(a) = h_J + (1 - a)h_U.$$

Proposition 11.1. *The mentor requirement is $N_t \geq N_s h(a)/H$, so the student-to-mentor ratio is at most $H/h(a)$. The function $h(a)$ is decreasing in a and satisfies $h(a) \geq h_J$ for all $a \in [0, 1]$, so that no degree of automation of supervised activity raises the student-to-mentor ratio above H/h_J . The maximum feasible ratio is H/h_J , attained only when $a = 1$ or $h_U = 0$.*

Proof. Total human hours demanded is $N_s h(a)$ and the supply is $N_t H$, whence the inequality. Since $h_U \geq 0$ and $a \leq 1$, $h(a) \geq h_J$ with equality iff $a = 1$ or $h_U = 0$. $\square$

Numerically, take $H = 25$ contact hours, $h_J = 1$ and $h_U = 4$ hours per week. With no automation ($a = 0$), $h = 5$ and the ratio is 5. With $a = 0.75$, $h = 2$ and the ratio is 12.5, so that a country with 10^6 students requires 80,000 mentors. In the limit $a = 1$ the ratio is 25 and 40,000 mentors are required. Complete automation of supervised work halves the requirement relative to $a = 0.75$, but it cannot go lower, and the whole of the saving is bounded by the factor $h(0)/h_J = 5$. The proposition thus exhibits the binding role of the non-delegable component.

A second computation concerns the size of the benefit that motivates the effort. If the achievement of control students is normally distributed and the treatment shifts the mean by d standard deviations, a treated student of average ability would be at the $\Phi(d)$ quantile of the control distribution. For $d = 2$ this is $\Phi(2) \approx 0.977$, and for $d = 0.8$ it is $\Phi(0.8) \approx 0.788$. The program uses the lower value for planning, since it is better supported, and treats the higher as the result of conditions that may not be reproducible at scale. Whatever value is used, the benefit is reported together with its measurement protocol, namely retention and transfer assessed without assistance after a stated delay.

Chapter 12

Curriculum, Access, and Translation

Whether an education can be universal depends on more than the capacity to teach. It depends on whether instruction reaches persons in a language they understand, in a form they can use, at a cost they can bear, and with a content that acknowledges their circumstances. The failures of access that remain in the world are largely of these kinds. Children in many regions are taught in a language they do not speak at home, adults who have never been taught to read have no way to use text-based services, persons with sensory or cognitive disabilities encounter materials not designed for them, and households without reliable electricity or connectivity cannot rely on any service that assumes both. This chapter examines what capable systems can contribute to these problems, what they cannot, and what distinguishes a program of access that expands human choice from one that merely distributes a product.

Translation is the most obvious contribution and the one for which the evidence is most developed. Machine translation of text has improved to the point that for widely used language pairs it is serviceable for many practical purposes, and for languages with large digital corpora the quality approaches what a competent bilingual reader would accept in non-critical settings. The qualification, which the interview presses in its discussion of the dependence of language technology on the texts available, is that the quality is distributed very unequally. Of the several thousand languages in the world a small number account for nearly all of the data on which systems are trained, and for the rest the systems are poor or unavailable. This asymmetry reproduces the asymmetry of power that produced it: the languages with the most data are those of the societies with the most institutions, and the speakers of the others receive the least benefit and bear the greatest risk of being represented incorrectly. The program accordingly distinguishes between translation as a convenience for speakers of well-resourced languages and translation as an instrument of inclusion for speakers of others, and it assigns to the latter a priority which the market does not.

The priority is not achieved by simply collecting data from the communities concerned. The collection of the speech and writing of a minority community for the training of a commercial system raises questions of consent, ownership, and benefit that have been the subject of

sustained criticism, in particular where a language is embedded in cultural practices that the community regards as not to be shared. A defensible approach treats the community as a party to the arrangement with authority over the use of its language data, grants that authority in a form that persists, ensures that the products made from the data are available to the community on terms it finds acceptable, and supports the community's own capacity to build and maintain the tools. Such an approach is slower and more expensive than extraction, and the program accepts the cost on the ground that a service built on the dispossession of its beneficiaries is not an instrument of access in the sense the book intends.

A second question concerns curriculum. The content of education is not a neutral selection from a body of settled knowledge, and universal education requires decisions about what is taught first, what is taught to all, and what is left to specialization. The mathematics of a stylized model helps to show the structure of these decisions without settling them. If the frequency of words in a language declines roughly in inverse proportion to their rank, as the empirical regularity associated with Zipf [40] describes, then a small vocabulary covers a large share of ordinary text, and the share gained by each additional word declines steadily. In the formalization, a vocabulary of one thousand words in a language of one hundred thousand covers roughly sixty per cent of running text under this stylization, and doubling it adds only about six percentage points. The lesson, which holds more generally for skills, is that a basic competence covering the bulk of everyday demands can be taught to everyone quickly, that the remaining demands are served at ever higher cost, and that the point at which universal provision should stop and specialization begin is a judgment about benefit and cost which a community must make. A capable system can reduce the cost of moving beyond the basic level for an individual who wants to, and it can personalize the order and pace of learning; it cannot decide what the community owes every child.

Adaptive pacing deserves discussion, because it is both the great promise of automated instruction and a source of subtle difficulty. A system that adjusts the difficulty of material to the performance of the student can keep each student in a zone where tasks are demanding but achievable, and the advantage over a class paced to the average is evident. The difficulty is that the adjustment is governed by a measure of performance, and the measure is not the same thing as the educational aim. A system optimized to maximize correct answers will tend to select easy material, and a system optimized to maximize engagement will tend to select material that holds attention, which is not always the same as material that educates. The remedy is to define the objective in terms of learning measured independently of the system's own measurements, to audit the behavior of the system for drift toward easy or entertaining tasks, and to allow teachers to inspect and override the sequence. It is a specific application of the general principle that the objective of an optimizing process should be set by those accountable for it and checked against an independent standard.

Access in the physical sense must also be provided for, and here the engineering of delivery matters as much as the sophistication of the software. A system that requires continuous high-bandwidth connectivity excludes the households that most need it. A design suited to the real

distribution of circumstances stores content locally, operates on inexpensive hardware, tolerates intermittent power and connection, synchronizes when it can, and degrades gracefully when it cannot. The design should also provide for shared use, since in many settings a device is used by a family or a classroom and not by an individual. Provision for persons with disabilities is likewise a matter of design from the beginning and not an adjustment at the end: captions, audio description, alternative input methods, and content adapted to differences in cognitive style are requirements which, once met, tend to benefit all users. The program's measure of success is the proportion of the target population that can use the service in practice, assessed by sampling among those least likely to be reached, and not the number of installations or the volume of traffic.

The last matter is the relation between universal education and the maintenance of a public culture. A curriculum delivered through individualized systems risks replacing the common body of knowledge and the shared experiences that allow citizens to deliberate together with a set of private educations that do not overlap. The risk is real in proportion to the degree of personalization. It can be addressed by preserving a common core, taught and discussed in groups, in which students encounter one another's views, and by reserving personalization for the pace and manner of learning and not for the content of the shared core. The reasons for this are those of the earlier chapters: a society directs itself only if its members can understand one another's claims, and an education that fragments the basis for mutual understanding weakens the capacity it was meant to build.

Formalization

Let a language have vocabulary of size V and suppose the relative frequency of the word of rank r is $f_r = r^{-1}/H_V$, where $H_n = \sum_{k=1}^n k^{-1}$ is the n -th harmonic number. The *coverage* of the K most frequent words is $C(K) = H_K/H_V$.

Proposition 12.1. *C is increasing and concave in K in the sense that the marginal coverage $C(K+1) - C(K) = [H_V(K+1)]^{-1}$ is decreasing in K . For large K and V , $C(K) \approx (\ln K + \gamma)/(\ln V + \gamma)$, where $\gamma \approx 0.5772$ is Euler's constant, so that to double the number of words multiplies the uncovered share by approximately $1 - \ln 2/(\ln V + \gamma)$ and adds $\ln 2/(\ln V + \gamma)$ to coverage independent of K .*

Proof. The marginal coverage follows from $H_{K+1} - H_K = (K+1)^{-1}$. The asymptotic formula follows from $H_n = \ln n + \gamma + O(1/n)$, and the increment for doubling is $[\ln(2K) - \ln K]/(\ln V + \gamma) = \ln 2/(\ln V + \gamma)$. $\square$

For $V = 10^5$ one finds $C(100) \approx 0.429$, $C(500) \approx 0.562$, $C(10^3) \approx 0.619$, $C(2 \times 10^3) \approx 0.676$, and $C(5 \times 10^3) \approx 0.752$. Each doubling adds about $\ln 2/(\ln 10^5 + 0.577) \approx 0.057$ to coverage, in agreement with the increment from 10^3 to 2×10^3 words. The pure Zipf form is a stylization,

and real corpora depart from it, particularly in the tail, but the qualitative conclusion that a modest vocabulary yields the bulk of coverage and that further coverage becomes steadily more expensive is robust. It supports the staging of curricula from a universal core to optional extensions, with the stopping point of the core fixed by a collective decision that weighs the marginal benefit $[H_V(K+1)]^{-1}$ against the marginal cost of teaching the $(K+1)$ -th item.

A second computation addresses the evaluation of adaptive systems. Let a student's mastery after a sequence of tasks be m_T and let a system choose task difficulties d_t to maximize the proportion of correct responses $\sum_t \mathbf{1}[\text{correct}_t]$. If the probability of a correct response is a decreasing function $p(d_t - m_t)$ of the gap between difficulty and mastery, while the gain in mastery from a task is $g(d_t - m_t)$ with g maximized at a positive gap $\delta^* > 0$, then the proportion-correct objective is maximized by the easiest available tasks (the smallest admissible d_t), while the gain objective is maximized at $d_t = m_t + \delta^*$. The two objectives agree only if g is itself maximized at the easiest task. This elementary observation is the formal reason for the requirement that the objective be defined as independently measured learning, since otherwise the optimization drives the system to the task level at which the student learns least.

Chapter 13

An Architecture of Authority

The principle that organizes the program receives its most concrete expression in the proposal for robotic surrogates directed by human operators. In this arrangement a physical or virtual embodiment is equipped with local autonomy for bounded functions: maintaining balance, avoiding collisions, limiting applied force, preventing injury, and executing emergency stops. These functions are well suited to automation because they are fast, their success conditions are specifiable, and their failure modes can be tested against explicit criteria. Decisions concerning teaching, caregiving, emotional development, and interpersonal interaction remain with qualified human operators, who direct the embodiment and who are responsible for what it does. A single operator might supervise several embodiments, although the safe number depends on the complexity of the task, the latency of the communication link, and the degree of continuous attention required. The allocation follows the tradition of human supervisory control [9], and the warning of Scott [25] about schemes of legibility imposed from above applies to any such allocation that is designed without those who live under it. The arrangement is notable because it specifies an architecture of authority and does not rely on an abstract promise of beneficial behavior: the allocation of control between machine and person is explicit, and so is the allocation of responsibility. The classification of automation by stages and levels, which separates the acquisition of information, its analysis, the selection of an action, and its implementation, gives the allocation its standard vocabulary [50], and the ladder of citizen participation [51] states the corresponding distinction between consulting a public and sharing power with it.

The architecture applies, with differing limits on delegation, to education, eldercare, rehabilitation, remote medical assistance, and hazardous industrial work, and the differences among these domains are themselves informative about the principle. Where the hazards are physical and the human operator's role is to select goals, as in hazardous industrial work, the supervisory ratio may be high and the embodiment may carry substantial autonomy over motion. Where the activity is relational, as in teaching and caregiving, the value of the activity resides substantially in the quality of human attention, and the ratio must be lower. In the case of infants and very young children the relevant constraint is not a ratio at all. Remote robotic

childcare should not be assumed equivalent to physically present human caregiving, since the developmental and protective functions of embodied human presence are not established to be reproducible by mediated control, and the program therefore treats physical human presence as a requirement for that class of care, with teleoperated assistance permitted only as an adjunct. Safeguarding, privacy, accountability, and auditability provisions follow from the same reasoning: because the operator is the bearer of responsibility, the operator's identity, qualifications, and actions must be recorded and reviewable, and the data collected through an embodiment in a child's environment must be governed by strict limits on retention and use.

The institutional question that this architecture answers is how increasingly capable systems might change the economics of public services without making access to those services depend on the removal of human professionals. A universal and tuition-free educational system extending from early childhood through advanced study, professional retraining, and lifelong learning is a plausible objective for such a program. Capable systems could assist with curriculum development, accessibility, translation, scheduling, and instruction in scientific subjects, and the supervised surrogate model could extend the reach of qualified educators to places and populations they currently cannot serve. In each case the improvement is a multiplication of access to human judgment rather than a replacement of it, which is the sense in which the program expands human capability without transferring authority.

A further distinction separates human direction from human accountability, and the architecture is incomplete without it. A person can authorize an action without understanding it, and an institution can formally retain responsibility for systems it cannot evaluate, so that authorization by a human is necessary but not sufficient for control in any meaningful sense. Direction is meaningful only if the directing party has adequate information about what is being proposed and what it is likely to do, has real alternatives among which to choose, including the alternative of not proceeding, can refuse without incurring a penalty that makes refusal infeasible, has access to verification performed independently of the proposer, is protected by limits that are enforced technically and not merely stated in policy, and retains the ability to stop or reverse a deployment wherever that is technically possible. The program treats these six conditions as part of what it means to say that authority remains with persons, and an arrangement in which a human signature is the only remaining human contribution fails them.

Formalization

Let the embodiment's state evolve as $x_{t+1} = f(x_t, u_t, w_t)$ on a state space X , where $u_t = \pi(x_t, \xi_t)$ is the action chosen by a local controller π , $w_t \in \mathcal{W}$ is a disturbance drawn from a specified modeled set, and $\xi_t \in \Xi$ is the communication condition, including loss of link and latency up to a stated bound. Let $S \subseteq X$ be a *safe set* specified in advance by human decision.

Definition 13.1 (Robust invariance). *A local controller π is robustly invariant for S from*

an initial set $X_0 \subseteq S$ over horizon H if, for every $x_0 \in X_0$, every disturbance sequence $(w_t) \subseteq \mathcal{W}$, and every communication sequence $(\xi_t) \subseteq \Xi$, the resulting trajectory satisfies $x_t \in S$ for $0 \leq t \leq H$.

Definition 13.2 (Delegation set). *The delegation set $\mathcal{D}$ is the set of local behaviors π such that (a) π is robustly invariant for S , supported by a certificate (a proof, a verified simulation envelope, or a tested bound) that is itself subject to independent verification, and (b) π performs no task with $\kappa(\alpha) = J$. Behaviors outside $\mathcal{D}$ require operator authorization.*

The delegation set thus encodes the principle that what can be delegated without surrendering what ought to remain under human direction is automated: condition (a) restricts automation to behaviors whose safety can be guaranteed, and condition (b) removes the judgment tasks regardless of whether they could be performed safely.

Remark 13.3 (Reversibility). Let $\theta(\alpha)$ denote the time after which the effects of action α can no longer be undone, t_r the operator response time, t_x the time needed to execute a reversal, and $\Delta\ell$ the worst-case uncertainty in communication latency. For classes of action for which reversal forms part of the safety case, reversibility is imposed as an additional, conjunctive requirement $\theta(\alpha) > t_r + t_x + \Delta\ell$, and it never substitutes for invariance. An action whose consequences can leave S under an admissible disturbance, as when an embodiment applies force to a person, is excluded from $\mathcal{D}$ whether or not the resulting injury could later be treated.

Definition 13.4 (Meaningful direction). *A directing party H exercises meaningful direction over a deployment if six predicates hold jointly: Info(H), that H has access to an evaluation of the proposal and its likely effects whose error rates are bounded at a stated level; Alt(H), that H can choose among at least two admissible options, one of which is not to proceed; Ref(H), that refusal is within H 's authority and is not penalized in a way that makes it infeasible; Ver(H), that the verification predicates φ_j relevant to the decision are operated independently of the proposer; Lim(H), that the constraints defining A are enforced technically and not only by policy; and Stop(H), that H can halt or reverse the deployment wherever this is technically possible.*

Authorization by H , meaning that H signs off, is necessary but not sufficient for meaningful direction, since it entails none of the six predicates.

For the supervisory ratio, suppose each of n embodiments places an attention demand $X_i \in [0, 1]$ on the operator, as a fraction of the operator's attention, with mean $m > 0$ and standard deviation $s \geq 0$, and with pairwise correlation $r \in [0, 1]$. The operator's capacity is 1. The sum $\Sigma_n = \sum_i X_i$ has mean nm and variance $\sigma_n^2 = ns^2(1 + (n-1)r)$. For a tolerance $\delta \in (0, \frac{1}{2}]$ let $k_\delta = \sqrt{(1-\delta)/\delta}$ and let $z_{1-\delta} \geq 0$ be the $(1-\delta)$ quantile of the standard normal distribution.

Proposition 13.5. (a) *If $nm + k_\delta \sigma_n \leq 1$ then $\mathbb{P}(\Sigma_n \leq 1) \geq 1 - \delta$, with no distributional assumption beyond the stated mean and variance.* (b) *Under a normal approximation to Σ_n , the constraint $\mathbb{P}(\Sigma_n \leq 1) \geq 1 - \delta$ holds approximately when $nm + z_{1-\delta} \sigma_n \leq 1$.* (c) *For $r = 0$*

and either multiplier $\lambda \in \{k_\delta, z_{1-\delta}\}$, the largest integer n satisfying $nm + \lambda s\sqrt{n} \leq 1$ is $\lfloor u^2 \rfloor$ with

$$u = \frac{-\lambda s + \sqrt{\lambda^2 s^2 + 4m}}{2m}.$$

(d) For $r > 0$ the ratio $\sigma_n/n \rightarrow s\sqrt{r}$ as $n \rightarrow \infty$, so the constraint is asymptotically $n \lesssim 1/(m + \lambda s\sqrt{r})$, which equals $1/(m + \lambda s)$ only when $r = 1$, that is, for perfectly correlated demands.

Proof. (a) Cantelli's inequality gives $\mathbb{P}(\Sigma_n - nm \geq t) \leq \sigma_n^2/(\sigma_n^2 + t^2)$ for $t > 0$. With $t = 1 - nm \geq k_\delta \sigma_n$ the right side is at most $1/(1 + k_\delta^2) = \delta$, and the case $\sigma_n = 0$ is trivial. (b) is the corresponding statement for the normal quantile; because each X_i is bounded and the approximation may be poor in the tails and for small n , (a) is the safer statement. (c) Substituting $u = \sqrt{n}$ gives $mu^2 + \lambda su - 1 \leq 0$. Since $m > 0$ and $\lambda s \geq 0$ the quadratic has one positive root, $nm + \lambda s\sqrt{n}$ is increasing in n , and the feasible integers therefore form an initial segment ending at $\lfloor u^2 \rfloor$. (d) We have $\sigma_n^2/n^2 = s^2(1/n + (1 - 1/n)r) \rightarrow s^2r$, and dividing the constraint by n gives the stated limit. $\square$

The role of the correlation parameter is central. When emergencies in separate embodiments are triggered by a common cause, such as a fire alarm or a network failure, r is large, the pooling gain that makes the $\sqrt{n}$ scaling of case (c) available disappears, and a ratio justified under independence can be unsafe under common-cause demand. For the class of infant care add the constraint $n_{\text{present}} \geq 1$, a requirement that a qualified human be physically present, which cannot be satisfied by any value of the supervision ratio.

Chapter 14

Childcare, Eldercare, and Presence

Care for those who cannot care for themselves, whether because they are young, old, ill, or disabled, is the domain in which the temptation to delegate is strongest and the grounds for restraint are most evident. It is strongest because the demand is large and growing: populations are aging in many regions, the number of persons requiring assistance with the activities of daily life is rising, and the supply of paid and unpaid carers is not rising with it. It is a domain of restraint because what is being provided is not only a service but a relation, and the relation is not equivalent to the service. The distinction is easily stated and difficult to maintain under pressure. A person who is bathed, fed, moved, and medicated has received services. A person who is bathed, fed, moved, and medicated by someone who knows them, notices a change in their expression, converses with them, and regards them as a person has received care. The principle of the book places the first within the range of what may be assisted by machines and the second beyond the range of what may be replaced by them.

To ground the principle in the details of practice, it is useful to distinguish the tasks involved in care by their relation to the three categories introduced in Chapter 13. Some tasks are physical and well defined, such as lifting a person from a bed to a chair, which poses a substantial risk of injury to the carer. Assistance with such tasks by a machine that is able to bear load and that limits its own force is consistent with the principle, and is among its clearest applications, since the strain of lifting is a leading cause of injury and departure from the profession. Some tasks are monitoring tasks, such as watching for falls, detecting wandering, or recording vital signs, and sensors with software that flags anomalies can extend the reach of a carer without replacing the carer's response. Some are tasks of companionship and conversation, in which the matter is delicate: an automated conversational partner may relieve loneliness in some cases, and there is evidence of benefit in particular settings, but the substitution of such a partner for human contact is a different thing from its supplementation, and the temptation of the institution that is short of staff to treat the one as the other must be explicitly resisted. And some tasks are decisions about the person's life: where they will live, what treatment they will accept, whom they will see, how their money is used. These are not delegable to a system,

and the more vulnerable the person the more care is owed to the process by which they are made and to the person's participation in it.

Children raise the same questions with additional urgency, since early relations form the basis of later capacities. The research literature on early development emphasizes the importance of responsive caregiving, in which an adult attends to the child's signals and responds contingently and consistently, and the absence of such care in early life has lasting effects. Devices that entertain or soothe children may be useful as aids, but they do not provide contingent response in the relevant sense, and a household or institution that relies on them for care in the absence of adults is in effect providing less care. The program accordingly sets a floor for the staffing of early childhood settings in terms of the number of adults per child, independent of any technology the setting adopts, and it regards technology as legitimate to the extent that it relieves adults of tasks other than attending to children. The same reasoning applies to the use of automated systems in the supervision of children online, where filtering and monitoring tools may reduce exposure to some harms but cannot replace the attention of a parent or teacher who knows the child.

A technical matter of some importance arises when care is provided at a distance. The architecture of Chapter 13 contemplates a human operator who directs a robotic embodiment remotely, a configuration that makes it possible for a skilled carer in one place to assist a person in another, and for a specialist to examine and treat a patient far from a hospital. The advantages are real, in particular for rural and underserved regions. The limits, however, are partly physical and partly physiological. Any communication channel has a delay, and a feedback loop with a delay can become unstable if the gain is too high: an operator who responds to what the remote embodiment appears to be doing, when it is actually doing something different by the time the response arrives, is liable to overcorrect, and the oscillation may grow. The formalization states the criterion for a simple loop. The practical consequence is that the functions requiring fast response, such as the limitation of force on contact with a person, the maintenance of balance, and the stopping of motion in the event of loss of signal, must be performed by controllers local to the embodiment, which run at rates far above those at which a human can respond, and the human operator must be confined to the slower, higher-level choices of what to do and when. This is the technical expression of the allocation of authority in the architecture, and it is a case where the engineering constraint and the ethical principle point in the same direction.

The sustainability of care work also requires attention, because the demand for care will not be met by the replacement of carers. It will be met, if at all, by changes in the conditions that determine who becomes a carer and how long they remain. Care work is at present poorly paid, physically demanding, emotionally exhausting, and weakly protected in many jurisdictions, and a substantial share of it is performed without pay by family members, disproportionately women. The introduction of machines that reduce the physical burden and administrative load can improve the working conditions of carers and make the work more attractive, but there is no guarantee that productivity gains will be used for this purpose and not to reduce staffing. The

use of the gains is a matter of institutional design, and it is governed by the same consideration as in the earlier chapters: who decides, and who is accountable. A public body that purchases care technology should attach conditions about staffing levels and working conditions to the purchase and should publish the outcomes, so that the effects on care recipients and carers alike are visible.

Finally, the protection of dignity and privacy in care settings sets requirements that technology can either meet or violate. Monitoring technologies that observe a person in the bedroom or bathroom create a record of intimate life, and the person's consent to the monitoring is meaningful only if the person is capable of giving it and has a real alternative. The default for persons who lack capacity should be the least intrusive means consistent with safety, applied with the involvement of a guardian or advocate, with regular review. Data from care settings should be held under the strict limits applicable to health information, should not be used for purposes other than care without consent, and should not be sold. A technology that cannot operate within these limits does not belong in the setting, whatever its advantages in efficiency.

Formalization

Consider a remote loop in which the operator's command acts on the embodiment through a channel with round-trip delay $\tau > 0$. The simplest model of the resulting position error $x(t)$ under proportional correction with gain $K > 0$ is the delay differential equation

$$\dot{x}(t) = -Kx(t - \tau).$$

Proposition 14.1. *The equation is asymptotically stable if and only if $K\tau < \pi/2$. For $K\tau = \pi/2$ it has a sustained oscillation of angular frequency K , and for $K\tau > \pi/2$ it has oscillating solutions of growing amplitude. The largest stable gain is $K_{\max} = \pi/(2\tau)$.*

Proof. Substituting $x = e^{st}$ gives the characteristic equation $s + Ke^{-s\tau} = 0$. At the stability boundary $s = i\omega$ with $\omega > 0$ (the root $s = 0$ is excluded since $K > 0$). Then $i\omega + K(\cos\omega\tau - i\sin\omega\tau) = 0$, whose real and imaginary parts are $K\cos\omega\tau = 0$ and $\omega = K\sin\omega\tau$. The first gives $\omega\tau = \pi/2 + n\pi$; the second requires $\sin\omega\tau > 0$, so n is even and the smallest root is $\omega\tau = \pi/2$ with $\omega = K$, so $K\tau = \pi/2$. For $K\tau < \pi/2$ no root has nonnegative real part: at $K = 0^+$ the single root is $s \approx -K < 0$ and the roots in the infinite chain originate at $\operatorname{Re} s = -\infty$; roots can cross the imaginary axis only at $K\tau = \pi/2 + 2n\pi$, the smallest of which is $\pi/2$. For $K\tau > \pi/2$ the crossing is from left to right, since $d\operatorname{Re} s/dK > 0$ at the crossing, which gives growing oscillations. $\square$

The consequences are quantitative. With round-trip delay $\tau = 0.2$ s, which is typical of intercontinental links with processing and display, $K_{\max} = \pi/0.4 \approx 7.85 \text{ s}^{-1}$, which corresponds to a correction time constant not shorter than about 0.13 s. With delay $\tau = 0.05$ s the maximum

gain rises to about 31.4 s^{-1} . A contact-force limiter, which must respond within milliseconds to prevent injury, would require K of the order of 10^2 – 10^3 s^{-1} , and by the proposition it cannot be placed in a loop that includes a human operator at any plausible delay. The division of control between local reflex layers and remote supervision is thus determined by $K\tau < \pi/2$ together with the response time that safety demands, and does not depend on an appeal to values.

A second quantity concerns staffing. Let N persons need care, each requiring c hours of attendance per week, of which a fraction $\theta \in [0, 1]$ is attendance of the relational kind that is not delegated. The weekly human hours required are at least $N\theta c$, and if the proportion of carers' working hours spent on non-relational tasks is reduced by a labor-saving device that reduces these tasks by a fraction β of the total, the hours released are βNc and may be reallocated to relational time only if βNc is allowed by the institution to be so used. In an institution with fixed staffing, relational time per person rises from θc to $\theta c + \beta c$ if the released time is reallocated, and remains θc if the staffing is reduced by $\beta Nc/H$, with H the hours per carer per week. The difference between the two outcomes is the entire benefit of the technology to the persons cared for, which is the quantitative content of the requirement that purchase conditions secure the use of the gain.

Chapter 15

Public Services and Human Professions

The preceding chapters of this part have considered education and care as domains in which the allocation of tasks between machines and persons is dictated by the nature of the activity. This chapter turns to the broader structure of public services, and to the professions through which they are delivered, and it takes up the question that most strongly affects the political reception of any program of automation, which is what happens to the persons who do the work. The question is not secondary. A program that proposes to increase the capacity of society to direct its own future cannot rest on the premise that a substantial part of society will lose its livelihood and its standing, and the history of technological transitions provides abundant reason to doubt the comfortable assurance that displaced workers will find equally good work elsewhere. The assurance has sometimes been borne out in the aggregate and has often failed for particular groups and regions, with consequences that lasted a generation.

An old observation in economics provides a useful starting point. Baumol and Bowen [10], writing in 1966 about the performing arts, noted that a string quartet requires the same number of performers and the same amount of time to play a given piece today as in the nineteenth century, so that the productivity of the activity has not risen, while the wages of musicians have risen with wages in the rest of the economy, since otherwise musicians would leave for other occupations. The cost of live performance therefore rises relative to the cost of goods produced in sectors where productivity grows. The pattern, since called the cost disease, applies to any activity in which the labor is itself the product, which includes much of education, health care, and personal service. Its implication is that as the economy becomes more productive in manufacturing and information processing, the relative cost of services of this kind increases, and a growing share of income and of employment is devoted to them. The pattern is observed in the national accounts of wealthy countries, where the shares of health and education in expenditure have risen for decades. It is not a pathology to be cured. It is a consequence of the fact that productivity growth in some sectors makes goods cheaper than

services, and the appropriate response is to ensure that the services remain accessible, which is a matter of public finance and not of technology alone.

The introduction of capable systems alters the pattern by making some tasks within the service sectors susceptible to productivity growth, and a naive reading would conclude that the cost disease will therefore be cured. A more careful reading reaches a more qualified conclusion. Services are bundles of tasks, and the tasks differ in their susceptibility to automation. Insofar as a service contains a component that is nondelegable by the principle of this book, such as the relation between teacher and student or the judgment of a physician about what to recommend to a particular patient, that component retains a fixed labor requirement, and the cost of the service cannot fall below the cost of that component. The mathematics is the same as that of Chapter 11: if a fraction of the labor in a service is irreducible, the cost reduction achievable by automating the rest is bounded by the reciprocal of that fraction. If, say, forty per cent of the labor is irreducible, the cost of the service cannot be reduced by more than a factor of two and a half, however productive the automated component becomes. The cost disease is therefore attenuated for the services affected and persists in its core.

Several consequences follow for the structure of public services. First, the nondelegable core of each service will account for a rising share of the wage bill, and of public budgets, and a society that is to afford the services it considers essential must either accept a rising share of national income devoted to them or let their quality and availability decline. The first is the better course, but it requires an institutional means of capturing the gains from productivity elsewhere in the economy and directing them to the core. Second, the professions involved in the core will change in character. The role of the physician will move toward the interpretation of automated analyses in the light of the patient's circumstances and wishes, the role of the teacher toward mentoring and the oversight of instructional systems, the role of the judge toward the explanation of decisions to those affected by them. These are roles of greater responsibility, and they require training that differs from the present training. Third, the roles that are most susceptible to displacement are those in which the work consists mainly of the tasks that can be automated, such as routine processing of claims and documents, and the persons concerned are often employed in the middle ranks of organizations, in occupations that have for decades offered stable livelihoods. The program must therefore address the transition of these persons with measures commensurate to their numbers: retraining supported by income during the retraining, placement into the expanding care and mentoring functions, and where necessary supplementary income, financed by the gains that automation produces.

The profession is not only an economic category; it is a body of knowledge, a code of conduct, and an institution of mutual accountability, and the integrity of these goods deserves protection. The licensing of professionals exists in part to ensure that persons who take responsibility for the lives and affairs of others are competent and answerable. If the profession's characteristic functions are performed by systems whose operators are not licensed, whose training is undocumented, and whose failures are not subject to professional discipline, the protection is lost. The program therefore requires that any system performing a function reserved to a

profession operate under the supervision of a licensed member of that profession who is personally responsible for the result, that the record of the system's contribution be preserved, and that the standards of the profession be extended to cover the use of such systems. The professional bodies have an important role in this extension, as they do in the setting of standards of competence and ethics, and the program regards their participation as necessary and not as an obstacle.

A few words are needed about the public interest in the adoption of such systems by government. Governments already use algorithmic tools to determine eligibility for benefits, to assess risk in policing and sentencing, to detect fraud, and to allocate resources, and the record of such use includes instructive failures in which errors affected thousands of persons before they were detected, and in which the persons affected were poorly placed to contest the decisions. The principles stated in Chapter 13 bear directly on this: a decision that affects the rights or benefits of a person is a judgment in the relevant sense and must be made by a person who can be asked to explain it and who is accountable for it, and the affected person must have access to the grounds and an effective means of appeal. The use of automated tools to assist officials is compatible with these requirements; the use of them to replace the official, with the possibility of review reduced to a formality, is not. The program recommends that government adoption of such systems be preceded by public impact assessment, accompanied by publication of the system's purpose, data, and measured error rates, and followed by independent audit.

Formalization

Two models are used, the first for the sectoral pattern of the cost disease and the second for the limit on cost reduction from partial automation.

Consider an economy with two sectors. In the progressive sector, output is $Y_p = A_p(t)L_p$ with $A_p(t) = A_0e^{gt}$, $g > 0$. In the stagnant sector output is $Y_s = BL_s$ with constant $B > 0$. Labor is mobile, so the wage w is common, and prices equal unit labor costs: $p_p = w/A_p(t)$ and $p_s = w/B$.

Proposition 15.1. *The relative price of the stagnant sector's output is $p_s/p_p = A_0e^{gt}/B$, increasing at rate g , and doubles every $\ln 2/g$ units of time. If the economy holds the ratio of real outputs fixed at $Y_s/Y_p = \varrho > 0$, then the employment share of the stagnant sector is*

$$\frac{L_s}{L} = \frac{1}{1 + B e^{-gt}/(\varrho A_0)},$$

which is increasing in t and tends to 1 as $t \rightarrow \infty$.

Proof. The first claim follows from the pricing rule. For the second, $Y_s/Y_p = BL_s/(A_0e^{gt}L_p) = \varrho$ implies $L_p/L_s = B/(\varrho A_0e^{gt})$, and $L_s/L = 1/(1 + L_p/L_s)$. $\square$

With $g = 0.02$ per year, the relative price rises by the factor $e^{0.02 \cdot 35} \approx 2.01$ in thirty-five years and by $e \approx 2.72$ in fifty. If instead the demand for the stagnant sector's output is such that the nominal expenditure share is fixed, the employment share rises with the relative price. In either case, the sector whose productivity does not grow absorbs a rising share of labor, and the proposition identifies what a public finance system must do to keep its services accessible.

For partial automation, let a service require, per unit of output, ℓ_J hours of nondelegable labor and ℓ_A hours of automatable labor, and let automation raise the productivity of the automatable part by the factor $1 + \gamma$, $\gamma \geq 0$, so that the labor requirement becomes $\ell(\gamma) = \ell_J + \ell_A/(1 + \gamma)$. Define the nondelegable share $s_J = \ell_J/(\ell_J + \ell_A)$.

Proposition 15.2. *The cost ratio $\ell(0)/\ell(\gamma)$ is increasing in γ and bounded above by $1/s_J$, with $\ell(0)/\ell(\gamma) \rightarrow 1/s_J$ as $\gamma \rightarrow \infty$. To attain a cost reduction by the factor $\kappa < 1/s_J$ requires $1 + \gamma = (1 - s_J)\kappa/(1 - s_J\kappa)$.*

Proof. Normalize $\ell_J + \ell_A = 1$. Then $\ell(\gamma) = s_J + (1 - s_J)/(1 + \gamma)$, which decreases in γ to s_J . Setting $1/\ell(\gamma) = \kappa$ and solving for $1 + \gamma$ gives the stated expression. $\square$

For $s_J = 0.4$ the supremum of the cost reduction is 2.5. To attain a reduction by $\kappa = 2$ requires $1 + \gamma = 0.6 \cdot 2/(1 - 0.8) = 6$, a sixfold productivity gain in the automatable part, and to reach $\kappa = 2.4$ requires $1 + \gamma = 0.6 \cdot 2.4/(1 - 0.96) = 36$. The strongly diminishing return shows that the final portion of the reduction demands increases in productivity that are not plausibly obtained, and it is the quantitative justification for the planning assumption that public services will retain a large wage bill in their nondelegable core.

Part IV

The Terrestrial Program

Chapter 16

Standards of Specification for Megascale Programs

The program includes a family of proposals at planetary and astronomical scale, among them continental-scale freshwater transport corridors, systems for supporting ecological recovery, geothermal energy infrastructure, momentum-exchange tethers, deployable and expandable structures in orbit, space-elevator concepts, and stellar-scale energy collection. These proposals span very different levels of technological readiness, and the program's treatment of them depends on keeping the differences in view. A body of theoretical literature on orbital tethers has existed since the 1970s, and tether concepts have been examined in considerable engineering detail, whereas a terrestrial space elevator requires materials with specific strength beyond what has been demonstrated at the necessary scale and faces unresolved problems of dynamics, debris, and construction, and a Dyson-scale collector involves quantities of mass and energy that place its construction at a different order of difficulty again. Presenting all such proposals as equally near-term would mislead, and the program therefore requires that each be classified by readiness, and that any proposal that depends on an unresolved step be marked as such. Three classes are used: projects that are scientifically understood but economically or institutionally difficult, projects that require genuine scientific or engineering breakthroughs, and projects whose feasibility remains conjectural.

For each proposal the program prescribes a standard of specification before evaluation. A proposal for geothermal infrastructure integrated with mass acceleration, for example, is at present a speculative concept, and the requirement is that it supply an explicit mechanism, an energy balance, the geological constraints on the site and the depth, and the material limits of the components. A proposal for continental freshwater corridors must state elevation changes, pumping energy, evaporative and seepage losses, effects on ecological connectivity, and the governance arrangements across jurisdictions, and it must be compared against alternatives that do not require the construction of the corridor. A proposal for forest restoration must distinguish the restoration of an ecosystem from its replacement by an artificial substitute, and

it must avoid afforestation that is inappropriate to the local ecology, since planting where a different biome belongs can damage rather than repair. Some proposals use terminology whose technical meaning has not been fixed, in which case the first task is a definition adequate to evaluate the physical function. The common requirement is that a proposal state what would have to be true for it to work and what would show that it does not.

A further consideration concerns the relation of these programs to present obligations. The long time horizon and the very large scale of megascale projects make them vulnerable to a characteristic form of reasoning in which the magnitude of the eventual benefit is invoked to justify expenditure and sacrifice in the present. The program resists this by a sequencing rule: measures intended to satisfy present obligations, such as clean water, energy access, health, and education for existing persons, must depend only on projects of the first readiness class, so that they do not hinge on breakthroughs that may not occur. The rule is a screening condition and not a guarantee, because a project in the first class can still fail for ordinary reasons of financing, governance, construction, opposition, or environmental effect, and the relevant distinction is between dependence on an unresolved scientific or engineering breakthrough and exposure to implementation risk, which every project carries. Programs in the second and third classes may be pursued as research, with resources proportioned to their prospects and subject to independent review, but they may not be made prerequisites for the fulfillment of what is owed now.

The chapters that follow in this part and the next apply the standard to the principal proposals, taking the terrestrial programs first. Each proceeds in the same order: the concept is stated in the author's own terms where a prior statement exists and otherwise by a working definition that is flagged as provisional, the physical relations on which its feasibility depends are written out, the dependencies on other programs and on governance are listed, the readiness class is assigned with reasons, and a calculation is supplied that either supports or undermines the concept's claim to be taken seriously. Two cautions govern the reading of the calculations. A calculation that shows a concept to be consistent with the laws of physics shows only that it has survived the first test, and the concept remains in the second or third readiness class until its other conditions are met. A calculation that shows a concept to be infeasible under its stated assumptions shows only that the assumptions require revision, and does not by itself refute the underlying idea.

Formalization

Let $\mathcal{P}$ be a finite set of megascale programs with a readiness function $r : \mathcal{P} \rightarrow \{1, 2, 3\}$ whose values denote, in order, understood but costly, requiring breakthrough, and conjectural. Let $\mathcal{O}$ be a set of present-obligation measures with a dependency relation $\prec \subseteq \mathcal{O} \times \mathcal{P}$, where $o \prec p$ means that measure o depends on program p .

Definition 16.1 (Readiness-respecting plan). *A plan is readiness-respecting if for every $o \in \mathcal{O}$*

and every p with $o \prec p$ (including transitive dependence) one has $r(p) = 1$.

Proposition 16.2. *In a readiness-respecting plan the fulfillment of every present-obligation measure is independent of the outcome of every program with $r(p) \geq 2$. Conversely, if some $o \prec p$ with $r(p) \geq 2$, the fulfillment of o inherits the uncertainty of a breakthrough that the success of programs with $r = 1$ does not determine.*

Proof. In a readiness-respecting plan the programs on which any o depends all have $r = 1$, so the fulfillment of o is a function of the outcomes of those programs alone. In the converse case the outcome of p with $r(p) \geq 2$ is not determined by the outcomes of programs of class 1, and o inherits its uncertainty. $\square$

Remark 16.3 (Screening, not guarantee). The proposition is close to definitional, since it follows from how dependence is defined, and it should be read as a design constraint on plans and not as an empirical claim. Let I_p denote the event that program p with $r(p) = 1$ is successfully implemented. If fulfillment of o requires every program on which it depends, then $\mathbb{P}(o \text{ fulfilled}) \leq \mathbb{P}(\bigcap_{o \prec p} I_p)$, which may be well below 1 because of financing, governance, construction error, political opposition, or unforeseen environmental effects. Readiness-respect is therefore a necessary screening condition for plans whose obligations are not hostage to unresolved breakthroughs, and it is not a sufficient guarantee of delivery. Implementation risk must be analyzed separately for each program.

Three physical relations illustrate the quantitative character of the specifications; the first is developed in Chapter 18, the second in Chapter 24, and the third in Chapter 25. For the transport of water through a rise of height h with a pipeline of length L , diameter D , flow speed v , and friction factor f , the head loss is $h_f = f \frac{L}{D} \frac{v^2}{2g}$ (the Darcy–Weisbach relation), and the minimum energy per unit volume of lifting against the total head is

$$\frac{E}{V} = \frac{\rho g (h + h_f)}{\eta},$$

where ρ is the density of water, g the gravitational acceleration, and $\eta \in (0, 1]$ the combined efficiency of pumping. For $h = 1000$ m, $h_f = 0$, and $\eta = 1$ this gives $\rho g h \approx 9.8 \times 10^6$ J per cubic meter, about 2.7 kWh per cubic meter, which sets a floor that no design can undercut and against which alternatives such as desalination can be compared on a common basis.

For a tether of constant working stress σ and density ρ_m extending from the surface at radius R to geostationary radius r_g , define the effective potential $\Psi(r) = -GM/r - \frac{1}{2}\omega^2 r^2$, where G is the gravitational constant, M the mass of the Earth, and ω its rotation rate. Then the cross-sectional area of a tether of uniform stress satisfies $\sigma dA/dr = \rho_m A \Psi'(r)$, whence

$$\frac{A(r_g)}{A(R)} = \exp \left[\frac{\rho_m}{\sigma} (\Psi(r_g) - \Psi(R)) \right],$$

and the taper ratio is determined by the specific strength σ/ρ_m of the material: because $\Psi(r_g) -$

$\Psi(R) > 0$, the ratio grows exponentially as specific strength falls, which is the quantitative reason that readiness for this concept depends on materials that have not been demonstrated. Finally, a collector capturing a fraction f of the solar luminosity $L_{\odot} \approx 3.8 \times 10^{26}$ W yields power $P = fL_{\odot}$, so that the scale of such a program is set by the choice of f and by the mass required to intercept it.

Chapter 17

Orthodromic Rivers: Geometry

The first terrestrial program to be treated is the family of freshwater corridors that the working vocabulary of this book calls orthodromic rivers. The term refers to engineered channels or conduits that follow, as nearly as terrain and the boundaries of the oceans permit, the arcs of great circles, which are the shortest paths between points on a sphere and are called orthodromes in the vocabulary of navigation. The concept belongs to a broader scheme set out in an earlier work by the author, the Orthodromic Infrastructure Blueprint, in which nine great circles, the mirror planes of the octahedral symmetry group, are used to organize a planetary network of infrastructure in a manner that is symmetric and free of privileged regions. The present book takes the hydraulic layer of that scheme as its subject, and it treats the geometry and the water separately, in this chapter and the next. It adopts as a working definition that an orthodromic river is a corridor for the conveyance of fresh water laid along an arc of one of these circles, subject to the modifications that the terrain and the law require, and it flags the definition as provisional pending the author's more detailed statement of the water system's parameters.

The attraction of great-circle routes is evident from the elementary geometry of the sphere, but the attraction needs to be examined and not assumed. On a sphere the shortest path between two points is the arc of the great circle through them, and the saving over other natural routes can be substantial: two points at latitude 60° and separated by 90° of longitude are 4,605 km apart along the great circle, whereas the path along the parallel of latitude is 5,004 km, longer by eight per cent. Shorter conduits cost less to build, lose less water to seepage and evaporation, and require less energy for pumping against friction. These advantages are real. They are, however, advantages of shortness, and the question whether shortness is the property to be minimized depends on the cost structure of the terrain. A route of minimum length may cross a mountain range which a longer route avoids, and the saving in length is more than lost in the cost of tunnelling or pumping. The correct generalization is the principle that governs optimal paths in any medium with variable cost, which is that the optimal route is a geodesic of the metric weighted by the cost of passage. A great circle is optimal only where that cost is

uniform, and in all other cases it serves as the reference against which the penalty of the actual route is measured. This is the reading of the great-circle network that the book adopts: it is a reference configuration that supplies a lower bound on length and a symmetric organization of the globe, and the corridors that would actually be built would deviate from it to the extent that terrain, jurisdictions, and ecology demand.

The octahedral scheme has a combinatorial structure that can be set out exactly. The full symmetry group of the octahedron contains forty-eight elements and nine mirror planes, three of which are the coordinate planes and six of which are the planes bisecting pairs of coordinate axes. Each plane meets the sphere in a great circle, and the nine circles divide the sphere into forty-eight congruent spherical triangles with interior angles of 90° , 60° , and 45° . The vertices at which the circles meet are of three kinds: six at the ends of the coordinate axes, where four circles meet, eight at the directions of the cube's vertices, where three circles meet, and twelve at the midpoints of the edges of the octahedron, where two circles meet. Each circle passes through exactly eight of the twenty-six vertices and is therefore divided into eight arcs, so that there are seventy-two arcs in all. The arcs are of three lengths, corresponding to angular lengths of 35.26° , 45° , and 54.74° , twenty-four of each, which on a sphere of the radius of the Earth are about 3,921 km, 5,004 km, and 6,086 km. The total length of the nine circles is $9 \times 2\pi R \approx 3.6 \times 10^5$ km. These numbers fix the scale of the network that the book considers: twenty-six nodes joined by seventy-two arcs of a few thousand kilometers each, and a total length that is about nine times the circumference of the Earth.

The Blueprint itself distinguishes this concurrent, cubic-symmetric realization from the general case. In general position, nine great circles intersect pairwise in two antipodal points and so produce $2\binom{9}{2} = 72$ vertices, 144 edges, and 74 faces, a count that satisfies $72 - 144 + 74 = 2$ and is the upper bound for any arrangement of nine circles; in the Blueprint's general formulation, five of the circles are fixed as reference orthodromes and four are inclined and flexible, so that the network can be adjusted to the geography without losing the protocol. The cubic-symmetric case, in which the nine circles are the mirror planes of full octahedral symmetry, is the special arrangement in which many circles meet at common points and the vertex count falls to 26. The count has a familiar mnemonic, which the author has used in physical models of the arrangement: the 26 nodes correspond to the 8 corners, 12 edges, and 6 face centers of a cube, which is also the number of visible pieces of the puzzle cube. The author has built such models, including a globe with the circles drawn on it and a lattice of the Hoberman type hung over a globe, and these serve as physical checks on the geometry. The present book adopts the cubic-symmetric realization as its reference configuration, because it is exactly specified and its numbers can be computed, and it treats the flexible circles of the general case as the means by which the reference configuration would be adapted to terrain.

A very large share of such a network could not be built as a surface channel, because most of the surface is ocean. The land area of the Earth is about twenty-nine per cent of the whole, and on average the same proportion of a great circle lies over land, so that of the 3.6×10^5 km of the nine circles perhaps 1.0×10^5 km would pass over land, and the rest across water, where

a corridor would have to be a submerged pipeline or would not exist at all. Moreover, the orientation of the reference configuration relative to the Earth's axis is a free parameter, as are the inclinations of the flexible circles in the general case, and different choices yield different land fractions and different intersections with mountain ranges, deserts, and the territories of the states that would have to participate. The selection of an orientation is a constrained optimization problem over the group of rotations, and its objective is not geometric alone. It must weigh the water demand of the regions traversed against the cost of traversal, the number of international borders crossed, and the sensitivity of the ecosystems along the route. The book states the problem and leaves its solution to the stage at which regional data are available. It treats the preliminary conclusion as one that follows from the geometry: the network, in its pure form, is not a plan for the whole planet but a plan for those arcs, or portions of arcs, that fall on land where freshwater demand and supply are mismatched, and its value must be judged arc by arc.

A second geometrical point is that freshwater conveyance is constrained by elevation in a way that has no counterpart in the flat geometry of navigation. A great circle takes no account of the relief it crosses. Water flows downhill without cost and must be lifted at cost, so the desirable corridor is one whose profile of elevation lets gravity do as much of the work as possible and requires lift only where the demand lies above the source. Because the energy of lift scales with the height gained, as the next chapter shows, a corridor that crosses a ridge of one kilometer must expend energy of the order of 2.7 kWh per cubic meter to carry the water over it, and a route that follows contours and keeps a constant gentle gradient may spend none. The great-circle reference therefore must be supplemented by a second reference, the profile of elevation along the route, and an assessment of a corridor consists of the joint consideration of both.

The geometry also bears on questions of governance that are treated in the next chapter. A corridor of thousands of kilometers crosses many jurisdictions, and its symmetric layout is not a source of legitimacy. A scheme that treats the planet as a uniform surface to be organized by a single geometric rule has the character of the abstract plans against which the literature on large-scale state projects has warned, in which a legible and elegant design imposed from above fails to take account of the local knowledge and the existing arrangements of the people affected. The response is not to abandon geometry, which supplies lower bounds and avoids arbitrary favoritism, but to treat it as a language in which the proposals are stated and compared and not as a source of authority. Where a route deviates from the great circle in order to avoid a sacred site, a settlement, or a wetland, the deviation is a reasonable decision, and its length penalty is simply a price to be reported.

Formalization

The spherical geometry is fixed first. Let the Earth be modelled as a sphere of radius $R = 6371$ km; the departure of the true figure from a sphere is a flattening of about $1/298$, which alters distances by at most a few tenths of one per cent and is neglected here. The angular distance θ between points with latitudes φ_1, φ_2 and longitude difference $\Delta\lambda$ satisfies $\cos \theta = \sin \varphi_1 \sin \varphi_2 + \cos \varphi_1 \cos \varphi_2 \cos \Delta\lambda$, and the orthodromic distance is $R\theta$.

Proposition 17.1. *Let Γ be the group generated by reflections in the nine planes with normals e_1, e_2, e_3 and $(e_i \pm e_j)/\sqrt{2}$, $i < j$. Then $|\Gamma| = 48$; the nine great circles divide the sphere into 48 congruent spherical triangles with angles $\pi/2, \pi/3, \pi/4$, each of area $4\pi R^2/48 \approx 1.06 \times 10^7$ km²; the vertices number 26 (of valencies 4, 3, 2, in numbers 6, 8, 12); each circle contains 8 vertices; and the arcs number 72, of which 24 have angular length $\arccos(1/\sqrt{2}) = 45^\circ$, 24 have $\arccos(\sqrt{2/3}) \approx 35.26^\circ$, and 24 have $\arccos(1/\sqrt{3}) \approx 54.74^\circ$.*

Proof. The group generated by these reflections is the full octahedral group O_h , of order 48. A spherical triangle with angles $\pi/2, \pi/3, \pi/4$ has area $R^2(\pi/2 + \pi/3 + \pi/4 - \pi) = R^2\pi/12 = 4\pi R^2/48$ by Girard's theorem, so 48 such triangles tile the sphere. The vertices are the points of the sphere fixed by a nontrivial rotation subgroup: the 6 axis directions $(\pm 1, 0, 0)$ and permutations, the 8 directions $(\pm 1, \pm 1, \pm 1)/\sqrt{3}$, and the 12 directions $(\pm 1, \pm 1, 0)/\sqrt{2}$ and permutations, giving 26. Counting incidences, a circle with normal n contains the vertices orthogonal to n , and direct enumeration gives 8 for each of the nine normals. Each circle is thus divided into 8 arcs, giving 72. The Euler relation $V - E + F = 26 - 72 + 48 = 2$ is satisfied. The three angular lengths are the angles between adjacent vertices of the three types: between an axis and an edge-midpoint, $\arccos(1/\sqrt{2}) = 45^\circ$; between a cube-vertex and an edge-midpoint, $\arccos(\sqrt{2/3})$; between an axis and a cube-vertex, $\arccos(1/\sqrt{3})$. Each type of edge occurs 24 times, and the angles sum to $24(45 + 35.26 + 54.74) = 3240^\circ = 9 \times 360^\circ$, as required for nine full circles. $\square$

The sums have been verified by direct enumeration of the vertices and edges. On the sphere above, the arc lengths are 5003.8, 3921.2, and 6086.3 km, and the total length of the nine circles is $9 \cdot 2\pi R = 360,272$ km.

The second result states the sense in which a great circle is optimal. Let a corridor be a curve γ on the sphere joining two points, and let the cost of construction and operation per unit length at a point x be $c(x) > 0$, a continuous function. Define the cost $C(\gamma) = \int_\gamma c(x) ds$.

Proposition 17.2. *A minimizer of C among curves with fixed endpoints is a geodesic of the conformally rescaled metric $c(x)^2 g$, where g is the round metric. If c is constant, the minimizers are arcs of great circles, and $C = cR\theta$. In the plane approximation, at an interface across which c jumps from c_1 to c_2 , the minimizer satisfies the refraction relation $c_1 \sin \alpha_1 = c_2 \sin \alpha_2$, where α_i is the angle between the path and the normal to the interface on side i .*

Proof. The first statement is the definition of geodesics of the rescaled metric, whose length functional is C . For constant c it reduces to the length functional of g , whose geodesics are great circles. For the interface, take the plane approximation with a straight interface and endpoints at fixed distances from it; the cost of a path crossing at a point y is $c_1 \sqrt{a_1^2 + y^2} + c_2 \sqrt{a_2^2 + (d - y)^2}$, and setting its derivative in y to zero gives $c_1 \sin \alpha_1 = c_2 \sin \alpha_2$. $\square$

The refraction relation shows how the great circle is bent by terrain. If the cost per kilometer on one side of a ridge zone is three times that on the other, a path approaching the interface from the cheap side at α_1 enters the costly zone at an angle α_2 satisfying $\sin \alpha_2 = \sin \alpha_1/3$, so that the path is steered toward the normal and crosses the costly zone by the shortest passage. The penalty of departing from the great circle is then $C(\gamma^*) - c R \theta \geq 0$, which is the quantity to be reported for each arc in the assessment.

Chapter 18

Orthodromic Rivers: Water and Governance

The geometry of the preceding chapter determines where a freshwater corridor might run. The hydraulics determines what it would cost to run water through it, and the governance determines whether it could be built and kept in operation. This chapter treats the second and third. The hydraulics is dominated by two relations that every proposal for long-distance conveyance must satisfy, the energy required to lift water and the energy dissipated by friction, and the chapter shows that the second, which is less discussed, is the more severe constraint over the distances that the concept contemplates. The governance is dominated by the fact that the water of a continent is claimed by many parties, and that the claims are not settled by engineering. The assessment concludes that the concept is in the second readiness class in its hydraulic aspects and, as to governance, depends on arrangements whose feasibility is open to doubt. Baseline data on scarcity, withdrawal, and transboundary basins are taken from the standing international assessments [52, 53], and the numerical parameters of this chapter should be read as stipulated illustrations against that background.

Begin with lift. The work required to raise a cubic meter of water through a height of one kilometer is about 9.8×10^6 joules, which is 2.7 kWh, and it is independent of the length of the route over which the rise is accomplished. This figure is not a design parameter but a floor, and it supplies a standard of comparison. Reverse-osmosis desalination of seawater at present consumes about three to four kilowatt-hours per cubic meter in large plants, including pretreatment and intake, and the thermodynamic minimum for seawater of typical salinity is a little over one kilowatt-hour per cubic meter. A corridor that lifted water by a kilometer would therefore consume about as much energy per unit volume as a desalination plant that supplied the same water at the coast, before any allowance for friction, and a corridor that lifted it by more than a kilometer would consume more. For a conduit with a uniform mean uphill gradient S , so that the rise is proportional to the distance traveled, there is a distance at which the energy of lift equals the energy of desalination, and the formalization computes

it as about 10,900 km for $S = 10^{-4}$ and 3.5 kWh per cubic meter at a pumping efficiency of 0.85. Within that distance, transport by uphill gradient is cheaper in energy than desalination at the coast; beyond it, it is dearer. The comparison is limited to energy and ignores capital cost, which is the larger part of the cost of either, but it shows that the distances at issue are not so great as to make lift prohibitive when the gradient is gentle.

Friction, however, is another matter. Water moving through a conduit loses head in proportion to the conduit's length and to the square of its velocity, and inversely to its diameter. The loss must be made up by pumping when the conduit does not descend as fast as the friction demands, and it must be made up by slope when the conduit is gravity-fed. In the first case it is a continuing energy cost, and in the second it is an elevation that must be available. The formalization shows that, for a given discharge Q , the friction slope varies as Q^2/D^5 , which means that the diameter is the dominant design variable: doubling it reduces the friction by a factor of thirty-two. A pipeline of ten meters in diameter, carrying about 157 m³/s at 2 m/s, with a friction factor 0.012, loses 1,223 meters of head over 5,000 km, which costs 3.9 kWh per cubic meter at 85 per cent efficiency, more than the whole cost of desalination. Widening the pipe to twelve meters reduces the figure to 1.6 kWh, to fourteen meters to 0.73 kWh, and to sixteen meters to 0.37 kWh. The sensitivity illustrates why the concept cannot be assessed by a single number and why it is a problem of design in which the capital cost of larger conduits is balanced against the energy cost of friction. Open channels dissipate energy by a different law, expressed by the Manning equation [34], in which the velocity is proportional to the square root of the slope. A concrete-lined channel one hundred meters wide and ten meters deep, on a gradient of 10^{-4} , would carry about 3,400 m³/s, or about 108 cubic kilometers a year, and would fall 500 meters in 5,000 km, so that a gravity-fed channel of such length must descend from a source at least that much above the point of delivery. The geographical requirement is a severe one; the sources of fresh water on land are not usually higher than the regions short of water by the necessary amount, and the opposite relation is common.

Open channels expose the water to the atmosphere and therefore to evaporation and, in the absence of a lining, to seepage. Evaporation from the channel just described, in a region where the open-water evaporation rate is two meters per year, would amount to about one cubic kilometer a year over 5,000 km of length, or about nine-tenths of one per cent of the flow. This loss is not a serious one for a channel of such capacity, but it increases in proportion to the surface and decreases in relation to the discharge, so that small channels over long distances are lossy and large channels are not. Seepage depends on the lining and on the substrate, and a failure of the lining over an arid region may deliver the water to the aquifer and not to the destination, which may be a benefit in a recharge program and a defect in a delivery program. These considerations point to a conclusion that is not new but is worth repeating: the efficient scale of a long conduit is large, and large works concentrate risk, so that the cost of failure and the difficulty of repair are also large.

Ecological effects are the third class of physical consequence. A channel that connects drainage basins previously separate creates a path for the movement of organisms, and the history of

canals contains instructive cases. The Suez Canal opened a corridor through which species of the Red Sea have invaded the Mediterranean, with substantial effects on its fisheries and ecology, and proposals to separate the Great Lakes from the Mississippi system are motivated by the fear that species of carp will cross between them through the Chicago waterway. A freshwater corridor of continental length would connect basins that have been isolated for millennia, and the movement of pathogens, plants, and animals through it would have to be prevented by design, which is difficult, or accepted as a cost, which may be unacceptable. The corridor also intercepts the flow of surface and subsurface water across its path and may disrupt wetlands, migrations, and sediment transport. An assessment of these effects is a prerequisite for any decision to proceed and could not be left to the later stages of design.

Governance determines more than the physical aspects what can be done. Water is claimed by those who live on a river and by those who live far from it, by the upstream state and by the downstream state, by farms and cities and the ecosystems on which they depend, and the law of transboundary waters has developed principles, notably equitable and reasonable utilization and the obligation not to cause significant harm to other states, that are widely accepted and unevenly applied. The Indus Waters Treaty of 1960 has endured conflict between the parties and is often cited as a rare success, and the failure of the parties on the Nile to agree on the allocation of its waters is often cited as the contrary case. A corridor that transfers water between basins is a physical proposition that cannot be separated from these claims. It would alter the allocation among the parties, and the parties' consent would have to be obtained for a project whose benefits and burdens fall on different persons. The design principles of Ostrom for durable common-pool arrangements imply that the proposal should provide for monitoring by the users, for graduated sanctions for violation of the rules, for accessible mechanisms for the resolution of disputes, and for the recognition by higher authorities of the right of the users to organize. A project designed by a central body and imposed on the users, however elegant its geometry, would fail these tests.

The assessment under the standard of Chapter 16 follows. The mechanism is explicit and the energy balance has been written out. The relation of lift to the alternatives has been stated, and the friction analysis shows that the designs for which the energy cost is competitive with desalination require conduit dimensions much larger than any in service, which is the main reason that the concept is placed in the second readiness class and not the first. Dependencies include the production of materials in the required quantity, the availability of sources at sufficient elevation, and the agreement of the states concerned. The comparison with alternatives that do not require the construction of the corridor, including demand management, wastewater reuse, local recharge, and desalination, is mandatory, and for the present these are the measures on which present obligations to supply water may legitimately be based. The corridor remains a candidate for study as a supplement to them and not as a substitute.

Formalization

Let water of density $\rho = 1000 \text{ kg m}^{-3}$ be moved at discharge Q through a conduit of length L , diameter D , friction factor f , with net rise Δz and pumping efficiency $\eta \in (0, 1]$. The friction slope is $S_f = f v^2 / (2gD)$ with $v = 4Q / (\pi D^2)$, and the energy required per unit volume, ignoring minor losses, is

$$\frac{E}{V} = \frac{\rho g (\Delta z + S_f L)}{\eta}.$$

Proposition 18.1. (i) At fixed Q , $S_f = 8fQ^2 / (\pi^2 g D^5)$, so that S_f is proportional to D^{-5} and a factor $\lambda > 1$ in diameter reduces friction by λ^{-5} . (ii) For a uniform mean uphill gradient S , so that $\Delta z = SL$, and neglecting friction, the energy of lift equals a reference energy E_d per unit volume (for example, that of desalination) at the break-even distance

$$L^* = \frac{\eta E_d}{\rho g S},$$

and lift is energetically cheaper than the reference for $L < L^*$. (iii) If annual cost per unit length is $aD^\beta + bQ^3 D^{-5}$, where the first term is capital cost with $\beta > 0$ and the second is the cost of the energy to overcome friction, then the minimizing diameter is $D^* = (5bQ^3 / (\beta a))^{1/(\beta+5)}$, and at the optimum the energy term equals $\beta/5$ times the capital term.

Proof. (i) Substituting $v = 4Q / (\pi D^2)$ into S_f gives $S_f = f 16Q^2 / (2g\pi^2 D^5)$. (ii) Setting $\rho g S L^* / \eta = E_d$ and solving. (iii) The pumping power to overcome friction is $\rho g Q S_f L / \eta \propto Q^3 D^{-5}$, so the cost per unit length has the stated form. Setting the derivative $\beta a D^{\beta-1} - 5bQ^3 D^{-6}$ to zero gives $D^{\beta+5} = 5bQ^3 / (\beta a)$ and at that point $bQ^3 D^{-5} = (\beta/5)aD^\beta$. $\square$

Numerical values follow. With $E_d = 3.5 \text{ kWh m}^{-3} = 1.26 \times 10^7 \text{ J m}^{-3}$ and $S = 10^{-4}$: $L^* = 1.2844 \times 10^7 \text{ m} \approx 12,844 \text{ km}$ at $\eta = 1$ and $\approx 10,917 \text{ km}$ at $\eta = 0.85$. The lift energy for $\Delta z = 1000 \text{ m}$ is $\rho g \Delta z = 9.81 \times 10^6 \text{ J m}^{-3} = 2.725 \text{ kWh m}^{-3}$ at $\eta = 1$. For $Q = 157 \text{ m}^3/\text{s}$, $f = 0.012$, $L = 5000 \text{ km}$ and $\eta = 0.85$, with $\Delta z = 0$ (a level conduit), the friction energy is 3.92 kWh m^{-3} at $D = 10 \text{ m}$, 1.58 at 12 m , 0.73 at 14 m , and 0.37 at 16 m . The friction factor is held constant for simplicity; in practice it varies weakly with the Reynolds number and the relative roughness, and this variation alters the figures by tens of per cent but not their scaling with D . For the open channel, the Manning equation $v = n^{-1} R_h^{2/3} S^{1/2}$, with $n = 0.012$, $R_h = 1000/120 = 8.33 \text{ m}$ and $S = 10^{-4}$, gives $v = 3.43 \text{ m/s}$ and $Q = 3425 \text{ m}^3/\text{s}$, which is $108 \text{ km}^3 \text{ yr}^{-1}$, and the evaporative loss over 5000 km at 2 m yr^{-1} and width 100 m is $1.0 \text{ km}^3 \text{ yr}^{-1}$, a fraction 0.93 per cent. The rule of (iii), which parallels the classical law for the choice of the cross-section of an electrical conductor, is useful in practice: at the optimal design the friction energy is a fixed fraction of the capital cost, so that if the diameter is chosen optimally and the cost of energy is higher than the proposal assumes, the design should be revised toward wider conduits and not narrowed.

Chapter 19

Rainforest Generators and Terra Preta Rain

The second terrestrial program concerns the support of biological production on land, and it is stated in the author's own vocabulary as a pair of systems that are designed to operate together. A rainforest generator, in this vocabulary, is a support architecture: a tensile trellis or hanging cable grid carried on smart anchors, with light wells, misters, nutrient fog, and microclimate control, whose function is to enlarge the physical state space available to rainforest growth. It is not a machine that produces a forest from nothing, and the distinction is central to its assessment. A forest is an ecological community whose composition and persistence depend on seed sources, soil organisms, pollinators, and disturbance regimes that no structure supplies, and the generator is intended to supply only what the structure can supply, namely vertical surface on which epiphytes, lianas, and climbers can establish, a controlled supply of moisture and nutrients to the canopy, and a modified light and temperature regime beneath it. Terra preta rain is the paired system for the supply of water and nutrients. Rainwater is collected in elevated basins, including the basins of calderas, filtered through gravel, biochar, and sand and through biochar-lined berms, enriched with nutrients, minerals, microbes, and suspended carbon, and redistributed under gravity over the constructed canopy as a circulating medium and not a static amendment. The name recalls the Amazonian dark earths, and it signals a hypothesis that the properties of those soils can be reproduced in a flowing form. The agronomic evidence on charcoal-amended soils is reviewed in the literature on biochar [54], and the extension of those results to a flowing, atmospheric form is the book's hypothesis and is not established by that literature.

The precedent deserves a measured account. The dark earths of the Amazon basin, known by the Portuguese name terra preta de índio [14], are patches of soil, often of a few hectares and occasionally larger, that contain high concentrations of charcoal, ceramic fragments, organic matter, and nutrients, and that remain markedly more fertile than the surrounding weathered soils centuries after their formation. They are generally attributed to the deliberate and

incidental activities of pre-Columbian populations, who deposited the residues of cooking, waste, and burning, and the persistence of their carbon, which has been attributed to the stability of charred material and to the community of microorganisms associated with it, has made them a model in the literature on soil carbon and on sustainable agriculture in the humid tropics. Several features of the precedent must be noted. The dark earths accumulated over generations and not in a season, which suggests that the process at issue is slow. They are a soil amendment, a body of material in place, and the flowing form proposed here is a different thing whose behavior has to be demonstrated and not inferred. And the mechanisms by which the earths sustain fertility are not fully settled, so that an engineering system modelled on them inherits the uncertainty. The book therefore treats the extension from static soil to circulating medium as a hypothesis for which the program must supply its own evidence.

Three criteria govern the assessment, and they follow from the principle stated in the author's earlier work that the quantity of biomass is not a measure of the health of the biosphere. The first is habitat complexity: the structure should support a diversity of niches and the organisms that occupy them, not merely a large mass of foliage, and a trellis that carried a monoculture of fast-growing climbers would increase biomass while reducing the quality of the habitat. The second is nutrient continuity: the system should maintain a supply of nutrients matched to the demand of the community and should not produce pulses that favor a few opportunistic species or that leave the system through drainage. The third is ecological autonomy: the community that develops should become able to sustain itself without the support of the structure, so that the structure is scaffolding in the literal sense, needed during establishment and removable thereafter. The third criterion has a consequence for the evaluation of persistence which the book adopts as a general principle for interventions of this kind. The risk of withdrawing the support is not a fixed quantity. It is proportional to the fraction of the supported biomass that has not yet become autonomous, and that fraction should be tracked continuously and not asserted once. A system that reports its fraction of autonomous biomass at stated intervals, and that publishes the figures, supplies the evidence on which a decision to continue or to terminate can rationally rest.

The physics of the trellis is that of a hanging cable under load, and a calculation shows that the structural demand is moderate for spans of the order of a hundred meters. A cable grid with parallel cables five meters apart, carrying a distributed load of twenty kilograms per square meter of epiphytes, lianas, water, and accumulated debris, loads each cable with about 1 kN per meter. Over a span of one hundred meters with a sag of ten meters the horizontal tension is 123 kN and the tension at the anchors is 132 kN, which requires a steel cross-section of about 2.6 cm^2 at a working stress of 500 MPa, a wire rope of roughly eighteen millimeters in diameter. These magnitudes are well within the practice of cable-supported structures. The assumed load is a parameter of the design and not an established figure, since the mass that a constructed canopy would eventually carry depends on what grows on it; mature tropical forest holds on the order of tens of kilograms of above-ground biomass per square meter, of which only a fraction would be borne by a trellis, and the allowances for wind, for the weight

of saturated vegetation, and for the dynamic loading of storms would raise the design load considerably. The structural problem is therefore not the difficult one. The difficult problems are ecological and institutional, and the structural calculation shows only that the scale of the structure does not by itself preclude the concept.

The nutrient dynamics of the circulating system have a property that deserves emphasis, because it distinguishes a design that conserves nutrients from one that leaks them. Water delivered to the canopy gives up a fraction of its nutrient content to uptake on each pass and loses another fraction to drainage and leaching, and the remainder returns to the basin. At steady state the fraction of the makeup input that reaches biomass is the ratio of the uptake fraction to the sum of the uptake and loss fractions, and this ratio, which is independent of the rate of circulation, is the efficiency of the loop. If uptake per pass is five per cent and losses are one per cent, the efficiency is 83 per cent and a nutrient atom makes about seventeen passes before it is lost; if losses rise to five per cent the efficiency falls to 50 per cent and the number of passes to ten. The result implies that the engineering effort should be directed at reducing loss, by the design of the biochar-lined berms and the capture of drainage, more than at increasing the circulation, and it implies that the monitoring of losses at the boundary of the system is the essential measurement. A system that cannot show that it retains its nutrients is exporting them to the watershed, and the effect on downstream waters could be harmful.

The remaining questions are those of siting and consent. Rainforests grow where rain falls, and the regions suited to the concept are largely in the tropics of Latin America, Africa, and Southeast Asia, much of which is the territory of indigenous and local communities with customary claims, often unrecognized in law. A program that proposed to construct large structures on such land without the free, prior, and informed consent of the communities would reproduce the pattern of extraction against which the book defines itself, and the placement of the structures would be a question of equity and not merely of ecology. The communities hold knowledge of the forests, including in some regions of the dark earths themselves, that no external designer possesses, and the Ostrom principles of Chapter 18 apply here as they did to water: monitoring by the users, graduated sanctions, and recognition of their rights to organize. It is consistent with the program that a community should decline, and a program that could proceed only if no community declined would not be a program of direction.

The assessment under the standard of Chapter 16 places the trellis in the first readiness class in its structural aspects and the system as a whole in the second or third. The mechanism is explicit. The energy balance is simple, since the circulation is gravity-driven and the principal energy cost is the lift required to return water to the basin where rain does not. The material limits are known for the cables and anchors and unknown for the biochar filtration at the scale and flow rates proposed. The comparison with alternatives is required and unflattering to the concept in the near term: the protection of existing forest, the restoration of degraded land by assisted natural regeneration, and the support of indigenous land management all have a better-documented record and lower cost per hectare. The generator and the rain are candidates for supplementing these in places where they are not sufficient, and for demonstrating, at pilot

scale and under independent monitoring, that the hypotheses on which they rest are true.

Formalization

Four results are given, covering resource limitation, the structure, nutrient retention, and graded persistence.

Let a community grow at a rate determined by resources $R_1, \dots, R_m$ (water, light, nitrogen, phosphorus, and so on) each with requirement $K_i > 0$, and let $a_i = R_i/K_i$ be the availability ratio. The *minimum law* states that the relative growth rate is $g = \min\{1, a_1, \dots, a_m\}$.

Proposition 19.1. *Under the minimum law, $\partial g / \partial R_i = 0$ whenever $a_i > \min_j a_j$, so an increase in a non-limiting resource leaves growth unchanged. The marginal value of a resource is positive only for the limiting resources, and if S is the set of indices attaining the minimum with $\min a_j < 1$, raising all of S together by a common factor λ raises growth to $\min\{1, \lambda \min_j a_j, \min_{i \notin S} a_i\}$.*

Proof. The function $\min$ is nondecreasing in each argument and, at a point where the minimum is attained only by index j , independent of every other argument locally. The factor formula follows by substituting λa_i for a_i , $i \in S$. $\square$

For availability ratios $(N, P, \text{water}, \text{light}) = (1.4, 0.6, 1.1, 2.0)$ growth is 0.6 and is limited by phosphorus; doubling nitrogen changes nothing; raising phosphorus to 0.9 gives growth 0.9; raising it again to 1.1 gives growth 1.0. The implication for design is that the nutrient supply must be matched to the limiting element, which will change as each is relieved, and that a program supplying a standard nutrient mix is wasteful in all but one element.

Second, for a cable of horizontal span L carrying a uniform load w per unit horizontal length with sag d , the parabolic approximation gives horizontal tension $H = wL^2/(8d)$, tension at the supports $T_{\max} = \sqrt{H^2 + (wL/2)^2}$, and cable length $\approx L[1 + \frac{8}{3}(d/L)^2]$. The required cross-section for working stress σ is $A = T_{\max}/\sigma$. With $w = 981 \text{ N m}^{-1}$, $L = 100 \text{ m}$, $d = 10 \text{ m}$: $H = 122.6 \text{ kN}$, $T_{\max} = 132.1 \text{ kN}$, and at $\sigma = 500 \text{ MPa}$, $A = 2.64 \text{ cm}^2$ and diameter 18.3 mm for a solid section of that area. Since $H \propto L^2/d$, halving the sag doubles the tension, which is why the sag is a primary design choice.

Third, consider a circulating loop with basin content M of a nutrient, circulation flow Q of water of concentration $c = M/V$, uptake fraction f , loss fraction ℓ per pass, and external makeup m per unit time. Then $\dot{M} = m - (f + \ell)Qc$.

Proposition 19.2. *The steady state is $M^* = mV/((f + \ell)Q)$ and is globally asymptotically stable, with time constant $\tau = V/((f + \ell)Q)$. The steady uptake is $fm/(f + \ell)$, so the retention efficiency is $\epsilon = f/(f + \ell)$, independent of Q and V , and the mean number of passes before loss is $1/(f + \ell)$.*

Proof. The equation is linear: $\dot{M} = m - \frac{(f+\ell)Q}{V}M$, whose solution relaxes exponentially to M^* with the stated time constant. Uptake is $fQc^* = fm/(f + \ell)$. The number of passes before a unit of nutrient is removed is geometric with parameter $f + \ell$. $\square$

With $f = 0.05$ and $\ell = 0.01$ one has $\epsilon = 0.833$ and 16.7 passes, and with $\ell = 0.05$ one has $\epsilon = 0.5$ and 10 passes. The energy to return water over a head h is ρgh per unit volume, which is 0.82 kWh m^{-3} for $h = 300 \text{ m}$, an amount that is avoided entirely where rain refills the basin.

Fourth, let B be the supported biomass and $A(t) \in [0, 1]$ the autonomous fraction. If support were withdrawn at time t the expected loss is $L(t) = (1 - A(t))B$, the *termination exposure*. If autonomy grows logistically, $A(t) = [1 + e^{-k(t-t_0)}]^{-1}$, then the exposure falls below a tolerance ϵB at $t_\epsilon = t_0 + k^{-1} \ln[(1 - \epsilon)/\epsilon]$. For $k = 0.3 \text{ yr}^{-1}$, $t_0 = 10 \text{ yr}$ and $\epsilon = 0.05$ this gives $t_\epsilon \approx 19.8 \text{ yr}$. The functional form is a modelling choice, and the point of the statement is that the date at which the structure can be removed is an output of the monitoring, not an input to the design.

Chapter 20

Intervolsorial Pediments and Marine Biomass

The marine counterpart of the terrestrial program of the preceding chapter couples the cultivation of kelp to the energy of the tide and to the processing of the harvest, and it is organized around structures that the author's earlier notes call volsorial and intervolsorial pediments. The descriptions given there are not mutually consistent in every detail, and the book uses the following reading of them. A volsorial pediment is a large tidally powered structure, described as a thousand meters tall, that serves as a gravitational battery: tidal energy is used to raise mass, the mass stores the energy as gravitational potential, and its descent returns the energy on demand. In oceanic versions the pediment is a platform that also carries kelp cultivation, with kelp-based parachutes and cables tethered by mycelial networks, and in some descriptions the platforms are mobile. Intervolsorial pediments are the suspended terraces or habitable levels between and among such structures, which the notes envision as housing on the order of two hundred thousand people each and as multifunctional sites for the recovery of resources, and which in the later orbital proposals serve as landing hubs for nested expandable towers. A further and more speculative idea, the hurricane mitt, would deploy an array of some ten thousand pediments with kelp parachutes to engage a tropical cyclone. The book treats the energy-storage function and the kelp coupling as the core of the concept, both of which can be analyzed with elementary mechanics, and treats the habitation and storm-engagement functions as conjectural elaborations whose assessment awaits specification. The definition used here is therefore a reading of scattered descriptions and is to be superseded by the author's fuller specification. The chapter analyzes three quantities: the energy of the tide that charges the battery, the storage that a gravitational battery of a stated height can hold, and the supply of nutrients that the tidal exchange delivers to the cultivation. The condition of marine and terrestrial biodiversity against which any such program would be judged is summarized in the global assessment of the intergovernmental platform [55].

The energy of a tidal lagoon is a matter of elementary mechanics. A basin of area A that fills

at high tide and is emptied through turbines at low tide, when the level outside has fallen by the tidal range R , releases the potential energy of the water that it holds above the outside level, and if the water is released as the level falls the available energy per tide is $\frac{1}{2}\rho g A R^2$. The dependence on the square of the range explains the geographical concentration of tidal power: it is practicable where the range is several meters, which occurs in a small number of estuaries and bays of the world, and impracticable elsewhere. The best-known operating example, the barrage on the Rance estuary in France, encloses a basin of about 22 km² in a region whose mean range is about eight meters, and a calculation with these figures gives about 7×10^{12} J, or 1.97 GWh, per tide, which if realized once in each tidal cycle of 12.42 hours corresponds to a mean power of about 158 MW. The actual mean output of the station, of the order of 500 GWh per year, is about 57 MW, roughly a third of that figure. The shortfall is a typical one, resulting from the partial use of the range, the efficiency of the turbines, the requirement of operating within a variable tide, and the need to hold water during parts of the cycle. A reasonable planning assumption is that an installation will deliver a quarter to a third of the energy that the formula gives for the full range, and the estimate of any proposal should be accompanied by the corresponding derating.

The coupling to cultivation depends on a different property of the exchange. Kelp, like all photosynthetic organisms, requires nutrients, among which nitrogen is commonly the limiting one in temperate coastal waters, and in an enclosed lagoon the supply of nitrogen is set by the rate at which the water is exchanged with the sea and by the concentration of the nutrient in the incoming water. Because the biomass takes up nutrient in proportion to its growth, the exchange rate places a ceiling on the production that the lagoon can sustain, whatever the density of the plants and the quantity of light. The formalization derives the ceiling: production per unit volume cannot exceed the product of the exchange rate and the external nutrient concentration, divided by the nutrient content of the biomass. For an exchange of 0.2 volumes per day, an external concentration of 10 millimoles of nitrogen per cubic meter, and a nitrogen content of two per cent of dry mass, the ceiling is 1.4 grams of dry biomass per cubic meter per day, which for a lagoon five meters deep corresponds to about 26 tonnes of dry mass per hectare per year. The figure is illustrative, but it indicates the magnitude: the nutrient supply of an ordinary coastal exchange can support production of the order that cultivated kelp achieves, and cultivation at greater density than the supply can bear would simply starve. The result also shows what the tidal energy and the cultivation have in common: both are proportional to the exchange, and both are maximized by a design that moves a large volume of water through the structure, so that the interests of the two functions are aligned in the dimension that matters most. Two kinds of precedent should be kept apart in reading what follows. The practical case for farming kelp together with shellfish in three-dimensional ocean plots, as a working livelihood and not as a laboratory proposal, is made from experience in Smith's account of regenerative ocean farming [61]. The imaginative precedent, in which a living marine ecosystem is treated as part of the infrastructure of a civilization and not as a crop, belongs to the futurist and science-fiction tradition, of which Brin's novels are a leading example [62]; such works supply a way of posing the question and no evidence about productivity, sequestration,

or ecological consequence, which must come from the oceanographic literature.

The ecological assessment is governed by the standard of the author's earlier work, which is that the objective is diverse, geographically available, seasonally synchronized, and recoverable ecological production and not the maximum of biomass. Kelp cultivation can serve that objective when it is carried out as the cultivation of a known species in a defined place, with measurable effects on the food web, on the oxygen and carbon chemistry of the water, and on the organisms that depend on the habitat it creates. It can also fail the objective, through the escape of cultivated genotypes into wild populations, the transmission of disease, the shading of the benthos, and the concentration of grazers. The experience with the Sargassum belt in the Atlantic supplies the contrary case. Sargassum in the open ocean is productive habitat, but the same biomass stranded on a coast blocks light, consumes oxygen, and releases hydrogen sulfide and ammonia as it decomposes. The belt reached the highest abundance in the satellite record in 2025 and remained near that level afterward, with several factors, among them river nutrient discharge, atmospheric deposition, upwelling off West Africa, and changes in wind and current, implicated and none established as the sole cause. Biomass increased, and the function of the ecosystem in the places where it was deposited reversed. The lesson is that the quantity of biomass is not a measure of health, and that biomass the receiving system cannot incorporate is a burden and not a resource.

The processing of the harvest is the place where the argument returns to the energy balance. Kelp is wet: freshly harvested it is typically about nine-tenths water by mass. Any process that requires dry feedstock must first remove the water, and the latent heat of evaporation of water, 2.26 MJ per kilogram, is large. For a tonne of kelp with eighty-eight per cent water the drying energy is about 1.99 GJ, or 552 kWh, whereas the dry matter, taken at a heating value of 12 MJ per kilogram, contains 1.44 GJ. The energy required to dry the feedstock thermally is therefore about 1.4 times the energy it contains, and a conversion pathway that begins with thermal drying cannot be energetically self-sustaining unless the heat is obtained from a source other than the feedstock. This is the quantitative reason for the interest in processes that operate on wet material, among which hydrothermal conversion is the principal class. The Caldera Reactor, an oceanic thermopneumatic biomass-processing system developed in the author's earlier work, is a proposal of this kind. It is described as using geothermal steam to lift a compression plate, drawing seawater into the chamber by the vacuum produced as the steam condenses, compressing layered kelp, peat, and sediment, recovering energy through turbines, and yielding biocrude and feedstock for biochemicals and bioplastics, with the conversion handled by hydrous pyrolysis for wet feedstock and anhydrous pyrolysis for drier streams. It is a system whose individual elements are plausible and whose integration has not been demonstrated at scale, and it is correctly assigned to the second readiness class. The author's design rule that cultivated kelp should be the normal feedstock, with excess Sargassum intercepted before it reaches the shore as a secondary and disturbance-responsive input, is a sound one. A processing system that depended structurally on the bloom would have a financial interest in the continuation of a pathology, which is the situation of any enterprise that profits from the

damage it is presented as remedying.

Finally, the program is subject to the constraints of governance that apply to any use of the coast. Coastal waters are used for fishing, navigation, recreation, conservation, and the cultural practices of coastal communities, many of whom have claims of long standing. A structure that encloses a lagoon excludes some of these uses and alters others, and the effects on migrating fish and on sediment transport along the coast can extend far beyond the structure. The assessment must therefore include the effect on sediment budgets and on the fisheries of the region, the provision of passage for migratory species, and the mechanism by which the affected communities participate in decisions. The experience of tidal barrages, which have altered the ecology of the estuaries in which they were built and are nonetheless regarded by some as successes, indicates that the effects can be substantial and the judgments contested.

Formalization

Three results are given, on the energy of a lagoon, the nutrient ceiling on production, and the energetics of drying.

For a lagoon of horizontal area A (taken constant over the range), water density ρ , and a tidal range R , suppose the lagoon fills at high tide, and then is emptied through turbines while the outside level falls from the high-tide level to the low-tide level, with generation continuing until the lagoon level reaches the outside level. In the idealization in which the lagoon is emptied at the moment of low tide and all potential energy of the retained water is converted, the energy per tide is the work to lift the retained volume AR by its mean height $R/2$.

Proposition 20.1. *The energy released per tide by complete emptying is $E_{\text{tide}} = \frac{1}{2}\rho gAR^2$. If the lagoon is emptied once in each tidal period T , the mean power in the idealized cycle is $\bar{P} = \frac{1}{2}\rho gAR^2/T$. In practice the delivered energy is ηE_{tide} with $\eta < 1$ accounting for the partial use of the range, the turbine efficiency, and constraints of operation.*

Proof. The mass of water retained above the low-tide level is ρAR , whose centre of mass is at height $R/2$ above that level. The potential energy is $\rho AR \cdot g \cdot R/2$. With one emptying per semidiurnal tidal period $T = 12.42$ hours, the mean power is E_{tide}/T . $\square$

With $\rho = 1025 \text{ kg m}^{-3}$, $A = 22 \text{ km}^2$, and $R = 8 \text{ m}$, $E_{\text{tide}} = 7.08 \times 10^{12} \text{ J} = 1.97 \text{ GWh}$, and with one emptying per period $T = 12.42 \text{ h} = 4.47 \times 10^4 \text{ s}$ the mean power is 158 MW. A mean output of 57 MW corresponds to $\eta \approx 0.36$.

For the gravitational battery, a mass M raised through a height H stores $E = MgH$, with round-trip efficiency η_{rt} . The energy per tonne at $H = 1000 \text{ m}$ is $9.81 \times 10^6 \text{ J}$, which is 2.725 kWh, so that the mass required to store an energy E_{st} is $M = E_{\text{st}}/(gH\eta_{rt})$. A settlement of 2×10^5 persons with a mean demand of 1 kW per person, which is 200 MW, requires 4.8 GWh to bridge a full day, which at $H = 1000 \text{ m}$ and $\eta_{rt} = 1$ is 1.76×10^6 tonnes, comparable in

order of magnitude to the mass of the largest masonry monument, and at $\eta_{rt} = 0.8$ is 2.2×10^6 tonnes. For comparison, at a representative 200 kWh per tonne for a lithium-ion pack the same energy would be stored in 2.4×10^4 tonnes, about 1% of the gravitational mass, so that the gravitational battery trades a vastly larger mass for the absence of chemical degradation and for the use of inexpensive material, a trade whose merit is an engineering and economic question.

For the nutrient ceiling, let the lagoon have volume V , exchange rate e (volumes per unit time), external nutrient concentration N_{ext} , internal concentration N , and uptake of y units of nutrient per unit of biomass produced. Let P denote production of biomass per unit volume per unit time.

Proposition 20.2. *At steady state, $e(N_{\text{ext}} - N) = yP$. Hence $P \leq eN_{\text{ext}}/y$, with equality if and only if the internal nutrient concentration is driven to zero. The ceiling is proportional to the exchange rate e and cannot be raised by increasing the density or the light supply.*

Proof. A nutrient balance on a well-mixed lagoon gives $V\dot{N} = eV(N_{\text{ext}} - N) - yPV$, and at steady state the bracket vanishes. Since $N \geq 0$, $yP = e(N_{\text{ext}} - N) \leq eN_{\text{ext}}$. $\square$

With $e = 0.2 \text{ d}^{-1}$, $N_{\text{ext}} = 10 \text{ mmol m}^{-3}$, and $y = 0.02/14 \text{ mol g}^{-1} = 1.43 \text{ mmol g}^{-1}$ of dry mass, the ceiling is $P = 1.4 \text{ g m}^{-3} \text{ d}^{-1}$. Over a depth of 5 m this is $7 \text{ g m}^{-2} \text{ d}^{-1}$, or $25.6 \text{ t ha}^{-1} \text{ yr}^{-1}$ of dry mass. The corresponding sustainable harvest policy is obtained from a logistic growth law with harvest rate h and effective growth rate r_{eff} (the rate of growth under the nutrient conditions): the equilibrium biomass is $K^* = K_{\text{max}}(1 - h/r_{\text{eff}})$ and the yield hK^* is maximal at $h = r_{\text{eff}}/2$, where it equals $r_{\text{eff}}K_{\text{max}}/4$. The maximum-yield policy sits at half the carrying capacity and is the boundary of the region in which a perturbation of r_{eff} can cause collapse, so that a policy chosen with margin sets h below $r_{\text{eff}}/2$.

For the drying energetics, let wet biomass of mass m have a water mass fraction w and let the dry matter have heating value q per unit mass. The thermal drying energy is $\ell_v w m$, where $\ell_v = 2.26 \text{ MJ kg}^{-1}$, and the energy content is $q(1 - w)m$.

Proposition 20.3. *The ratio of drying energy to energy content is $\mathcal{R} = \ell_v w / (q(1 - w))$. It exceeds 1 when $w > q/(q + \ell_v)$, a threshold of 0.842 for $q = 12 \text{ MJ kg}^{-1}$ and 0.869 for $q = 15 \text{ MJ kg}^{-1}$, and it increases without bound as $w \rightarrow 1$. For kelp at $w = 0.88$ and $q = 12$, $\mathcal{R} = 1.38$.*

Proof. Direct computation: $\mathcal{R} > 1$ iff $\ell_v w > q(1 - w)$, i.e. $w(\ell_v + q) > q$. For $w = 0.88$, $\mathcal{R} = 2.26 \cdot 0.88 / (12 \cdot 0.12) = 1.381$. $\square$

The result shows why wet-processing pathways are a requirement and not an optimization for feedstocks of high moisture content, and why the heating value is of secondary importance: even at $q = 15 \text{ MJ kg}^{-1}$ the ratio is 1.10.

Chapter 21

Geothermal Mass Accelerators

The last of the principal terrestrial concepts is the one that connects the terrestrial program to the orbital one. The geothermal mass accelerator appears throughout the author's earlier notes as a ground-based system that uses the heat of the Earth as its source of energy to accelerate a payload, so that a rocket may dispense with some or all of its own propulsion in the first stage of ascent. In those notes the concept has several recurring features. It is to provide the initial thrust for vehicles that carry no engine of their own or only a small one, often through staged rings that accelerate the payload in sequence. It is to operate at low acceleration, so that it can serve passengers and fragile freight as well as hardened payloads, and the notes speak of rocketless launch and of slow freight. It is to be sited near the equator, where the rotation of the Earth contributes to the velocity of an eastward launch, and in one version on a floating structure described as an international space bridge. And it is to be integrated with the orthodromic network of Chapter 17, by retrofitting the great-circle corridors with accelerator track, and with the skyhook and expandable towers of the next part, which would receive the payload at altitude. The concept is accordingly a node in a larger architecture, and it is assessed here as such, with the assumption that the geothermal element supplies energy and the accelerator is an electromagnetic or equivalent device for converting that energy into the kinetic energy of the payload. The resource base for geothermal energy and its engineering limits are assessed in the standard interdisciplinary study [56]; no source cited there addresses the accelerator, which remains the book's own proposal.

The first quantity to settle is how much energy is required, because the answer governs the whole assessment and, as it turns out, is favorable. To place a kilogram in a circular orbit at four hundred kilometers altitude requires its kinetic energy at the orbital speed of about 7.67 km/s, which is 29.4 MJ, and its potential energy at that altitude, about 3.8 MJ, a total of 33.2 MJ or 9.2 kWh. If the payload is instead given the speed of about 8.03 km/s that a surface-grazing transfer orbit to that altitude requires, with a small circularization burn of about 0.12 km/s at the top, the kinetic energy is 32.2 MJ, or 8.95 kWh per kilogram. These figures are the energy delivered to the payload, and they exclude the losses of the accelerator

and of the passage through the atmosphere. At a conversion efficiency of one half, the delivery of ten thousand tonnes a year would require about 180 GWh a year, a mean electrical power of about 20 MW, which a single geothermal field of modest size supplies. A field of one gigawatt, operating at ninety per cent of capacity, could in principle supply the energy for something like 880 kilotonnes a year at perfect efficiency, which exceeds present annual launch mass to orbit by some two orders of magnitude. The energy is thus not the binding constraint, and this should be said plainly because discussions of launch often concentrate on cost per kilogram as though it were an energy cost; at the price of industrial electricity the energy for a kilogram is of the order of a dollar, and the present cost of launch is dominated by the vehicle, its propellant, and its operations.

The constraints that do bind are the track and the atmosphere. The track length required to reach a speed v at a constant acceleration a is $v^2/(2a)$, and for $v = 8.03$ km/s it is 33 km at $100g$, which is acceptable for hardened payloads and not for people, 3,286 km at $1g$, 1,095 km at $3g$, and 32,900 km at $0.1g$. The last figure is of the order of the circumference of the Earth, and it explains the interest of the notes in rings and in the equator: a gentle launch implies a very long track, and the only structures that long on Earth are those that follow its curvature. Launching eastward from the equator reduces the velocity that the track must supply by the equatorial rotation speed of about 0.465 km/s and the track length by about eleven per cent, a useful but modest gain. On the other hand the power demanded is proportional to the acceleration, because the instantaneous power of a track delivering force ma at speed v is mav , which has the mean value $\frac{1}{2}mva$ for a launch from rest, so that a gentle launch also needs a small power plant. For a payload of one tonne, the mean power at $3g$ is about 118 MW (the peak is twice as great), within the reach of a large geothermal plant and a small storage buffer, and at $0.1g$ the mean is about 4 MW, whereas at $100g$ it is 3.9 GW, which would have to be drawn from storage. The compatibility of geothermal baseload with low acceleration is a real feature of the concept: the form of the energy source and the form of the launch match.

The atmosphere imposes the severest constraint. A body moving at 8 km/s at sea level meets air with a dynamic pressure $\frac{1}{2}\rho v^2$ of nearly 40 MPa, which is about four hundred times atmospheric pressure, and the stagnation enthalpy $\frac{1}{2}v^2$ of 32 MJ per kilogram would, if converted to heat in the air, correspond to temperatures of tens of thousands of kelvins before dissociation and radiation intervene. Both are conditions of reentry, run in reverse, and they would destroy any payload not protected by a shield of a mass comparable to that of reentry vehicles, only here the exposure occurs at the lowest and densest part of the atmosphere, where the loads are greatest. The dynamic pressure falls to 23.7 MPa at five kilometers, 2.9 MPa at twenty, and 0.13 MPa at forty, so that high speed is bearable only at great altitude. There are two responses, both of which are present in the notes. One is to evacuate the track and extend the evacuated tube to high altitude, so that the payload leaves the tube above most of the atmosphere; the mass and strength of a tube a few tens of kilometers tall exceed anything built, and its association in the notes with expandable towers reflects this requirement. The

other is to reduce the exit speed, so that the accelerator supplies only the first part of the ascent and a skyhook, a rocket, or both supply the rest. At an exit speed of 2 km/s the dynamic pressure at sea level is 2.5 MPa and at five kilometers 1.5 MPa, which is about what high-speed projectiles and missiles endure.

The reduced-speed option has an attractive consequence in the mathematics of the rocket. The velocity increment that a rocket must supply to reach orbit, including losses to drag and gravity, is about 9.4 km/s. With a specific impulse of 350 s, corresponding to an exhaust velocity of 3.43 km/s, the mass ratio is $e^{9.4/3.43} \approx 15.5$. If the accelerator supplies 2 km/s of the increment the mass ratio falls to $e^{7.4/3.43} \approx 8.6$, and the fraction of initial mass that is not propellant rises from 6.5 per cent to 11.6 per cent, a factor of 1.79. The accelerator supplies only $(2/7.67)^2 \approx 7$ per cent of the orbital kinetic energy, but because propellant must lift the remaining propellant, the benefit in payload fraction is out of proportion to the energy share. This is the logic of the staged arrangement: a modest accelerator, fed with inexpensive energy, makes a difference to the economics of the rocket, and a more ambitious one makes a larger difference at disproportionately higher difficulty.

The siting of the device raises the geothermal question. Geothermal resources are concentrated along tectonic boundaries and volcanic arcs, and the high-temperature fields lie in Iceland, East Africa, the Andes, Central America, Indonesia, the Philippines, and the western United States. Few are near the equator at low altitude, and those that are, such as the fields of the East African Rift and of the volcanic arcs of Ecuador and Indonesia, are inland and mountainous, which is favorable for exit altitude and unfavorable for the transport of the vehicles and for the length of a straight track. A floating equatorial installation, as in the notes, would not be located on a geothermal resource at all and would receive its energy by cable. The designation geothermal is thus best read as a statement of the energy source, which can be transmitted over distance, and not of the location of the accelerator.

A final and serious consideration is that the device is symmetric with respect to its use. An apparatus that can accelerate a mass to orbital speed can also accelerate a mass to hypersonic speed along a trajectory that does not enter orbit, and a kinetic projectile of ordinary material carries energy comparable to that of conventional explosives per unit mass at a speed of only 3 km/s. A mass accelerator is accordingly a dual-use technology of a particularly stark kind, and its governance cannot be separated from its engineering. The institutional requirements include transparency of the facility's operations, the inspection of payloads, the registration of launches, and a treaty framework that does not exist at present. The notes describe the geothermal mass accelerators in one narrative as having been dormant and then becoming tools of sustainable growth and not weapons or instruments of control, and the statement is a good statement of the requirement, not a description of the means by which it would be met. The assessment under the standard of Chapter 16 is that the energy balance is favorable and the principal difficulties, the atmospheric exit and the length of the gentle track, are unsolved; the concept belongs to the third readiness class and must not figure in any plan on which present obligations depend.

Formalization

Let a payload of mass m be accelerated at constant acceleration a along a track to a speed v relative to the track, the track itself moving at an inertial speed v_0 along the direction of launch (the equatorial rotation, for an eastward launch).

Proposition 21.1. (i) The track length is $L = (v - v_0)^2/(2a)$ with v the inertial exit speed, and the duration is $t = (v - v_0)/a$. (ii) The work done on the payload by the track is $W = \frac{1}{2}m(v^2 - v_0^2)$, and the instantaneous power is $P(s) = ma(v_0 + as)$ at time s , whose time average is $\bar{P} = W/t = \frac{1}{2}ma(v + v_0)$ and whose peak is mav . (iii) For orbital insertion at radius r_a from a surface transfer ellipse, the exit speed at the surface is $v_p = \sqrt{\mu(2/R - 1/a_t)}$ with $a_t = (R + r_a)/2$, and the apogee circularization increment is $\Delta v = \sqrt{\mu/r_a} - \sqrt{\mu(2/r_a - 1/a_t)}$.

Proof. (i) Kinematics in the frame of the track: $v - v_0 = at$ and $L = \frac{1}{2}at^2$. (ii) The inertial speed of the payload at time s is $v_0 + as$, the force is ma , and power is force times speed. Integration over $[0, t]$ gives $W = ma(v_0t + \frac{1}{2}at^2) = \frac{1}{2}m(v^2 - v_0^2)$. The average is W/t , and $v_0 + at = v$ gives the peak. (iii) The vis-viva equation $v^2 = \mu(2/r - 1/a_t)$ evaluated at the perigee radius R and at the apogee radius r_a , with the circular speed $\sqrt{\mu/r_a}$. $\square$

Numerically, with $\mu = 3.986 \times 10^5 \text{ km}^3 \text{ s}^{-2}$, $R = 6371 \text{ km}$, and $r_a = 6771 \text{ km}$, one finds $v_p = 8.029 \text{ km/s}$, apogee speed 7.555 km/s , circular speed 7.673 km/s , and $\Delta v = 0.118 \text{ km/s}$. The kinetic energy at v_p is 32.2 MJ kg^{-1} , which is 8.95 kWh kg^{-1} . The track lengths at $v_0 = 0$ are $32,865 \text{ km}$ at $0.1 g$, $6,573 \text{ km}$ at $0.5 g$, $3,286 \text{ km}$ at $1 g$, $1,095 \text{ km}$ at $3 g$, 329 km at $10 g$, and 32.9 km at $100 g$, with durations $136, 27, 13.6, 4.5, 1.4$ minutes and 8.2 s respectively; at the equatorial $v_0 = 0.465 \text{ km/s}$ the lengths are reduced by the factor $(1 - v_0/v)^2 = 0.887$. The mean power $\frac{1}{2}mav$ for $m = 1000 \text{ kg}$, $v = 8.03 \text{ km/s}$, $v_0 = 0$ is 3.9 MW at $0.1 g$, 39 MW at $1 g$, 118 MW at $3 g$, and 3.9 GW at $100 g$, and the peak mav is twice these values.

The pressure loads are $q = \frac{1}{2}\rho v^2$, equal to 39.5 MPa at sea level ($\rho = 1.225 \text{ kg m}^{-3}$) for $v = 8.03 \text{ km/s}$, 23.7 MPa at 5 km (0.736), 2.87 MPa at 20 km (0.0889), 0.59 MPa at 30 km (0.0184), and 0.13 MPa at 40 km (0.0040).

For the benefit of partial acceleration, let the required ideal velocity increment be Δv_{tot} , the accelerator supply Δv_a , and the rocket exhaust velocity c .

Proposition 21.2. The ideal mass ratio of the rocket stage is $\mathcal{M} = \exp[(\Delta v_{\text{tot}} - \Delta v_a)/c]$, and relative to no accelerator the ratio of payload fractions is $\exp(\Delta v_a/c)$, which is independent of Δv_{tot} . The fraction of orbital kinetic energy supplied by the accelerator is $(\Delta v_a/v_{\text{orb}})^2$ when the payload starts at rest, so the payload-fraction gain exceeds the energy share whenever $\exp(\Delta v_a/c) > (\Delta v_a/v_{\text{orb}})^2$.

Proof. The first statement is the rocket equation, and the ratio follows by division. The second is the ratio of $\frac{1}{2}\Delta v_a^2$ to $\frac{1}{2}v_{\text{orb}}^2$. $\square$

For $\Delta v_{\text{tot}} = 9.4$ km/s and $c = 3.43$ km/s: $\Delta v_a = 0, 1, 2, 3$ km/s gives mass ratios 15.5, 11.5, 8.6, 6.4 and payload-fraction factors 1, 1.34, 1.79, 2.40 (ideal, ignoring structural mass), against energy shares of 0, 1.7, 6.8, and 15.3 per cent. The disproportion between energy share and payload gain is the mathematical content of the claim that a modest accelerator is worthwhile if it can be built at acceptable cost, and the same proposition shows why the claim weakens once structural mass, which scales with the stage and not with the propellant alone, is included.

Chapter 22

Heat, Radiation, and Resilient Infrastructure

The terrestrial programs described in the preceding chapters share three physical constraints that are easily overlooked in the enthusiasm for each, and the present chapter, which closes the fourth part, treats them together. The first is heat: every use of energy ends as heat, and a civilization that multiplies its energy use will eventually meet the limit set by the planet's capacity to radiate it. The second is radiation, in the ordinary sense of ionizing radiation from nuclear processes and in the sense of the radiation environment that a long-lived structure experiences. The third is resilience, the capacity of an infrastructure to withstand shocks and to restore itself, which is the property that determines whether the programs deserve the label of infrastructure or are merely demonstrations. The three share a mathematical character, since in each there is a simple model whose parameters are measured and whose conclusions are sharper than the discussion usually allows. The mitigation pathways and the infrastructure constraints against which the heat budget is read are those of the sixth assessment of the intergovernmental panel [57].

The planetary heat budget supplies the first example. The Earth absorbs from the Sun about 1.2×10^{17} W, which is a quarter of the solar constant times one minus the albedo times the surface area, and radiates the same amount to space in equilibrium. Human use of primary energy is about 1.8×10^{13} W, which is 0.015 per cent of the absorbed flux, or about 0.036 W m^{-2} averaged over the surface. The direct heating is accordingly negligible by comparison with the radiative forcing from greenhouse gases, which exceeds two watts per square meter, some sixty times larger. The comparison reassures about the present and misleads about the future, because the heat of human activity scales with the energy used whereas the capacity of the planet to shed it is fixed, and the limit would be reached by a growth of some orders of magnitude. If the climate responds to a forcing F with an equilibrium warming F/λ , where the feedback parameter λ is of the order of $1.3 \text{ W m}^{-2} \text{ K}^{-1}$, then a direct heating that was permitted to cause a warming of 0.1 K corresponds to a total of 66 TW, about 3.6 times present

use, and a warming of 0.3 K to 199 TW, about eleven times. These figures are indicative, since the response to a heat flux differs from the response to a forcing of another kind, but they show that a planetary energy supply of the scale contemplated by some of the more ambitious proposals, a thousand times the present, could not be used on the surface without a major change in the climate. The conclusion has consequences for the orbital programs of the fifth part: energy generated in space and used on the surface is eventually dissipated on the surface, and it is not a way around the limit; and processes that need very large amounts of energy are better located where their waste heat can be radiated directly to space.

The time scale of the thermal response matters to the interventions of the polar and oceanic kind and to the dynamics of termination. The upper ocean has a large heat capacity, and a layer of one hundred meters has a capacity of about $4 \times 10^8 \text{ J m}^{-2} \text{ K}^{-1}$, so that with $\lambda = 1.3$ the time constant of the layer's response to a forcing is about ten years. The deep ocean has a far longer time constant. A forcing that is applied and then removed therefore produces a response that lags it by a decade for the upper layer and by centuries for the deep, and a program that reduces a forcing by a deliberate intervention and then ceases will be followed by a rebound whose rate is set by these time constants and by the rate at which the intervention is withdrawn. The termination analysis required by Chapter 16 for any such program must include the thermal response, and the formalization gives the relevant relation.

Radiation presents a different kind of limit. Several of the proposed systems involve nuclear energy, either as a power source for installations in remote locations or in the form of the refrigeration at the polar margins that appears in the author's earlier proposals, and any such system must be assessed against the physics of shielding and the institutions of safeguards. The attenuation of a narrow beam of gamma radiation by a material of thickness x follows the exponential law with an attenuation coefficient μ , which for photons of one million electron volts in ordinary concrete is about 0.15 cm^{-1} . The thickness that halves the beam is 4.7 cm, the thickness that reduces it tenfold is 15.7 cm, and a reduction by a factor of 10^6 requires about 94 cm. These are figures for an idealized beam; real shielding is complicated by scattering, by the build-up of secondary radiation, by neutrons that penetrate far more readily, and by activation of the shield itself, but the exponential law shows why a few meters of material suffice for a reactor and why the shielding of a structure that must be occupied is a question of mass and not of principle. The more difficult problems are not the shielding but the fuel cycle, the waste, and the institutions that guard against diversion. The weight of the arguments recorded in Chapter 21 about dual use applies here with equal force, and a nuclear installation in a polar region, where jurisdiction is contested, adds the complication that the body competent to inspect and license it may not exist.

The third constraint, resilience, is the property on which the preceding chapters most depend and which is hardest to demonstrate. An infrastructure is resilient to the extent that it continues to function under shocks, and it is restorable to the extent that it can be returned to function after failing, with limited help from outside. The two properties are different. A system with great redundancy may be robust to the failure of any component and yet

impossible to restart if it should fail altogether, because the knowledge, the spare parts, and the energy that a restart requires are held outside it. The author's earlier work on the Caldera Reactor proposes a layered architecture of redundancy, described as hyperpleonastic, with ten recursive layers of safety and fault tolerance. The attraction of such an architecture is evident from the arithmetic of independent failure, since if each of ten layers has a probability of failure of one in ten then the probability that all fail is one in ten billion. The arithmetic is misleading when the layers share causes of failure, a flood, a fabrication defect, a software error, an act of sabotage, that disable many of them at once. With a modest common-cause fraction of five per cent the probability of total failure is not 10^{-10} but 5×10^{-3} , set almost entirely by the common-cause term, as the formalization shows and as Chapter 29 develops. The lesson is that depth of redundancy cannot substitute for diversity, and that the layers of a defensive architecture should differ in physical principle, in supplier, in location, and in the people who maintain them.

The measure of restorability proposed here is an index of autonomy. For an installation that must restore itself, the relevant stocks are those of the items on which restoration depends: energy, spare parts, consumables, tools, and the hours of trained labor available. Each is consumed at some rate during the restoration, and the time for which the installation can continue is set by the stock that runs out first. The autonomy time is the minimum over the items of the stock divided by the consumption rate, and restoration is feasible without outside help only if the autonomy time exceeds the time required to complete it. The structure is identical to the minimum law of Chapter 19, and it has the same implication: the addition of stock to an item that is not the binding one does not improve autonomy, and the sensible policy is to identify the binding item, increase its stock until another item binds, and repeat. The practical value of the index is that it can be measured by exercises, in which an installation is deprived of outside supply and required to restart, and that the results can be published. The program requires that every installation which claims resilience supply the figure and the record of the exercises on which it rests.

Formalization

First, the heat balance. Let the surface temperature anomaly T obey $C dT/dt = F(t) - \lambda T$, where C is the heat capacity per unit area of the responding layer, F the forcing (in W m^{-2}), and $\lambda > 0$ the feedback parameter.

Proposition 22.1. *For constant F the solution with $T(0) = 0$ is $T(t) = (F/\lambda)(1 - e^{-t/\tau})$ with $\tau = C/\lambda$. If F is removed at time t_1 , then $T(t) = T(t_1)e^{-(t-t_1)/\tau}$ for $t > t_1$. If a standing cooling forcing $-F_c$ has held the system at the equilibrium $T = -F_c/\lambda$ and is removed abruptly, the initial rate of warming is F_c/C , and the rate is smaller the more gradually the forcing is withdrawn.*

Proof. The equation is linear with constant coefficients and the solutions are standard. For

the last statement, immediately after removal $C dT/dt = 0 - \lambda T = F_c$. $\square$

For a mixed layer of $h = 100$ m, $C = \rho c_p h = 1025 \times 3990 \times 100 = 4.09 \times 10^8$ J m⁻² K⁻¹, and with $\lambda = 1.3$ the time constant is $\tau = 3.15 \times 10^8$ s = 10.0 years. An abrupt termination of a cooling forcing of $F_c = 1$ W m⁻² implies an initial warming rate of $F_c/C = 2.4 \times 10^{-9}$ K s⁻¹ = 0.077 K per year in the layer, which is the quantity the termination plan must bound by staging the withdrawal over an interval comparable to or longer than τ .

For the energy ceiling, let $A = 5.1 \times 10^{14}$ m² be the surface area. A heat flux P spread uniformly produces a forcing P/A and an equilibrium warming $P/(\lambda A)$, so that for a tolerated warming ΔT the maximum sustained power is $P_{\max} = \lambda A \Delta T$. With $\lambda = 1.3$ this is 66 TW for $\Delta T = 0.1$ K and 199 TW for 0.3 K. The absorbed solar power is $S(1 - \alpha)A/4 = 1.21 \times 10^{17}$ W for $S = 1361$ W m⁻² and albedo $\alpha = 0.30$, and the present use of 18.5 TW is 1.5×10^{-4} of it.

Second, the attenuation law. A narrow beam of intensity I_0 traversing a thickness x of material with linear attenuation coefficient μ is attenuated to $I = I_0 e^{-\mu x}$. The half-value layer is $x_{1/2} = \ln 2/\mu$ and the tenth-value layer $x_{1/10} = \ln 10/\mu$; to achieve an attenuation factor 10^n requires thickness $n \ln 10/\mu$. With $\mu = 0.147$ cm⁻¹ (concrete of density 2.3 g cm⁻³ and mass attenuation coefficient 0.064 cm² g⁻¹ at 1 MeV), $x_{1/2} = 4.7$ cm, $x_{1/10} = 15.7$ cm, and $n = 6$ requires 94 cm. These are narrow-beam values for primary photons only, and a shield design requires transport calculations that include build-up and neutrons.

Third, the redundancy of layered defenses and the autonomy index. Let a defense have k layers, each failing with probability p and let a common cause with probability βp disable all layers, the remaining failures being independent with probability $(1 - \beta)p$ each.

Proposition 22.2. *The probability that every layer fails is $p_{\text{eff}} = \beta p + (1 - \beta p)((1 - \beta)p)^k$, which is bounded below by βp for every k , and which tends to βp as $k \rightarrow \infty$ when $(1 - \beta)p < 1$. Hence for $p = 0.1$, $\beta = 0.05$ and $k = 10$, $p_{\text{eff}} = 5.0 \times 10^{-3}$, although the independent model gives $p^k = 10^{-10}$.*

Proof. The total failure event is the union of the common-cause event, of probability βp , and, in its complement, the event that all k independent failures occur, of probability $((1 - \beta)p)^k$. Add the probabilities. $\square$

The autonomy time for items $i = 1, \dots, n$ with stocks S_i and consumption rates $c_i > 0$ is $T_a = \min_i S_i/c_i$, and restoration of duration T_r is feasible without external supply if and only if $T_a \geq T_r$. If T_r is required to be at most the autonomy time, then the cost-effective increment is to raise the stock of the item attaining the minimum until a second item attains it, and the proof is the same as that of the minimum-law proposition of Chapter 19.

Part V

The Orbital Program

Chapter 23

Skyhooks and Orbital Transfer

The fifth part of the book turns from the Earth to the space around it, and it begins with the concept that is the best developed of the orbital proposals and the one on which the others in this part lean. A skyhook, in the sense used here, is a long tether in orbit that rotates about its center of mass so that its lower tip, at the lowest point of each revolution, moves relative to the Earth's surface at a speed much less than the orbital speed, and its upper tip, half a revolution later, moves at a speed much greater. A payload launched to meet the lower tip at the modest speed of an accelerator or a small rocket can be caught there and carried around to the upper tip, from which it is released at a speed that takes it to a higher orbit or on an escape trajectory. The concept was developed by Moravec in 1977 [7] as a non-synchronous orbital skyhook and was elaborated in the following decades by Forward, Hoyt, and others, and a large literature of engineering analysis exists for it, which is why this program, among those of the book, has the strongest claim to the second readiness class. The author's notes combine the skyhook with the geothermal mass accelerator of Chapter 21 in an assembly in which the accelerator supplies the first part of the velocity and the skyhook the remainder, and the combination is the subject of the present chapter.

The first thing to understand about the skyhook is what it does and does not do. It does not create energy. The payload's gain of energy, and of momentum, is drawn from the orbital motion of the tether and its hub, which slows and drops, and unless the loss is made good the system decays after a few launches. The restoration of the hub's momentum can be accomplished in three ways. Electric propulsion can expend propellant to raise the orbit again; an electrodynamic tether can push against the magnetic field of the Earth, exchanging the hub's momentum with that of the planet's field at the expense of electrical power; and traffic in the opposite direction can return momentum to the hub, as when a payload descending from a higher orbit is caught and slowed. The third is the most elegant, since a system in which the mass launched upward is matched by mass returned downward conserves the hub's momentum on average and requires no external supply. It places a requirement on the economy that it served: the traffic must be balanced, or the imbalance must be paid for. The propellant cost of

the first method is calculable and is not trivial. With an electric thruster of exhaust velocity 29.4 km/s, corresponding to a specific impulse of 3000 s, restoring the hub after a payload has gained 3.5 km/s of velocity requires propellant equal to 12 per cent of the payload's mass, and after a gain of 7 km/s, 24 per cent. These fractions are small against the order-one fractions of rocket propellant, which is the reason the concept is attractive, but they are not zero, and the supply of propellant to the hub is a continuing logistic obligation.

The second point concerns the tether. The strength of a material is measured in the length of a column of it that would just support itself, or equivalently in the energy per unit mass at which the material fails, which for steel is about 0.1 MJ per kilogram, for aramid fibers about 2.5, for polyethylene and polybenzoxazole fibers about 3.5 to 3.7, and for carbon nanotubes, in the best laboratory samples of short length, many times more, though the strength of macroscopic yarns has been far below that of individual tubes. The strength per unit mass determines a characteristic velocity, $V_c = \sqrt{2\sigma/\rho}$, where σ is the working stress and ρ the density, and the mass of a rotating tether depends on the ratio of the tip velocity to this characteristic velocity through a function that increases extremely rapidly. At small ratios the tether is light, a modest multiple of the payload; at ratios above one, its mass grows faster than exponentially, and a tether of any practical mass can be built only if the tip velocity is kept comfortably below the characteristic velocity. Taking a working stress equal to half the tensile strength, the characteristic velocity of aramid is 1.6 km/s, of polybenzoxazole 1.9 km/s, and of a hypothetical nanotube yarn with a strength of 10 GPa 2.8 km/s. A tether with a tip velocity of 2 km/s then has a mass of ten times the payload for aramid, 4.6 times for polybenzoxazole, and 1.5 times for the hypothetical yarn. At a tip velocity of 3.5 km/s the ratios are 526, 86, and 10. The qualitative lesson is the one that the history of the concept has taught: the skyhook is feasible with materials of present strength at tip velocities of one to two kilometers per second, which is enough to reduce the burden on the lower stages of a launch but not enough to replace them, and it becomes a transportation system for escape trajectories only if materials of far greater strength become available. The assembly of the author's notes, in which an accelerator supplies several kilometers per second at the bottom, is a way of reducing the demand on the tether, and the quantitative case for the combination is that it moves the requirement from the material, which is unavailable, to the accelerator, which is merely very difficult.

The geometry of capture is the third point. The lower tip of the tether passes through the lowest point of its trajectory at a speed relative to the ground equal to the orbital speed minus the tip velocity, and the payload must arrive at the right place at the right time, within a window that is short because the tip moves fast. A tether of 100 km length rotating with a tip velocity of 3.5 km/s has a period of about three minutes. The payload must be guided to the tip with precision of meters, over a distance of hundreds of kilometers, by active guidance, and any error in the arrival is either corrected by the payload's own maneuvering or ends in a missed encounter or a collision. The loads on the payload at capture are those of an acceleration of the order of ten times gravity for a tether of this size, which is acceptable for

freight and unacceptable for passengers, and a gentler arrangement requires a longer tether at the same tip velocity: the acceleration is V^2/L , so doubling the length halves it. A tether of several hundred kilometers is therefore the natural scale for a system that is to carry people, and a tether of that length multiplies the hazards of the environment through which it passes.

The fourth point is survival. A long thin structure in low Earth orbit is exposed to micrometeoroids and to orbital debris, and the probability that it will be struck by an object large enough to cut it, over the lifetime of the system, is not negligible. The surviving designs avoid single-strand failure by multi-line construction, in which the load is carried by a network of lines with redundant cross-links so that a cut in one line does not part the whole, and the network is repaired by robotic agents that traverse it. The survival of the structure over its design life then depends on the rate of cuts, the capacity of the network to tolerate them, and the rate of repair, and the analysis, like that of redundancy in Chapter 22, depends on whether the events are independent. A collision with a large piece of debris may cut many lines at once, and the growth of the debris population in the orbits of interest, which is a consequence of past launches, is a common cause of exactly the kind that defeats redundancy. The orbital regime in which the tether must live is a commons in Ostrom's sense, and its governance is a precondition of the engineering.

The last point concerns orbital transfer itself, which is the subject of the second half of the chapter's title. The transfer between circular orbits of different radii by the most economical two-impulse maneuver, the Hohmann transfer [31], requires a total velocity increment that for a transfer from low Earth orbit at 400 km altitude to geostationary orbit is 3.86 km/s, composed of 2.40 km/s at the start and 1.46 km/s at arrival, and takes 5.3 hours. A skyhook that adds $2V_t$ to the payload's velocity with a tip velocity of 2 km/s supplies 4 km/s, which exceeds this requirement, so that a single tether of modest tip velocity, suitably placed, could deliver payloads from low orbit to geostationary transfer without propellant. This is the application for which the concept has been most seriously studied, and it is a more modest and more credible program than the launch of payloads from the ground. It turns the skyhook into infrastructure for traffic among orbits and not a means of reaching orbit, and in that role it connects to the orbital elevator of the next chapter, which would provide the reverse link.

The assessment under the standard of Chapter 16 follows. The mechanism is explicit and the energy balance is written out. The material limits are known, and they confine the near-term concept to tip velocities of one to two kilometers per second. The dependencies are the accelerator or rocket that supplies the payload to the tether, the guidance system, the orbital environment, and the supply of propellant for the hub. The concept is placed in the second readiness class for transfer between orbits, and in the third for the assembly with the accelerator reaching escape velocity. It has no bearing on the satisfaction of present obligations, and its development, if it proceeds, should be funded as a technology demonstration, beginning with a short tether that can validate the dynamics and the capture, and proceeding only upon successful demonstration.

Formalization

Consider a uniform-rotation tether of length L (from hub to tip) carrying a tip mass m_p , rotating at angular velocity ω , with tip velocity $V = \omega L$ relative to the hub. Let the material have working stress σ and density ρ , and let the cross-sectional area $A(r)$ be tapered so that the stress equals σ everywhere. Define $V_c = \sqrt{2\sigma/\rho}$ and $x = V/V_c$.

Proposition 23.1. *The tether mass (one arm) is $M_t = m_p \sqrt{\pi} x e^{x^2} \operatorname{erf}(x)$. The mass ratio M_t/m_p is increasing in x , behaves as $\frac{2}{\sqrt{\pi}} \sqrt{\pi} x^2 = 2x^2$ for small x , and grows as $\sqrt{\pi} x e^{x^2}$ for large x .*

Proof. Equilibrium of a tether element gives $\sigma dA/dr = -\rho\omega^2 r A$, hence $A(r) = A_0 e^{-kr^2}$ with $k = \rho\omega^2/(2\sigma)$. The tip condition is $\sigma A(L) = m_p \omega^2 L$. The mass is $M_t = \rho \int_0^L A(r) dr = \rho A_0 \frac{\sqrt{\pi}}{2\sqrt{k}} \operatorname{erf}(\sqrt{k}L)$, and $A_0 = A(L)e^{kL^2}$. Since $\sqrt{k}L = x$, $kL^2 = x^2$ and $\rho\omega^2 L/\sigma = 2x^2/L$,

$$\frac{M_t}{m_p} = \frac{\rho\omega^2 L}{\sigma} e^{x^2} \frac{\sqrt{\pi} L}{2x} \operatorname{erf}(x) = \sqrt{\pi} x e^{x^2} \operatorname{erf}(x).$$

For small x , $\operatorname{erf}(x) \approx 2x/\sqrt{\pi}$ gives $2x^2$; for large x , $\operatorname{erf}(x) \rightarrow 1$. □

Numerical values follow, with working stress equal to half the tensile strength: aramid (3.6 GPa, 1440 kg m⁻³) $V_c = 1.58$ km/s; polybenzoxazole (5.8 GPa, 1560) $V_c = 1.93$ km/s; a hypothetical yarn of 10 GPa and 1300 kg m⁻³, $V_c = 2.77$ km/s. For $V = 1, 2, 3, 3.5$ km/s the mass ratios M_t/m_p are, for aramid, 1.05, 10.3, 122, 526; for polybenzoxazole, 0.65, 4.6, 30, 86; for the hypothetical yarn, 0.28, 1.5, 5.4, 10. The centripetal acceleration at the tip is V^2/L , equal to 122 ms⁻² = 12.5 g for $V = 3.5$ km/s and $L = 100$ km, with a tip tension for a 1000 kg payload of 122.5 kN, and a tip cross-section of 61 mm² (a diameter of 8.8 mm if circular) at a working stress of 2 GPa.

For the hub's momentum, let a hub of mass $M_h \gg m_p$ impart a velocity gain Δv_p to a payload of mass m_p .

Proposition 23.2. *By conservation of momentum, the hub's velocity decrement is $\Delta v_h = m_p \Delta v_p / M_h$. Restoring it with a thruster of exhaust velocity v_e requires propellant mass $m_{\text{prop}} = M_h(1 - e^{-\Delta v_h/v_e}) \approx m_p \Delta v_p / v_e$, so the propellant fraction is $m_{\text{prop}}/m_p \approx \Delta v_p / v_e$, independent of M_h . If an equal mass descends and is slowed by the same amount Δv_p , the net decrement is zero.*

Proof. Momentum conservation gives $M_h \Delta v_h = m_p \Delta v_p$. The rocket equation [60] gives the propellant needed to change the hub's velocity by Δv_h , with $\Delta v_h/v_e \ll 1$ allowing the linearization. For the symmetric case, the momenta are equal and opposite. □

With $v_e = 29.4$ km/s, $\Delta v_p = 2, 3.5, 7$ km/s gives $m_{\text{prop}}/m_p = 6.8, 11.9, 23.8$ per cent.

For orbital transfer, with $\mu = 3.986 \times 10^5 \text{ km}^3 \text{ s}^{-2}$, $r_1 = 6771 \text{ km}$ and $r_2 = 42164 \text{ km}$, the transfer ellipse has semimajor axis $a = (r_1 + r_2)/2 = 24467 \text{ km}$; the circular speeds are 7.673 and 3.075 km/s; the transfer speeds at perigee and apogee are $\sqrt{\mu(2/r_1 - 1/a)} = 10.072 \text{ km/s}$ and $\sqrt{\mu(2/r_2 - 1/a)} = 1.618 \text{ km/s}$, so $\Delta v_1 = 2.399 \text{ km/s}$, $\Delta v_2 = 1.457 \text{ km/s}$, total 3.857 km/s, and the transfer time is $\pi\sqrt{a^3/\mu} = 5.29 \text{ h}$.

Chapter 24

Space Elevators and Expandable Structures

The space elevator is the oldest of the megascale concepts in this book and the one whose physics is simplest to state. A cable anchored on the equator and extending beyond geostationary altitude, with a counterweight at its far end, remains taut by the balance of gravitation and centrifugal force, and payloads can climb it without rockets. The idea originates with Tsiolkovsky [39] in the form of a tower and with Artsutanov [28] and Pearson [8] in the form of a cable, and the analysis of Pearson in 1975 established the central quantitative fact, which is that the taper of the cable required to carry its own weight is governed by the specific strength of the material. The engineering study of Edwards and Westling [13] is the standard modern account of the cable. The author's notes pair the elevator with a second idea, the Hoberman-inspired expandable structure, which they use for nested towers that extend by unfolding, for orbital rings and tunnels, and as a lattice of the kind drawn over a globe in the physical models of the nine-circle arrangement. The notes do not give a mechanical model of what the expandable linkage contributes to the problem of bearing load, and the present chapter supplies one, with a conclusion that is unwelcome to the concept and is nonetheless the one the physics supports: expandable linkages solve the problem of packaging and deployment, and they do not solve the problem of strength. The chapter therefore separates the two roles and assesses each.

The elevator proper is a tension structure. At each radius r along the cable the load it must carry is the net of the gravitational pull toward the Earth and the outward centrifugal force of the corotating frame, which is the derivative of the effective potential $\Psi(r) = -GM/r - \frac{1}{2}\omega^2 r^2$. The potential increases with radius up to the geostationary radius of 42,164 km, where the net force vanishes, and the tension in a cable of uniform stress accumulates according to the difference of the potential between the surface and that point. Evaluating the difference gives $\Delta\Psi = 4.84 \times 10^7$ J per kilogram. A cable designed so that the stress is the same at every height must have a cross-section that varies with height, and the ratio of its cross-section

at geostationary altitude to that at the surface is the exponential of the ratio of $\Delta\Psi$ to the specific strength of the material. The formula leaves little room for argument. At the specific strengths, taking the full tensile strength as the working stress to give the material the most favorable treatment, of aramid, 2.5 MJ per kilogram, the taper ratio is $e^{19.4} \approx 2.6 \times 10^8$; of polybenzoxazole, 3.7 MJ per kilogram, $e^{13.0} \approx 4.5 \times 10^5$; of steel at a working stress of 0.7 GPa, a number with more than two hundred digits; and of a hypothetical yarn with a strength of 50 GPa and a density of 1300 kg m^{-3} , $e^{1.26} \approx 3.5$. A taper of ten to the fifth or more is not a design but a refutation, since the cross-section at the surface must carry the working load of a payload while the cross-section at the top must be a hundred thousand times larger. The elevator is therefore a concept that depends entirely on a material, available in macroscopic lengths of tens of thousands of kilometers with a strength of tens of gigapascals at densities near that of carbon, that has not been produced; and a safety factor of two squares the taper ratios of every entry above, so that even the hypothetical yarn rises to 12 and the polybenzoxazole to 2×10^{11} . The reasons for the optimism of the concept's proponents are carbon nanotubes, whose individual strengths approach the required figure, and the reasons for the skepticism of its critics are that the strength of an assembly of nanotubes is limited by the weakest defect in a very long structure and by the imperfect transfer of load between the elements. The question of whether such a material can be made is a question of unresolved science and not of engineering, and the readiness class follows from it.

If the material were available, the remaining problems are of a different order but are not trivial. The cable would be exposed to debris and micrometeoroids along its length, in the region of low orbit where the debris is densest, and the survivability analysis of the preceding chapter applies. It would be subject to oscillation, since a tension structure that is tens of thousands of kilometers long has many modes, excited by the climbers, by the tides, and by the atmosphere at the base. It would require a moving platform at the base or a stabilizing counterweight at the top to manage the dynamics, and its traffic is limited by the power available to the climbers. A climber that moves at 200 km/h takes 7.5 days to reach the geostationary station, and a payload of one tonne at the surface requires a mechanical power of about 0.54 MW to maintain that speed, declining as the effective gravity falls with height. The energy to be supplied per kilogram is the difference of the potential, 48.4 MJ or 13.5 kWh, which is larger than the energy of orbital velocity in low orbit because the destination is higher; the elevator does not reduce the energy required to reach geostationary altitude but allows it to be supplied from the ground, by electricity, as in the geothermal arrangement of Chapter 21, rather than by the combustion of propellant.

The second half of the pairing concerns the structures of the Hoberman type. A scissor linkage consists of pairs of bars pinned at their midpoints, so that the bars can be rotated together from a compact folded state to an extended state, and the Hoberman sphere, devised by Chuck Hoberman in the 1990s, is an arrangement of such linkages in which a sphere expands uniformly to several times its folded diameter. The geometric property is the ratio between the stowed and deployed dimensions. For a stack of scissor cells whose bars make an angle θ with the

horizontal, the height of each cell is proportional to $\sin \theta$, and the expansion ratio between a stowed angle of five degrees and the fully deployed angle of ninety degrees is $1/\sin 5^\circ = 11.5$; for a stowed angle of three degrees it is 19. Nested or telescoping arrangements, in which successive stages are stowed inside one another and extend sequentially, multiply the achievable extension by the number of stages. These are real and useful properties, which suit the structure to the transport of a large volume in a small package, as is required for anything built in orbit from components lifted by rockets, and they suit it for adjusting its dimensions in response to loads. They are the properties by virtue of which expandable structures have found their uses on spacecraft in the form of antennas, solar arrays, and booms.

The load-bearing function is another matter, and here the physics is conclusive. A locked scissor lattice is a truss with additional joints, and its strength in compression is limited by the buckling of its members and by the stiffness of its joints, which are less than those of a welded or bonded truss of the same mass. A tower is a column, and a column is limited by the self-buckling length computed by Greenhill in 1881 [30]: a uniform column of bending stiffness EI and weight per unit length q buckles under its own weight when its height exceeds $(7.84 EI/q)^{1/3}$. For a thin-walled tube of radius r this gives a limiting height of $1.58 (E/\rho g)^{1/3} r^{2/3}$, which is proportional to only the two-thirds power of the radius, so that large height demands radius out of all proportion. For steel, a tube of radius 5 m can stand 630 m, of radius 50 m 2.9 km, and of radius 500 m 13.6 km. To reach 20 km a steel tube would need a radius of 890 m, and to reach 100 km, ten kilometers. With a material of the modulus of an ideal carbon nanostructure, one terapascal at a density of 1300 kg m^{-3} , the radii are 160 m for 20 km and 1.8 km for 100 km. A self-supporting compression tower to the edge of space is therefore excluded for any material of known modulus, whatever the geometry of its joints; an actual tall structure is tapered, guyed, or braced against a wide base, which changes the numbers by a modest factor but not the conclusion, and the tallest structures built are under a kilometer. The proposals in the notes in which nested expandable towers extend to great heights, and carry within them linear ring tunnels for launching, must therefore be interpreted either as structures of tens of kilometers whose feasibility would depend on pneumatic or tensile support and has not been analyzed, or as structures assembled in orbit, where gravity does not load them in the same way. In orbit, the expandable lattice comes into its own: a ring or habitat that must be built from components lifted separately can be folded into a launch vehicle and deployed on station, and the strength it requires is that needed to hold its own shape against centrifugal and tidal loads, which is modest.

The ordering that the author has proposed for the use of expandable structures, which is to build first a ring with compressive strength of its own so that the scissor-like motion is confined to the macroscopic act of deployment, is the correct one, and it is the ordering that the argument above recommends. In it the load-bearing member is a rigid structure whose strength does not depend on its joints, and the expandable linkage serves as the means of carrying the structure into place and opening it, after which it is locked and plays no further part in bearing load. The question that remains is what loads the ring itself must bear, and

the answer depends sharply on how the ring moves. A thin ring of radius r that turns about the Earth's center at an angular rate Ω has a hoop stress equal to the density times the difference between the square of its own speed and the square of the local circular orbital speed, $\sigma = \rho(v^2 - v_{\text{orb}}^2)$: positive, a tension, if the ring moves faster than a circular orbit, zero if it moves at orbital speed, and negative, a compression, if it moves slower. A ring in free fall at orbital speed therefore bears no hoop load at all, and this is the condition in which an expandable ring is most easily built, deployed in orbit from components lifted separately and opened to its full radius. A ring that is to be stationary with respect to the ground, however, moves slowly and is crushed: at low altitude the compressive stress is the density times roughly $5.9 \times 10^7 \text{ J kg}^{-1}$ at 400 km altitude and 6.2×10^7 at the surface, which for steel is nearly 490 GPa and which for any material requires a compressive specific strength greater than that of the strongest known fibers in tension. The figure falls to 3.9×10^7 at a radius of 10,000 km, to 1.8×10^7 at 20,000 km, and to 8.5×10^6 at 30,000 km, and it vanishes at the geostationary radius, where a corotating ring is in free fall as well, and beyond which it becomes a tension. The geostationary ring is thus the one place at which a ring that stays fixed above the ground bears no hoop load, and it is where the elevator's cable is balanced, so that the two structures meet naturally there. The conclusion for the ring-first architecture is that a rigid ring should be built in free fall or at geostationary radius, where the compressive requirement is small, and that a rigid ring that is fixed close to the ground is excluded by the same arithmetic that excludes the tower. It also follows that the ring is a hub and not a road: a ring in low orbit moves past the ground at nearly eight kilometers per second, and the connection to the ground must be made by the skyhooks and accelerators of the preceding chapters, or by cables that descend from a ring at geostationary altitude, which return the argument to the tensile material on which the elevator depends. The scale of a ring is also considerable: its circumference at 400 km altitude is 42,500 km, and even at a mass of one kilogram per meter it contains 4.3×10^4 tonnes, at one hundred kilograms per meter 4.3 million tonnes. A free-fall ring is, moreover, subject to Maxwell's instability, so that it must be actively stabilized, and the cost of that stabilization is a continuing obligation. The ring-first architecture is thus a coherent program of orbital construction, and it is placed in the second readiness class as such, while its connection to the ground by anything but skyhooks and accelerators remains in the third.

The assessment under the standard of Chapter 16 is that the elevator is in the third readiness class until a material exists, that the expandable linkage is in the first class for packaging and the second for orbital assembly of large structures, and that the combination of the two as a tower to space is excluded by the argument above. The book records this as an impossibility and not as a speculative possibility, in keeping with its commitment to state the failures of proposals as carefully as their promise. The hub structures that serve in the notes as landing places for the expandable towers, the intervalorial pediments of Chapter 20, can be treated separately as large offshore platforms, which is a feasible engineering category, and the question of what arrives on them is the question of the preceding chapters. What remains of the concept after the exclusion is a program of materials science, directed at the production

of high-strength fiber in macroscopic lengths, which is worth pursuing for its own sake, and of orbital construction, which is the subject of the following chapters.

Formalization

Two results are given, the first for tension structures and the second for compression columns.

Let a cable of density ρ have working stress σ at every cross-section, and let $A(r)$ be the cross-sectional area at geocentric radius $r \in [R, r_g]$, with $\Psi(r) = -GM/r - \frac{1}{2}\omega^2 r^2$ the effective potential of the corotating frame and $r_g = (GM/\omega^2)^{1/3}$ the geostationary radius.

Proposition 24.1. *Equilibrium of a cable element requires $\sigma A'(r) = \rho A(r) \Psi'(r)$, so $A(r) = A(R) \exp[(\rho/\sigma)(\Psi(r) - \Psi(R))]$. Since $\Psi'(r) = GM/r^2 - \omega^2 r > 0$ for $r < r_g$, the area increases monotonically to r_g , where $\Psi' = 0$. The taper ratio $A(r_g)/A(R) = \exp(\Delta\Psi \rho/\sigma)$ with $\Delta\Psi = \Psi(r_g) - \Psi(R)$. A climber of mass m moving at speed u at radius r requires mechanical power $\mu u \Psi'(r)$, and the energy to ascend from R to r_g is $m\Delta\Psi$.*

Proof. For the cable element between r and $r + dr$, the tension at the outer end exceeds that at the inner end by the net of the gravitational and centrifugal forces on the element, which is $-\rho A \Psi' dr$ directed outward-minus-inward, giving $d(\sigma A) = \rho A \Psi' dr$ with σ constant. Integrate. The climber's force against the effective gravity Ψ' times speed gives the power, and integrating over r gives the energy. $\square$

With $GM = 3.986004 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$, $\omega = 7.2921 \times 10^{-5} \text{ s}^{-1}$ and $R = 6378 \text{ km}$, one obtains $r_g = 42,164 \text{ km}$, $\Delta\Psi = 4.842 \times 10^7 \text{ J kg}^{-1}$ (13.45 kWh kg^{-1}), and the taper exponent $\Delta\Psi \rho/\sigma$ equal to 543, 19.4, 13.0, 1.26, and 0.63 for working-stress-to-density ratios of 0.089 (steel at 0.7 GPa), 2.5 (aramid), 3.7 (polybenzoxazole), 38.5 (50 GPa at 1300 kg m^{-3}), and 76.9 MJ kg^{-1} (100 GPa). The surface effective gravity is 9.78 ms^{-2} , giving climber power 0.54 MW per tonne at 55.6 ms^{-1} , and the ascent time at that speed is 7.5 days.

For the compression column, let a uniform thin-walled tube of radius r_0 and wall thickness $t \ll r_0$ stand with its base clamped and its top free. The weight per unit length is $q = 2\pi r_0 t \rho g$ and the second moment of area is $I = \pi r_0^3 t$.

Proposition 24.2. *The column buckles under its own weight at the critical height $H_{\text{cr}} = (\lambda_1 EI/q)^{1/3}$, with $\lambda_1 \approx 7.84$ (Greenhill). For the tube, $H_{\text{cr}} = (\lambda_1/2)^{1/3} (E/\rho g)^{1/3} r_0^{2/3} \approx 1.58 (E/\rho g)^{1/3} r_0^{2/3}$, independent of the wall thickness. To reach a height H requires $r_0 = [H/(1.58(E/\rho g)^{1/3})]^{3/2}$.*

The substitution is immediate: $EI/q = Er_0^2/(2\rho g)$. The value of λ_1 is the first root of the Bessel-function eigenvalue condition for the self-buckling of a clamped-free column and is quoted without derivation. Numerically, for steel ($E = 200 \text{ GPa}$, $\rho = 7850$) $H_{\text{cr}} = 633 \text{ m}$, 2.94 km, 13.6 km at $r_0 = 5, 50, 500 \text{ m}$, and $r_0 = 887 \text{ m}$ and 9.9 km for $H = 20 \text{ km}$ and 100 km;

for a 1 TPa, 1300 kg m^{-3} material $H_{\text{cr}} = 1.97, 9.2, 42.5 \text{ km}$ at the same radii, and $r_0 = 161 \text{ m}$ and 1.8 km for $H = 20$ and 100 km . The $2/3$ power of the dependence on radius means that increasing height tenfold requires increasing radius by a factor $10^{3/2} = 31.6$, which is the quantitative content of the exclusion of compression towers to the edge of space.

For the expandable lattice, a scissor cell whose bars of total length $2b$ cross at their midpoints at an angle θ to the horizontal has height $2b \sin \theta$, so a stack of n cells has height $H(\theta) = 2nb \sin \theta$ and expansion ratio $H(\pi/2)/H(\theta_s) = 1/\sin \theta_s$, equal to 19.1, 11.5, 5.8, 2.9 for $\theta_s = 3^\circ, 5^\circ, 10^\circ, 20^\circ$. The ratio depends on the stowed angle alone and is independent of the strength of the bars, which is the sense in which the expansion ratio and the load capacity are separate properties of the structure.

For the ring-first architecture, consider a thin ring of density ρ and radius r moving tangentially at speed $v = \Omega r$ about the Earth's center, in the plane of the equator.

Proposition 24.3. *The hoop stress in the ring, positive in tension, is $\sigma = \rho(v^2 - GM/r)$. It vanishes for $v = v_{\text{orb}} = \sqrt{GM/r}$ (free fall), is compressive for $v < v_{\text{orb}}$ with magnitude $\rho(v_{\text{orb}}^2 - v^2)$, and is tensile for $v > v_{\text{orb}}$. For a ring corotating with the Earth, $v = \omega r$, the stress is $\sigma = \rho(\omega^2 r^2 - GM/r)$, which is compressive for $r < r_g$, zero at $r = r_g = (GM/\omega^2)^{1/3}$, and tensile for $r > r_g$. A ring of compressive specific strength s_c can be corotating at radius r only if $s_c \geq GM/r - \omega^2 r^2$.*

Proof. A ring element of length $r d\varphi$ and cross-section A has mass $\rho A r d\varphi$. Its centripetal requirement is $\rho A r v^2 / r d\varphi$ directed inward, supplied by gravity $\rho A r d\varphi GM/r^2$ and by the net inward component of the hoop force $2N \sin(d\varphi/2) \approx N d\varphi$. Hence $N = \rho A(v^2 - GM/r)$, and $\sigma = N/A$. $\square$

Numerically, the required compressive specific strength $GM/r - \omega^2 r^2$ is 62.3 MJ kg^{-1} at $r = 6378 \text{ km}$, 58.6 at 6771 km , 39.3 at $10,000 \text{ km}$, 17.8 at $20,000 \text{ km}$, 8.5 at $30,000 \text{ km}$, and 0 at $42,164 \text{ km}$. For steel at $r = 6378 \text{ km}$ the stress is $7850 \times 6.23 \times 10^7 = 489 \text{ GPa}$. For comparison, the specific strengths of the strongest fibers in tension are of the order of 4 MJ kg^{-1} (polybenzoxazole) to a few tens of MJ kg^{-1} (hypothetical nanotube yarns), and compressive specific strengths are lower. The circumference of a ring at 6771 km radius is $42,543 \text{ km}$, so its mass at linear density λ is $4.25 \times 10^4 \lambda \text{ kg}$, which is 4.3×10^4 tonnes for $\lambda = 1 \text{ kg m}^{-1}$ and 4.3×10^6 tonnes for $\lambda = 100 \text{ kg m}^{-1}$. A planar circular scissor mechanism that deploys the ring from a stowed state has radial expansion ratio $1/\sin \theta_s$ for the same reason as the stack of the preceding paragraph, 11.5 at $\theta_s = 5^\circ$, and for this reason a ring that is to be deployed to a radius of thousands of kilometers must either be built in segments and joined or be a ring whose stowed form is itself a large structure; in neither case does the linkage carry the loads of the deployed ring.

Chapter 25

Dyson Rings and Solar Slingshots

The most extravagant of the concepts in the author's notes is the collection of stellar energy on a scale that dwarfs the whole of human use, and it is the concept with the longest history of misunderstanding, which begins with its name. Dyson's 1960 paper [4] proposed that an advanced civilization might be detected by the infrared radiation of structures that intercept the light of its star, and he proposed these structures as a swarm of independent bodies in orbit, a loose shell of many objects, and not a solid sphere. The solid sphere belongs to the literature of fiction. The notes use the terms ring and swarm, together with a gravitational slingshot around the Sun, and this chapter begins by distinguishing the architectures that the terms can denote, because the argument of the earlier chapters, that a proposal is incomplete unless its mechanism is stated, applies with special force to a concept of which a single name stands for physically very different objects. The closest experimentally grounded precedent for propulsion by light on a large sail is the analysis of laser-pushed lightsails [58], which concerns a different problem from the one considered here.

Consider first the rigid shell, a solid surface enclosing the star. The gravitational field inside a uniform spherical shell vanishes, a result of Newton, so that the shell exerts no force on the star and the star none on the shell, and the position of the shell relative to the star is in neutral equilibrium: any displacement leaves it displaced, and the slightest drift brings one side toward the star and the other away, with no restoring force. The compressive stresses in a shell of radius one astronomical unit under the star's gravity exceed the strength of any known material by many orders of magnitude, and the mass of the shell is enormous. Even a shell with an areal density of one kilogram per square meter, a film a millimeter thick of material of the density of water, has a mass of 2.8×10^{23} kg, which is eighty-five per cent of the mass of Mercury. The rigid shell is excluded for reasons of stability, of strength, and of mass.

The rigid ring is a smaller structure with a similar defect. A solid ring in orbit around a central mass is in unstable equilibrium, as Maxwell showed in 1859 [20] in his essay on the stability of the motion of Saturn's rings: a displacement of the ring toward one side increases the attraction on that side, and the displacement grows. Maxwell concluded that the rings

of Saturn must be composed of many small separate particles, and the conclusion has been confirmed by observation. A ring around a star can be held in place only by active control, with thrusters or by a continuing exchange of momentum, and a rigid ring of astronomical size is a structure that must be controlled at every moment, which compounds the difficulty. A third architecture is the statite shell, a surface of reflective sails held at rest by radiation pressure against gravity, and it has a precise criterion. A perfectly reflecting sail at any distance from the star experiences radiation force and gravitational force that both vary as the inverse square of distance, so the balance depends only on the areal density, and the critical value [21] is $\sigma_c = L/(2\pi cGM) = 1.53 \text{ g m}^{-2}$ for the Sun. A sail with a lower areal density is pushed outward, a sail with a higher one is pulled in, and a shell of such sails at one astronomical unit has a mass of $4.3 \times 10^{20} \text{ kg}$ or more, about 0.13 per cent of the mass of Mercury, if it is a complete shell at the critical density. The film must be thinner than a few micrometers of a material of ordinary density, and it must be reflective, thermally stable, and kept in position against the perturbations of the other bodies of the solar system.

The fourth architecture is the swarm, in which independent collectors in orbit around the star, each a sail, a mirror, or a platform with a radiator, are numerous enough to intercept a given fraction of the star's output without being rigidly connected. This is Dyson's own picture. It requires no structural strength beyond that of the individual collectors, since each is in free fall, and it requires control only for the avoidance of collision and the maintenance of orbit, tasks that scale with the number of collectors. A ring in the notes is best understood as a particular arrangement of the swarm, a band of collectors distributed around a circular orbit, whose width determines the fraction of the stellar output that it intercepts. For a band of width W at orbital radius r with its collectors oriented to face the star, the intercepted fraction is $W/(2r)$. A band of width one million kilometers at one astronomical unit intercepts 0.33 per cent of the output, $1.3 \times 10^{24} \text{ W}$, and has an area of $9.4 \times 10^{20} \text{ m}^2$ and, at a mass per unit area of one gram per square meter, a mass of $9.4 \times 10^{17} \text{ kg}$. The numbers show two things. The mass needed to intercept a given fraction is small, a small fraction of the mass of the asteroid belt, because collectors can be thin. And the power intercepted is vastly greater than anything that could be used: it exceeds present human use by nearly eleven orders of magnitude. The sizes at which the concept begins to be relevant to human purposes are accordingly many orders below the ring of a million kilometers. A swarm intercepting a petawatt, about fifty-five times present use, requires a collecting area of $2.4 \times 10^{12} \text{ m}^2$ at an efficiency of thirty per cent and, at one gram per square meter, 2.4×10^6 tonnes, a modest mass by the standards of terrestrial industry, but a mass to be placed in space, which is the whole difficulty.

The difficulty determines the sequence, and the quantity that determines it is the fraction of the energy that is useful in space. The energy collected cannot all be sent to the surface, because, as Chapter 22 showed, the surface of the Earth can shed only some tens of terawatts of additional heat without a measurable change of climate, so the energy would have to be used at or near the collectors for purposes such as manufacture, propulsion, and computation in space. This sets the purpose of the swarm: it is not an electric utility for the Earth, and a

proposal that presented it as such would founder on the heat budget. It is infrastructure for an economy in space, and its justification depends on the existence of that economy.

The material for the swarm would have to come from somewhere other than the Earth, and the notes propose the dismantling of Mercury, a planet of 3.3×10^{23} kg whose gravitational binding energy is 1.8×10^{30} J. This is the energy equivalent of 4,700 seconds of the Sun's output, and to disassemble the planet within a century at perfect efficiency would require 5.7×10^{20} W, which is 1.5×10^{-6} of the solar luminosity. The figure is of the order given in the literature on the subject. It shows that the process is limited by the engineering of mining, transport, and processing and not by the energy, and it shows also the scale of the intervention, the removal of a planet, about which Chapter 13 and Chapter 16 require that authority and consent be addressed before the engineering is.

The figure also invites a correction of scale. Whole-planet disassembly is the extreme case, and the uses that the book has reason to consider need a vanishing fraction of Mercury. A collecting swarm of one petawatt needs about 2.4×10^9 kg, which is 7×10^{-15} of the planet, and a heat shield or a fleet of battery factories needs less. The critics of the concept have been right about the planet and have not been addressing the uses. Knapp's objection of 2012 [15] is that the energy and effort of dismantling Mercury would repay themselves only on a timescale far beyond any benefit, and that the same resources would do more good elsewhere. The objection holds for a scheme that spends energy at the outset and collects the return later, since a static ledger of this kind is dominated by the binding energy. It does not hold for a scheme that bootstraps, in which the collectors that the first mining produces power the next round of mining. A film of aluminium of areal density one gram per square meter has an embodied energy of about 0.2 MJ m^{-2} at 200 MJ kg^{-1} . At the Earth's distance a collector at thirty per cent efficiency returns 408 W m^{-2} and repays the energy of its own film in about 490 s, and at Mercury's orbit, where the solar flux is $9,083 \text{ W m}^{-2}$, 6.7 times the flux at the Earth, it returns 2.7 kW m^{-2} and repays it in about 73 s. The film is not the whole cost, since each square meter of collector requires machinery to mine, refine, and deploy it. If the machinery is a hundred kilograms per square meter of collector at the same specific embodied energy, the payback is 1.55 years at the Earth's distance and 0.23 years at Mercury's, and the doubling times of an installation that reinvests all its output are 1.08 and 0.16 years respectively. Growth from one square kilometer to the $2.4 \times 10^{12} \text{ m}^2$ of a petawatt swarm requires about twenty-one doublings, which at Mercury is about three and a half years under these stipulated inputs. The bootstrapping answer is therefore valid, and it is a conditional one: the doubling time is set by the machinery's embodied energy and not by the Sun, the machinery must be able to reproduce itself from regolith with a closure that is not presently demonstrated, and a growth law of this kind is exactly the kind of process for which Chapter 16 requires a stop rule, because a process that doubles in a year is also a process that cannot be recalled after twenty doublings.

The gravitational slingshot around the Sun is a different kind of proposal: a means of sending a payload out of the solar system at a high speed. The principle is the Oberth effect [35]: a rocket burn made at high speed deep in a gravitational well changes the payload's kinetic

energy by an amount proportional to its speed at the time of the burn, so that an increment of velocity delivered at perihelion is worth more than the same increment delivered far away. If the payload passes the Sun at a perihelion speed v_p and burns to add Δv , the energy gained per unit mass is $v_p \Delta v + \frac{1}{2} \Delta v^2$. The cost of the strategy is the dive: to bring a payload from the Earth's orbit to a perihelion of ten solar radii requires cancelling most of the Earth's orbital velocity, a velocity change of 20.9 km/s, and the payload must survive the heat. At ten solar radii the solar flux is $6.3 \times 10^5 \text{ W m}^{-2}$, which is 462 times that at the Earth, and a blackbody surface in equilibrium with this flux, radiating from one side, would reach 1,825 K. The Parker Solar Probe has come to within about ten solar radii, at a speed of about 190 km/s, protected by a carbon-composite shield, and this is the heat load that a larger payload would face. The comparison with a direct departure is informative. A payload that is to leave with a hyperbolic excess speed of 47 km/s needs 33.4 km/s of velocity increment from Earth orbit if it goes directly, and 30.9 km/s, composed of 20.9 km/s for the dive and 10 km/s at perihelion, if it uses the slingshot. For a smaller excess speed the direct departure is cheaper; at 15 km/s the dive costs 25.9 km/s and the direct departure 15.1 km/s. The slingshot thus breaks even near an excess speed of forty kilometers per second and is advantageous only above it. It is a technique for very high speeds, and in the context of the book it would matter for missions that no program of present obligation requires. It is classified accordingly in the third readiness class, and its relation to the other orbital programs is that of a possible later extension.

Two uses of Mercury material bear on the slingshot and on the economy of the swarm, and both are proposals of the author's notes. The first is the heat shield. A payload that dives to ten solar radii must carry a shield, and the shield is a considerable fraction of its mass, as in the Parker Solar Probe, where the shield is of the order of a tenth of the spacecraft. If the shield is made from material mined and fabricated at Mercury, its mass need not be lifted from the Earth, and the dive can start from Mercury's orbit, where the orbital speed is 47.9 km/s and the dive to ten solar radii costs 25.7 km/s, against 20.9 km/s from the Earth's orbit. The perihelion speed on the dive ellipse is 184.6 km/s, and a burn of 15, 20, or 25 km/s there gives hyperbolic excess speeds of 40.9, 60.8, and 75.9 km/s at total increments of 40.7, 45.7, and 50.7 km/s. A direct departure from Mercury's orbit to the same speeds costs 31.2, 43.1, and 53.8 km/s, so that from Mercury the slingshot breaks even near 60 km/s, which is a higher speed than the forty kilometers per second from the Earth. The result is not what the proposal expects. Mercury staging helps by supplying shield material and solar power on site, and the dive is no cheaper there, because the orbital speed to be cancelled is larger. The shield is thus an argument for building at Mercury and not an argument that the slingshot has become attractive, and the classification of the slingshot in the third readiness class is unchanged.

The second use is the remote production and charging of batteries in orbit about Mercury. The solar flux there is 6.7 times that at the Earth, and the planet supplies the metals, silicon, and carbon of cells, so a battery factory powered by collectors in Mercury orbit is a plausible first customer of the swarm. The destination of the batteries sets the limit on the proposal. To move a battery from Mercury to the Earth by a Hohmann transfer costs 9.61 km/s at departure

and 7.53 km/s at arrival, 17.1 km/s in all, in a transit of 105 days, and the kinetic energy that this increment represents, $\frac{1}{2}\Delta v^2$, is 147 MJ kg^{-1} of payload. A battery of 250 Wh kg^{-1} holds 0.9 MJ kg^{-1} , so that moving the battery by rocket costs about a hundred and sixty times the energy it stores. Even escape from Mercury alone, 4.25 km/s, costs 9.0 MJ kg^{-1} for a mass driver, ten times the stored energy. The stored energy of a battery cannot, therefore, be exported by propulsion in any economic sense. The battery factories at Mercury have a use only if the energy is consumed where the batteries are: in tugs and miners that work in Mercury orbit, in the charging of vehicles that return only a short distance, or in swap stations in which a battery is exchanged between vehicles that already travel the route. Export of the energy itself as a beam is a different proposal with its own losses and hazards, and it faces the heat budget of Chapter 22. The two uses together give the result that the Mercury program is justified, if at all, as an industrial base for the orbital economy, which is the position the chapter has reached by the argument from scale.

Formalization

Four results are stated: the statite criterion, the swarm geometry, the Oberth gain, and the cost of the dive.

For a flat perfectly reflecting sail of areal density σ at distance r from a star of luminosity L and mass M , facing the star, the radiation force per unit area is $2L/(4\pi r^2 c)$ and the gravitational force per unit area is $\sigma GM/r^2$.

Proposition 25.1. *The sail is in force balance at every distance if and only if $\sigma = \sigma_c = L/(2\pi c GM)$. For $\sigma > \sigma_c$ the net force is toward the star and for $\sigma < \sigma_c$ away from it, at every r . For the Sun, $\sigma_c = 1.53 \text{ g m}^{-2}$ (or 0.77 g m^{-2} for a perfect absorber). A complete shell at radius r at the critical density has mass $4\pi r^2 \sigma_c$, which for $r = 1 \text{ AU}$ is $4.3 \times 10^{20} \text{ kg}$.*

Proof. Equating $L/(2\pi r^2 c)$ to $\sigma GM/r^2$ gives σ_c , independent of r . The sign of the net force follows from the sign of $\sigma_c - \sigma$. The shell mass is area times density, with the area $4\pi(1.496 \times 10^{11})^2 = 2.81 \times 10^{23} \text{ m}^2$. $\square$

For a rigid shell of areal density 1 kg m^{-2} the mass is $2.81 \times 10^{23} \text{ kg}$, which is 0.85 of the mass of Mercury and exceeds the critical value for a statite by a factor of 650, so such a shell cannot be supported by radiation.

For the swarm, let a band of collectors at orbital radius r and width W face the star, each with an effective collecting area per unit of band area α , and let the optical-to-useful conversion efficiency be η .

Proposition 25.2. *For $W \ll r$ the fraction of stellar output intercepted by the band is $f = W/(2r)$, the useful power is $\eta f L$, and the band area is $2\pi r W$, whose mass at areal density σ_a is $2\pi r W \sigma_a = 4\pi r^2 f \sigma_a$. Hence the mass per unit intercepted power is $4\pi r^2 \sigma_a / L$ per unit*

of fraction, independent of the geometry of the band, and equal to $\sigma_a/(\Phi\eta)$ per unit of useful power, where $\Phi = L/(4\pi r^2)$ is the flux at the band.

Proof. A band of width W at radius r presents a projected area $2\pi rW$ to the star, and the star's output passing through the sphere of radius r is distributed over area $4\pi r^2$, giving the fraction $2\pi rW/(4\pi r^2) = W/(2r)$. The remaining statements follow from the definitions. $\square$

With $L = 3.828 \times 10^{26}$ W and $r = 1.496 \times 10^{11}$ m, a band of $W = 10^9$ m intercepts $f = 3.34 \times 10^{-3}$, i.e. 1.28×10^{24} W, has area 9.4×10^{20} m², and at $\sigma_a = 1$ g m⁻² a mass of 9.4×10^{17} kg. To collect 10^{15} W at $\eta = 0.3$ and flux 1361 W m⁻² needs 2.4×10^{12} m² and 2.4×10^9 kg. The binding energy of a uniform sphere is $\frac{3}{5}GM^2/R$; for Mercury ($M = 3.301 \times 10^{23}$ kg, $R = 2.440 \times 10^6$ m) it is 1.79×10^{30} J, equal to 4.67×10^3 s of solar luminosity; a disassembly over 10^2 years at 100 per cent efficiency requires 5.7×10^{20} W, a fraction 1.5×10^{-6} of the output.

For the slingshot, let a payload pass the Sun at perihelion radius r_p with speed v_p in the heliocentric frame and make an impulsive burn Δv in the direction of motion.

Proposition 25.3. *The hyperbolic excess speed after the burn satisfies $v_\infty^2 = (v_p + \Delta v)^2 - v_{\text{esc}}^2(r_p) = 2v_p\Delta v + \Delta v^2 - (v_{\text{esc}}^2 - v_p^2)$, where $v_{\text{esc}}^2 = 2GM/r_p$. The specific energy gained is $v_p\Delta v + \frac{1}{2}\Delta v^2$ and is proportional to v_p , so for fixed Δv the gain is greater at small r_p . The velocity increment to reach a perihelion r_p from a circular orbit of radius r_1 is $\Delta v_{\text{dive}} = \sqrt{GM/r_1} - \sqrt{GM(2/r_1 - 1/a)}$ with $a = (r_p + r_1)/2$.*

Proof. Energy conservation gives $\frac{1}{2}v_\infty^2 = \frac{1}{2}(v_p + \Delta v)^2 - GM/r_p$. The dive formula applies the vis-viva equation at the aphelion of the transfer ellipse. $\square$

With $GM_\odot = 1.327 \times 10^{20}$ m³ s⁻², $r_p = 10R_\odot = 6.957 \times 10^9$ m, $r_1 = 1$ AU: $v_{\text{esc}}(r_p) = 195.3$ km/s, perihelion speed on the dive ellipse $v_p = 190.9$ km/s, $\Delta v_{\text{dive}} = 20.9$ km/s. A perihelion burn of 5, 10, 15, 20 km/s gives $v_\infty = 15.5, 47.2, 65.3, 79.6$ km/s at a total velocity increment of 25.9, 30.9, 35.9, 40.9 km/s. A direct departure from Earth orbit to the same v_∞ requires heliocentric speed $\sqrt{v_\infty^2 + 2GM/r_1}$ at 1 AU and thus increments of 15.1, 33.4, 47.9, 60.3 km/s; the ratio of direct to slingshot cost is 0.58, 1.08, 1.33, 1.47, with break-even near $v_\infty = 40$ km/s. The thermal load at $r_p = 10R_\odot$ is $S(1 \text{ AU})(r_1/r_p)^2 = 6.29 \times 10^5$ W m⁻² and the blackbody equilibrium temperature of a surface radiating from one side is $(S/\sigma_{SB})^{1/4} = 1825$ K.

For the Mercury uses, let $r_m = 5.79 \times 10^{10}$ m be Mercury's orbital radius and $v_m = \sqrt{GM_\odot/r_m} = 47.87$ km/s its circular speed.

Proposition 25.4. *Let E_a be the embodied energy per unit area of a collector including its machinery, $\eta\Phi$ the useful power per unit area at flux Φ , and $t_p = E_a/(\eta\Phi)$ the energy payback time. If the installed area is reinvested in its own replication, it grows as $A(t) = A_0 e^{t/t_p}$, with doubling time $t_d = t_p \ln 2$; the number of doublings to reach A_f is $\log_2(A_f/A_0)$. A transport of*

stored energy e_b per unit mass over a velocity increment Δv is energetically self-supporting by propulsion only if $e_b > \frac{1}{2}\Delta v^2$.

Proof. With reinvestment of the whole output, $dA/dt = A\eta\Phi/E_a = A/t_p$, which gives the exponential and, setting $A = 2A_0$, the doubling time $t_p \ln 2$. The condition on transport compares the stored energy with the minimum kinetic energy of the increment. $\square$

Numerically, with $\eta = 0.3$, the useful power is 0.408 kW m^{-2} at 1 AU and 2.72 kW m^{-2} at Mercury. A film of 1 g m^{-2} at 200 MJ kg^{-1} has $E_a = 0.2 \text{ MJ m}^{-2}$ and $t_p = 490 \text{ s}$ at 1 AU and 73 s at Mercury. With 100 kg m^{-2} of machinery at the same specific embodied energy, $E_a = 2.0 \times 10^{10} \text{ J m}^{-2}$, $t_p = 1.55 \text{ yr}$ and 0.23 yr , and $t_d = 1.08 \text{ yr}$ and 0.16 yr , respectively; and $\log_2(2.4 \times 10^{12}/10^6) = 21.2$ doublings. The time to a petawatt swarm is dominated by the machinery term, and $t_d \geq t_p \ln 2$ bounds any scheme whose output is the sole source of its own growth. The embodied-energy figures are stipulated inputs of the book and not measurements.

For the dive from Mercury's orbit to $r_p = 10R_\odot$, $\Delta v_{\text{dive}} = 25.70 \text{ km/s}$ and $v_p = 184.6 \text{ km/s}$ by the formula of the preceding proposition with $r_1 = r_m$; for $\Delta v = 15, 20, 25 \text{ km/s}$ the excess speed is $40.9, 60.8, 75.9 \text{ km/s}$ at totals $40.7, 45.7, 50.7 \text{ km/s}$, and a direct departure from r_m to the same v_∞ requires $\sqrt{v_\infty^2 + 2v_m^2} - v_m$, namely $31.2, 43.1, 53.8 \text{ km/s}$, so the slingshot breaks even near $v_\infty = 60 \text{ km/s}$. The Hohmann transfer from Mercury's orbit to the Earth's has $\Delta v = 9.61 + 7.53 = 17.14 \text{ km/s}$ in 105 days, with $\frac{1}{2}\Delta v^2 = 147 \text{ MJ kg}^{-1}$ against $e_b = 0.9 \text{ MJ kg}^{-1}$ for a cell of 250 Wh kg^{-1} , a ratio of 163; the escape energy $\frac{1}{2}v_{\text{esc}}^2 = 9.0 \text{ MJ kg}^{-1}$ for $v_{\text{esc}} = 4.25 \text{ km/s}$. The fraction of Mercury used by a one-petawatt swarm is $2.4 \times 10^9 / 3.30 \times 10^{23} = 7 \times 10^{-15}$.

Chapter 26

Sequencing Space Programs

The orbital programs of the preceding chapters are not independent, and the order in which they are undertaken matters more than the decision to undertake any of them. The accelerator of Chapter 21 supplies the first part of the velocity that the skyhook of Chapter 23 completes. The skyhook delivers material to orbits from which a fabrication industry could build the structures of Chapter 24 and the collectors of Chapter 25. The collectors supply the energy for the fabrication that builds them, and the whole depends on a supply of material that cannot come from the Earth in the quantities required. This circularity is the central fact of the sequencing problem, and it is the reason that the answer cannot be a calendar. The sequence is determined by dependencies, by the cost and information that each stage yields, and by the institutions that must be in place before each stage can be undertaken legitimately. This chapter sets out the principles by which the sequence is to be chosen, and it closes the fifth part by connecting the orbital programs back to the standard of Chapter 16 and the constraint of present obligations.

The first principle is that present obligations are not postponed for the sake of orbital programs. A reader who has followed the earlier chapters will see why the principle needs to be stated. The scale of the orbital programs invites a form of reasoning in which the magnitude of the eventual benefit, energy and material in quantities that dwarf the economy of the Earth, is invoked to justify the diversion of resources from nearer needs. The reasoning is that of the longtermist argument which Chapter 31 examines in detail, and the principle that blocks it is the readiness rule of Chapter 16: no measure that satisfies a present obligation may depend on a program of the second or third readiness class. The orbital programs may therefore be funded, but from resources that do not compete with the floor of present obligations, and the plan must show that this is so. A corollary is that the programs earn continued support only by passing gates, and not by the size of their promised payoff.

The second principle is that the sequence is determined by reversibility and information. A stage that is cheap, reversible, and informative should precede a stage that is costly, irreversible, and no more informative. A short tether in low orbit that validates the dynamics of

rotation and capture costs little relative to the programs it informs and can be abandoned. A pilot accelerator of modest speed, used to test the technology of the track and the handling of payloads, similarly costs a small fraction of the full installation, and its failure would teach what the design lacks. By contrast the dismantling of a planet is irreversible, and its justification cannot be settled by anything less than a long record of smaller operations in the same technologies. The structure that results is the stage-gate process familiar from the development of drugs and aircraft: stages are ordered so that the cheapest uncertainty-resolving steps come first, each stage ends in a gate at which an independent body decides whether to proceed, and each gate is defined by criteria stated in advance.

The calculation that underlies the stage-gate is backward induction, and its result is easily stated. At the final stage the decision is whether to spend the cost of the stage in return for the probability of success times the value of completion. At the preceding stage the value of completion is the value of the continuation, which is the maximum of zero and the value just computed, and so on back to the beginning. The value of a program with four stages with costs of one, three, ten, and thirty billion dollars, success probabilities of 0.5, 0.4, 0.5, and 0.6, and a final value of five hundred billion is 22.5 billion when each stage is undertaken only if the preceding one has succeeded and the continuation value justifies the cost. The same program pre-funded, with the whole forty-four billion committed at the outset, has an expected value of -14 billion, because the later and larger costs are incurred even in the many cases in which an early stage fails. The difference of 36.5 billion is the value of the option to stop, and it is the quantitative reason for gates. The calculation is also a test of honesty in proposals: a proposal that cannot specify the stages, the costs, and the probabilities on which it relies has not stated what it claims, and a proposal whose value is positive only if the probabilities are far higher than the evidence supports should be treated accordingly. The probabilities here are illustrative, and a real program would estimate them by reference to the history of comparable development efforts, adjusted by an independent review for the optimism that proposers display.

The third principle concerns the critical path. The dependencies among tasks form a directed graph, and the earliest completion of the program is the length of the longest path through it, since tasks off that path can be delayed up to their slack without delaying the whole. The implication is that effort should be concentrated on the critical path, and that a reduction in the duration of a task off the path does not shorten the program. A less obvious implication is that the critical path can change: when a technology advances, a task that had been fast becomes the bottleneck, and the plan must be revised. For the orbital programs the critical path is probably the production of material in space, since the other elements, from the accelerator to the collector, can be developed in parallel but all depend on it for their scale. The longest path is the bottleneck, and the most valuable early investment is the one that shortens it.

The fourth principle concerns authorization, and it is here that the argument of the book is most severely tested. The space beyond the Earth is governed by treaties of the 1960s and 1970s that were not drafted with these programs in view. The Outer Space Treaty of 1967 [36]

prohibits the national appropriation of celestial bodies, requires states to bear international responsibility for the activities of their nationals, and prohibits the placing of weapons of mass destruction in orbit. The Liability Convention of 1972 makes states liable for damage caused by their space objects. The Moon Agreement of 1979, which would have declared the resources of the Moon the common heritage of mankind, has been joined by few states and by none of the principal space powers. The result is a regime that says clearly that no state may claim a planet and says much less clearly who may mine one, who is entitled to the energy of the Sun, and who may refuse. A program that proposed to disassemble Mercury, or to intercept a significant fraction of the Sun's output, would be an intervention in the common circumstances of all persons, and no state or group of states can authorize it by itself. The book's position, derived from the architecture of authority of Chapter 13, is that such a program requires an authorization that is multilateral, that provides for the participation of those not parties to it, that allows a party to refuse participation without loss of its existing rights, and that is revocable. There are precedents in the treaties governing Antarctica [27] and the seabed, which have reserved areas of common interest to the benefit of all and subjected them to collective decision, though the compromises were negotiated when the resources in question were out of reach, and they would be tested severely if reach were extended.

The fifth principle concerns dual use, which was raised for the accelerator in Chapter 21 and applies to the others. A skyhook is a means of changing the orbits of objects at will; a swarm of collectors is a means of concentrating power; the capacity to move large masses in space is the capacity to direct them. The orbital programs create capabilities that would be of the highest strategic importance, and the possession of them by a single party would be a standing threat to others. The sequencing plan must accordingly include, at every gate, an assessment of the new capability's potential for misuse and a statement of the transparency and the verification regime proposed to address it, and the plan should be refused at any gate at which the regime has not been agreed.

The last principle is that the plan must be revisable. A sequence of decades that cannot be changed on the basis of what is learned is a prediction and not a plan. The program therefore requires, at each gate, a review that has the power to change the sequence, to defer later stages, and to reallocate resources, together with a standing process for recording the failures of the program, as Chapter 10 requires of any verification process. The documentation of a stage that failed is an asset of the program, and the institution that suppresses it harms the program more than the failure did.

Formalization

Let a program have stages $i = 1, \dots, n$, with cost $c_i > 0$ incurred at the start of stage i , success probability $p_i \in (0, 1]$ of stage i given that it is undertaken, and terminal value $V > 0$ on

success of all stages. Define the continuation values $W_{n+1} = V$ and

$$W_i = \max\{0, -c_i + p_i W_{i+1}\}, \quad i = n, n-1, \dots, 1,$$

interpreted as the value at the beginning of stage i , given that stages $1, \dots, i-1$ have succeeded and that the decision to undertake stage i is still open.

Proposition 26.1. *The policy "undertake stage i if and only if $p_i W_{i+1} \geq c_i$ " maximizes expected value among all policies that may stop at any gate. The value of the gated program is W_1 . The value of the pre-funded program, in which all stages are paid for regardless of outcomes, is $\Pi = -\sum_i c_i + V \prod_i p_i$. Then $W_1 \geq \max\{0, \Pi\}$, and $W_1 - \Pi \geq 0$ with equality only if every continuation decision is positive at every gate and the pre-funded program incurs no cost after a failure.*

Proof. The recursion is Bellman's principle: the value of optimal play from stage i is the better of stopping (value 0) and undertaking the stage, which costs c_i and with probability p_i yields the optimal value from stage $i+1$. Induction from n to 1 establishes optimality. For the comparison, any pre-funded schedule is a policy that does not stop, and the gated policy is optimal among stopping policies, so $W_1 \geq$ its value; and stopping at the start gives 0. For the equality case, the gated policy that never stops after a failure incurs expected cost $\sum_i c_i \prod_{j < i} p_j$, which is less than $\sum_i c_i$ whenever some $p_j < 1$, $j < n$, and $n \geq 2$. $\square$

For $c = (1, 3, 10, 30)$, $p = (0.5, 0.4, 0.5, 0.6)$, and $V = 500$ (in billions), $W_5 = 500$, $W_4 = \max\{0, -30 + 0.6 \cdot 500\} = 270$, $W_3 = -10 + 0.5 \cdot 270 = 125$, $W_2 = -3 + 0.4 \cdot 125 = 47$, and $W_1 = -1 + 0.5 \cdot 47 = 22.5$. The pre-funded value is $\Pi = -44 + 500 \cdot 0.06 = -14$. The expected expenditure of the gated program is $1 + 0.5 \cdot 3 + 0.2 \cdot 10 + 0.1 \cdot 30 = 7.5$ billion, against 44 billion for the pre-funded one, and its expected value is $0.06 \cdot 500 - 7.5 = 22.5$ billion, in agreement. The gated program is positive at every gate here; if the estimate of p_3 were revised at the third gate from 0.5 to 0.03 the continuation value would fall to $-10 + 0.03 \cdot 270 = -1.9$ and the policy would stop, avoiding the 40 billion of the remaining outlay in an event the pre-funded program cannot respond to.

For the critical path, let the tasks form a directed acyclic graph $G = (T, E)$ with durations $d_t \geq 0$. The earliest finish time F_t of task t satisfies $F_t = d_t + \max_{s \rightarrow t} F_s$, with the maximum over the empty set equal to 0.

Proposition 26.2. *The earliest completion time of the program is $\max_t F_t$, which equals the maximum over all paths of the sum of durations along the path. A task has slack $\ell_t = \text{LF}_t - F_t \geq 0$ (latest finish minus earliest finish), and increasing d_t by $\delta \leq \ell_t$ leaves the completion time unchanged, while reducing d_t for a task of positive slack does not reduce it. A reduction of δ of a task on every longest path reduces the completion time by at most δ , and by exactly δ if the next-longest path is at least δ shorter.*

The proof is the standard one for the critical path method and is omitted. An illustration: let the accelerator pilot (8 years) and the tether demonstration (6 years) both precede in-space fabrication (10 years), which precedes the collector pilot (12 years). The longest path is accelerator pilot, fabrication, pilot, of length 30 years, and the tether demonstration has slack 2 years, so a delay of up to two years in the tether program does not delay the whole, whereas a one-year gain in the accelerator pilot shortens the program by one year, up to the point where the tether path (28 years) becomes critical.

Part VI

Deep Time

Chapter 27

Biological Aging and Repair

The sixth part of the book takes up the proposal that most sharply distinguishes the program from the schemes of the preceding parts, which is that human lives might be extended far beyond their present span, in the most extreme version to astronomical durations. The proposal has two halves, a biological one and a reliability one. The biological half asks whether the processes that accumulate damage in living tissue can be slowed, arrested, or reversed, and by how much. The reliability half asks what would follow if they could, since a person who does not age is nonetheless exposed to accident, violence, infection, and catastrophe. This chapter treats the first, the next the question of what it would mean for a person to continue, and Chapter 29 the second. The order matters because the biological half is the only one on which there is an experimental literature, and a book that has been careful to grade the readiness of its engineering programs should apply the same discipline here. The state of the evidence is that the biology of aging is a serious and growing science, that interventions exist which extend life in laboratory animals, that none has been shown to extend human lifespan, and that the leap from modest extensions to the indefinite lives of the more ambitious proposals is a leap of extrapolation and not of observation. The evolutionary account of why organisms age, in which maintenance is traded against reproduction, is the disposable soma theory [59], and it frames the question of whether repair is in principle available.

Aging, in the biological sense, is the progressive loss of function and the rising probability of death that accompanies the accumulation of damage and the failure of maintenance. The review of López-Otín and colleagues [19, 32], published in 2013 and revised in 2023, organized the field around a list of hallmarks: genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, and altered intercellular communication, to which later versions added disabled macroautophagy, chronic inflammation, and dysbiosis. The list is a map and not a theory. The hallmarks interact, their relative importance is disputed, and for most it remains unclear whether the hallmark is a cause of aging, a consequence, or a correlate. What the formulation has done is to turn a vague phenomenon into a set of specific processes, each

of which can be studied and in principle modified, and the research on each is a research on a well-posed problem in the sense of Chapter 4: a target, a measurable acceptance condition, and a cost of evaluation, which in this case is very high because the relevant outcome is lifespan or healthspan, measured over years or decades.

The interventions with the strongest laboratory evidence fall into a few classes. Restriction of caloric intake, without malnutrition, extends life in yeast, worms, flies, and rodents and appears to improve some markers of health in primates, though the effect on primate lifespan has been disputed between the two long studies conducted. The drug rapamycin, which inhibits a nutrient-sensing pathway, extends lifespan in mice in several independent laboratories, and the finding has been replicated at multiple sites in a protocol designed for the purpose. The removal of senescent cells, which accumulate with age and secrete inflammatory factors, extends healthy lifespan in mice by genetic and by pharmacological means, and drugs of this class, called senolytics, have entered early clinical testing. Reprogramming of cells toward a more youthful epigenetic state by transient expression of certain factors has reversed some markers of age in cells and in animals, with the risk, evident in the same experiments, of loss of cellular identity and of cancer. And the study of species with unusual longevity or with negligible measured senescence, such as some rockfish, the naked mole rat, and certain bats, has identified candidate mechanisms of resistance to cancer and of tissue maintenance. The effects in the best-studied model, the mouse, are of the order of a tenth to a third in extension of lifespan, with effects often larger in worms and flies, smaller or absent in primates, and unproven in humans.

It is important to be clear about what such results do and do not support. They support the proposition that the rate of aging is not fixed by physical law and that it can be modified, within a species, by genetic and pharmacological means. They do not support the proposition that aging can be eliminated, nor that the modest effects seen in short-lived animals will scale to a species whose natural lifespan is eight or nine decades and for which the interaction among causes of death is more complex. The longest documented human lifespan, that of Jeanne Calment, who died in 1997 at the age of 122, has not been exceeded in the intervening decades, and demographic analyses suggest a plateau in the maximum age at death in spite of large reductions in mortality at younger ages. There is a debate about whether the plateau reflects a biological limit or the paucity of data from the very old, and the program takes no position in it, observing only that a claim of the possibility of lives of centuries or more is a claim of the prospective success of research not yet done.

A feature of the research enterprise bears directly on the principle of the book. The evaluation of an intervention against aging is hampered by the lack of an accepted surrogate endpoint. In the regulation of medicines, a drug is approved for a disease, with an endpoint that can be measured in a trial of reasonable length, and aging is not classified as a disease. The result is that trials in humans must use endpoints of the diseases of aging, or biomarkers whose relationship to mortality is imperfectly established, among them epigenetic clocks, which estimate biological age from patterns of chemical marks on the genome. An epigenetic clock that is shifted by an intervention provides evidence that something changed, but not that the

change corresponds to a longer or healthier life, and the risk of treating such a surrogate as the goal is the standard one of the substitution of a measure for the thing measured. The role proposed for capable systems in this domain is the one described in the second part. They may help to search the large space of candidate interventions, to model the networks of interacting pathways, and to design trials. They cannot shorten the interval over which lifespan is observed, and the verification of a claim of extension requires the participation of human subjects over long periods, under consent and oversight, in studies whose designers have no financial interest in the outcome.

The ethical and institutional questions are serious and are taken up in Chapter 30. Here it suffices to note two that bear on the biology itself. The first is that research on aging is likely to be pursued by private interests with an incentive to claim benefits before they are demonstrated, in a market for products whose claims are weakly regulated, and the history of anti-aging products is a history of exaggeration and fraud. The standard of the book, independent verification of claims before reliance on them, applies here with full force. The second is the question of distribution. A therapy that extends life by years would, if it were costly, enlarge the difference in life expectancy that already separates rich from poor in most countries, and the principle that the admissible range should be assessed by the persons affected requires that distribution be considered at the outset and not after the technology exists.

Formalization

The standard quantitative framework is the Gompertz–Makeham law of mortality [29, 33], in which the hazard rate at age x is the sum of an age-independent term and an exponentially increasing term, $\mu(x) = A + Be^{\gamma x}$, with $A, B, \gamma > 0$. The first term represents extrinsic causes of death (accidents, violence, some infections); the second represents senescence, with a mortality-rate doubling time of $\ln 2/\gamma$.

Proposition 27.1. *The survival function is $S(x) = \exp[-Ax - (B/\gamma)(e^{\gamma x} - 1)]$. (i) If senescence is removed ($B = 0$) the survival function is $S(x) = e^{-Ax}$, with median $\ln 2/A$ and mean $1/A$, so that lifespan is bounded in expectation by the extrinsic hazard. (ii) If aging is slowed by a factor $k > 1$ in the sense $\gamma \mapsto \gamma/k$ with the hazard at a reference age x_0 held fixed, the senescent term at age x becomes $B_k e^{\gamma x/k}$ with $B_k = Be^{\gamma x_0(1-1/k)}$, and the two senescent terms coincide at x_0 , and for $x > x_0$ the modified term is strictly smaller than $Be^{\gamma x}$. (iii) The median age at death m solves $Am + (B/\gamma)(e^{\gamma m} - 1) = \ln 2$, and for $B\gamma^{-1}e^{\gamma m} \gg Am$ it satisfies $m \approx \gamma^{-1} \ln[\gamma \ln 2/B + 1]$.*

Proof. Integrate the hazard: $S(x) = \exp(-\int_0^x \mu)$, which gives the formula; the case $B = 0$ is immediate. For (ii), the new term equals the old at x_0 by construction, and $B_k e^{\gamma x/k} = Be^{\gamma x_0} e^{(\gamma/k)(x-x_0)} < Be^{\gamma x_0} e^{\gamma(x-x_0)} = Be^{\gamma x}$ for $x > x_0$. For (iii), set $S(m) = \frac{1}{2}$ and neglect the term Am . $\square$

A numerical illustration with parameters chosen to reproduce the mortality of a high-income population: a mortality-rate doubling time of eight years ($\gamma = 0.0866 \text{ yr}^{-1}$), an extrinsic hazard $A = 5 \times 10^{-4} \text{ yr}^{-1}$, and a total hazard of 10^{-3} at age 30, whence $B = 3.72 \times 10^{-5}$. The survival to ages 50, 70, 80, 90, 100, 110 is 0.944, 0.803, 0.620, 0.336, 0.079, 0.003, the median age at death 84.5 years and the mean 81.1 years. Slowing aging by a factor $k = 1.5$ (doubling time 12 years) yields a median of 104.6 years and a mean of 99.7; slowing by $k = 2$ (doubling time 16 years) yields a median of 122.5 years and a mean of 116.4. Removing senescence entirely leaves a median of $\ln 2/A = 1386$ years and a mean of 2000 years: the extrinsic hazard sets the ceiling. To obtain a median of a million years would require $A \leq 6.9 \times 10^{-7} \text{ yr}^{-1}$, a figure that is the subject of Chapter 29. The illustration shows both what the biology could achieve if its most optimistic prospects were realized, which is a large change in the span of life but within the range of hundreds of years, and that the larger numbers can be reached only if the extrinsic hazard is reduced by orders of magnitude, which is a problem of environment, institutions, and engineering, and not of biology.

Chapter 28

Continuity of the Person

Suppose that the medical program of the preceding chapter, or the reliability program of the next, were to succeed to an extent that now seems fanciful, and that a human being could continue for ten thousand years or a million. A question that has been put off would then become practical: in what sense would the person at the end be the person at the beginning? The question has a long history in philosophy, where it is called the problem of personal identity, and it is not an idle one for a program that proposes to preserve persons. A program to preserve something must say what the something is. If the answer is the body, then the program is a program of biological repair. If the answer is the pattern of memory, character, and intention, then it may be a program that could in principle be carried out by other means, such as the replacement of tissue by engineered substitutes, or the transfer of the pattern to another substrate, and it raises the possibilities of copying, branching, and loss that the biological program does not. The answer given in this chapter is not a metaphysical one. It identifies the features of continuity that the institutions of the program rely upon, shows how those features behave over very long intervals, and states the demands that follow for the arrangements of responsibility and consent.

The classical positions are two. On the view associated with Locke, a person persists as long as there is continuity of consciousness, which later writers have generalized to psychological continuity: the connections of memory, intention, belief, desire, and character that link a person at one time with the person at another. On the competing view the person is an organism, and persists as long as the organism does, regardless of what happens to the psychology. The controversy is old and unresolved, and the cases that pull the two views apart, such as severe dementia, or the thought experiments of a teleporter that destroys a body and rebuilds it elsewhere, are cases in which intuitions differ. The most influential modern treatment, that of Parfit in 1984 [24], argued that what matters in survival is not identity but a relation, which he called Relation R, of psychological connectedness and continuity with the right kind of cause. Connectedness is the holding of direct psychological connections, such as a memory of an experience or the carrying out of an earlier intention; continuity is the holding of overlapping

chains of strong connectedness. Connectedness comes in degrees and diminishes with time. I am now strongly connected with myself of last week, weakly with myself of twenty years ago, and not at all with the infant who bears my name. Yet we speak of a single life, because the chain of overlapping connections is unbroken, and Parfit's observation is that it is the chain, not the end-to-end connection, that carries what we care about.

The relevance of this account to the program is that it is a model that can be made quantitative, and the quantities it yields are instructive. Take the simplest assumption, that the strength of connection between two stages of a person separated by a time Δ decays exponentially with a characteristic time τ , and that two stages are strongly connected if the strength exceeds one half. Then each stage is strongly connected with those within a window of $\tau \ln 2$ on either side, and the stages form a chain of overlapping windows that extends indefinitely, whatever the total length of the life. For $\tau = 20$ years the window is 13.9 years. A person who lives a million years consists of about 72,000 such windows, and is continuous with its earliest stage by a chain of 72,000 links, but the direct connectedness between the first and the last is $e^{-50,000}$, a number with no practical meaning, and it falls below one part in a thousand after only 138 years. The structure is that of a rope in which no fiber runs the full length. Parfit would say that the person is the same in the sense that matters, and a critic would say that the person is a succession of different people each of whom is connected to the next. The program need not decide the question. It needs only to observe that the practices of accountability, promise, and consent rest on the connectedness of stages within a window, and that it is these practices, and not the existence of a chain, that long lives test.

The memory that carries connectedness is itself finite. If a person acquires information at a rate r and can retain a total amount M , then the whole of the record is retained only for a time M/r , after which the person must discard or compress. For stipulated and merely illustrative values of $M = 10^{15}$ bits and $r = 10^{10}$ bits per year, the interval is 10^5 years, and beyond it the fraction of life retained falls inversely with the length of the life. The values are not established, since estimates of the capacity of the brain differ by several orders of magnitude, and the point of the calculation is the form, not the figure: an unaugmented brain would retain an ever-smaller fraction of a long life, and any augmentation of memory, whether by external records or by modification, would become part of what the person is. The practical consequences are those of the philosophical literature on the extended mind, applied at scale. A person of great age will depend on records, and the integrity, ownership, and privacy of those records become matters of the highest importance, since their alteration would alter the person.

Three consequences for the institutions of the program deserve statement. First, responsibility across long intervals requires a rule. A person who made a promise or committed a wrong five thousand years earlier is connected to the doer by a chain and not by memory or character, and the practices of law have in the human case limited responsibility over time by statutes of limitation and by the rule that the young are not bound by the commitments of their earlier years in the way that an adult is. The program does not prescribe particular limits, but

it requires that rules of this kind exist and be public, and that the duty-bearer relations of Chapter 2 be defined so as to remain well posed when a bearer may persist for millennia: the coverage and reassignment conditions must hold across the life of the bearer. Second, consent given by a stage cannot be taken to bind stages beyond its window of connectedness without renewal. An agreement entered into by a person at one age, with consequences that extend for centuries, is an agreement with respect to a future self that may be at the outset only distantly connected, and the sensible rule is that long-term commitments require periodic reaffirmation, with a right to withdraw at stated intervals. Third, the possibility of branching, of copies of a pattern that are equally continuous with the original, destroys the one-to-one relation on which law and property depend. Relation R can hold between a person and two later persons at once, and it is not transitive in the way that identity is. The program therefore prohibits, as a condition of authorization of any technology that could produce copies, the creation of copies without a prior determination of their standing, and treats the copy of a person as a person with its own claims, to be neither the property of the original nor the original's instrument.

A final consideration concerns what the program owes to those who decline to continue. A society in which many persons live very long lives will contain persons who prefer not to, and the freedom to choose a natural lifespan, to decline interventions, and to die, is a condition of the legitimacy of the program. The right to refuse, which Chapter 13 required of every authorization, applies here with particular force, because the interest at stake is the person's own life. The institutions must not make the declining of treatment a source of disadvantage, nor make the acceptance of treatment a condition of full membership in society.

Formalization

Let a life consist of stages indexed by times $t_0 < t_1 < \dots < t_N$ with spacing $\Delta = t_{i+1} - t_i$ and pairwise connectedness $c(i, j) = \exp(-|t_i - t_j|/\tau) \in (0, 1]$. Fix a threshold $\theta \in (0, 1)$, with $\theta = 1/2$ for Parfit's "strong" connectedness.

Definition 28.1 (Continuity). *Stages i and j are strongly connected if $c(i, j) \geq \theta$. A continuity chain from 0 to N is a sequence of stages $0 = i_0 < i_1 < \dots < i_m = N$ with consecutive stages strongly connected. The life is continuous if a chain exists.*

Proposition 28.2. (i) *A continuity chain from 0 to N exists if and only if $\Delta \leq \tau \ln(1/\theta)$. (ii) The minimum number of links in a chain is $m_{\min} = \lceil (t_N - t_0)/(\tau \ln(1/\theta)) \rceil$ when Δ divides the window. (iii) The direct connectedness of the first and last stages is $c(0, N) = e^{-(t_N - t_0)/\tau}$, and falls below ε when $t_N - t_0 > \tau \ln(1/\varepsilon)$; it is below θ whenever $m_{\min} \geq 2$.*

Proof. (i) If $\Delta > \tau \ln(1/\theta)$ then no two distinct stages are strongly connected, since $c(i, j) \leq e^{-\Delta/\tau} < \theta$ for $i \neq j$, and no chain exists; if $\Delta \leq \tau \ln(1/\theta)$ the consecutive stages form a chain. (ii) Each link spans at most $\tau \ln(1/\theta)$ in time, so at least $(t_N - t_0)/(\tau \ln(1/\theta))$ links are needed,

and the bound is attained by taking the largest permitted steps. (iii) Direct substitution; $c(0, N) < \theta$ iff $t_N - t_0 > \tau \ln(1/\theta)$ iff $m_{\min} \geq 2$. $\square$

For $\tau = 20$ years and $\theta = \frac{1}{2}$ the window is 13.86 years; a life of 10^6 years requires at least 72,135 links; and the end-to-end connectedness falls below 10^{-3} at 138 years and is $e^{-5 \times 10^4}$ at 10^6 years. The exponential law is a stipulation, and any decay law yields a finite window, so that the qualitative conclusion, that continuity is compatible with arbitrarily small end-to-end connection, depends only on the existence of a threshold window.

For memory, let the person acquire r bits per unit time and retain at most M bits, with uniform retention of the full record up to the capacity. The retained fraction of a life of length T is $f(T) = \min\{1, M/(rT)\}$, equal to 1 for $T \leq T_* = M/r$ and decreasing as $1/T$ beyond. For stipulated $M = 10^{15}$ bits and $r = 10^{10}$ bits yr $^{-1}$, $T_* = 10^5$ yr and $f(10^6) = 0.1$. If the person supplements biological memory by an external store of capacity M_e , the retained fraction is $\min\{1, (M + M_e)/(rT)\}$, so that the fraction of the long life that is accessible depends on a store whose custody is, by the argument of the chapter, constitutive of the person.

For standing, let n copies of a pattern be produced at a branching event, each strongly connected to the original stage and to no other. Then the relation of strong connectedness between the original stage and each descendant holds, but the relation among descendants fails, so strong connectedness is not transitive; no assignment of a single identity to the descendants is consistent with identity as an equivalence relation unless $n = 1$. This is the formal reason for the requirement that standing be determined before copies are created.

Chapter 29

Reliability over Astronomical Intervals

The most ambitious objective in the program is the extension of human life beyond centuries and millennia toward durations measured in millions or billions of years. The objective should be stated with exactness about what is and is not presently established. A million-year lifespan is not a demonstrated medical possibility, and a billion-year lifespan raises further astronomical and probabilistic difficulties, among them the fact that the interval is comparable to the remaining period over which the Earth is expected to remain habitable. The approach begins with biological aging, cellular repair, the prevention of cancer, regenerative medicine, the accumulation of molecular damage, and the limits of organismal maintenance. It then considers more speculative possibilities, including comprehensive tissue replacement, synthetic biological systems, and methods of preserving and restoring neural structure, with the explicit understanding that whether any of these maintains personal continuity is a question on which scientific and philosophical analysis remain unsettled, and that the answer may differ among approaches.

The essential observation is that extreme longevity is not only a problem of aging. Even if senescence were eliminated entirely, a person would remain exposed to accident, violence, infectious and environmental hazard, institutional failure, and the cumulative effect of small risks that are negligible over a decade and decisive over a million years. The mathematics of this exposure imposes quantitative requirements that are independent of any particular technology. Suppose the annual probability of fatal failure is a constant p and that years are independent. Survival to time T then has probability $(1 - p)^T$, which for small p is well approximated by e^{-pT} . For survival over a million years with probability near one-half, the annual risk would have to be below a figure on the order of one in a million, and for a billion years below a figure on the order of one in a billion. For comparison, the annual mortality of healthy young adults in high-income countries is on the order of one in a thousand, so that the required reduction in risk is three orders of magnitude for the first case and six for the second,

and it must be sustained and not merely achieved. Reductions of this size cannot be obtained by treating disease alone, and they require the management of every class of hazard, together with the capacity to repair damage faster than it accumulates and to maintain redundancy against failures that cannot be repaired.

The mathematics also exposes a limit on redundancy that is easily missed. Maintaining multiple independent backups can drive the probability of simultaneous failure very low, but the argument fails when failures share a common cause, such as an environmental catastrophe, a flaw in a shared design, or a failure of the institutions that maintain the system, and the probability of such common-cause failure imposes a floor that no degree of replication lowers. For this reason the question of longevity is inseparable from the questions of institutional durability and environmental stability that other parts of the program address. The final reflection concerns what becomes of education, law, inheritance, population, memory, institutions, and identity when individual lives might extend across geological epochs. Legal and social structures presuppose generational turnover, and a population in which individuals persist indefinitely would require new provision for the renewal of authority, the redistribution of resources, the revision of commitments made long ago, and the preservation of the capacity to change one's mind and one's way of life. These questions are not consequences to be deferred until the biology is solved; they bear on whether the objective is desirable in the form in which it is stated, and they belong among the admissibility constraints of the program.

The sources of the hazard deserve to be listed, since the requirement of the preceding paragraph is a requirement on their sum. Some are natural and astronomical. Impacts of asteroids and comets large enough to cause global catastrophe, of the order of a kilometer in diameter, are estimated to occur on average once in a few hundred thousand years to a million, and the impact that ended the Cretaceous period, some sixty-six million years ago, was larger. Volcanic eruptions of the supervolcanic class, which can alter the climate of the whole planet for years, recur at intervals of the order of hundreds of thousands of years, and the eruption of Toba some seventy-four thousand years ago is the best known. Nearby supernovae and gamma-ray bursts are rare but not negligible over geological time. On the longest scale the luminosity of the Sun rises by about one per cent in a hundred million years, and the Earth is expected to lose its oceans in a time that is estimated to be of the order of a billion years, which means that the aim of a billion-year life cannot be met on the Earth alone, whatever is done about every other hazard. Other hazards are of human origin: war, and in particular nuclear war; engineered or natural pandemics; the failure of the complex systems on which life depends; and the actions of a person or group intent on harm. The first class has an irreducible rate on the surface of a single planet, to be reduced only by dispersal; the second has a rate that depends on institutions and on the choices that societies make, and it is the one on which the program can act soonest.

The consequence for the design of a long-lived civilization is that its hazard must be reduced by structure and not merely by vigilance. The most powerful structural measure is dispersal, the maintenance of populations in locations whose catastrophes are not correlated, so that the

loss of any one does not end the whole. If k sites fail independently with annual probability p_s each, the probability that all fail in a given year is p_s^k , and the survival of a civilization over the age of the Earth becomes plausible with a modest k provided that the sites are truly independent, which, as the beta-factor analysis below shows, they never are entirely. The common causes are the shared artifacts of human origin, the conflicts that arise among the sites, and the dependence of each on the same institutions, so that the floor on the failure probability is set by the weakest of the cross-site links. This is the point at which the orbital programs of the fifth part and the longevity program of the sixth meet: settlements in space, if they could be maintained, would be the sites whose failures are least correlated with those of the Earth, and the case for the orbital programs, apart from their direct benefits, is that they are the means by which the hazard of the whole can be brought toward the required values. The case is conditional and is the strongest form of the longtermist argument, which Chapter 31 examines; it can be accepted as a reason to keep the options open and to prepare, and it cannot be accepted as a reason to override present obligations.

A second structural measure is the detection and response to hazards before they arrive. The survey of near-Earth objects, which has already cataloged most objects of a kilometer in size and found none on a collision course in the coming century, and the demonstration of a deflection technique by an impact on a small asteroid in 2022, are instances in which a natural hazard has been moved from the class of the unavoidable to that of the manageable. The same logic applies to pandemics, with surveillance and rapid response, and to the vulnerability of the systems of energy, communication, and food. Each reduction is modest, a factor of two or ten in a hazard that must be reduced by factors of millions, and the lesson of the calculation that follows is that a long life is the product of many such reductions, each of which must be maintained without lapse. The requirement of independence between verifier and generator, and the readiness classification, are in this sense not peripheral to the longevity program but constitutive of it: an institution that cannot maintain its own reliability over a century cannot be the guarantor of a person's survival over a million years.

Formalization

Let $h : [0, \infty) \rightarrow [0, \infty)$ be the hazard rate of fatal failure at time t (in years), with cumulative hazard $H(T) = \int_0^T h(t) dt$. The survival function is $S(T) = \exp(-H(T))$. For a constant discrete annual risk p , $S(T) = (1 - p)^T = \exp(T \ln(1 - p))$.

Proposition 29.1. *For constant annual risk $p \in (0, 1)$ and a survival target $q \in (0, 1)$ over T years, the required annual risk is $p^* = 1 - q^{1/T}$, and $p^* = -\ln q/T + O(T^{-2})$. In particular, for $q = \frac{1}{2}$, $p^* \approx (\ln 2)/T \approx 0.693/T$.*

Proof. Solving $(1 - p)^T = q$ gives $p = 1 - q^{1/T}$. Writing $q^{1/T} = \exp((\ln q)/T) = 1 + (\ln q)/T + O(T^{-2})$ gives the expansion, and $\ln \frac{1}{2} = -\ln 2$. $\square$

The following table gives $S(T) = (1 - p)^T \approx e^{-pT}$ for the stated values.

Annual fatal risk p	$T = 10^6$ years	$T = 10^9$ years
10^{-3}	$\approx e^{-1000}$ (negligible)	$\approx e^{-10^6}$ (negligible)
10^{-6}	≈ 0.368	$\approx e^{-1000}$ (negligible)
10^{-9}	≈ 0.999	≈ 0.368
10^{-12}	≈ 0.999999	≈ 0.999

Here $e^{-1} \approx 0.368$ occurs whenever $pT = 1$, which marks the characteristic scale: a risk p supports survival over about $1/p$ years with probability $1/e$.

For redundancy consider a single repair interval and k replicas, under the following explicit assumptions: a common-cause event occurs with probability q_c and fails every replica simultaneously; independently of it, each replica suffers an independent failure with probability q_i , independently across replicas; and failure of all replicas is the event of interest. Then

$$p_{\text{eff}} = q_c + (1 - q_c) q_i^k.$$

In the beta-factor parameterization, in which p_f denotes the per-replica failure probability (taken small, so that it is approximately $q_c + q_i$) and $\beta \in [0, 1]$ the common-cause fraction, $q_c = \beta p_f$ and $q_i = (1 - \beta)p_f$, so that

$$p_{\text{eff}} = \beta p_f + (1 - \beta p_f)((1 - \beta)p_f)^k \geq \beta p_f.$$

Increasing k drives the second term to zero but leaves the floor βp_f per repair interval, which accumulates over the intervals contained in a year. The achievable annual risk is therefore bounded below by a quantity proportional to βp_f , and the requirement p^* of the proposition can be met only if that floor does not exceed it, a condition on the correlation of failures that replication cannot substitute for. Over a population of N individuals the expected number of survivors is $N S(T)$, and if the cumulative hazard H varies across individuals the expectation of survival is $\mathbb{E}[e^{-H}] \geq e^{-\mathbb{E}[H]}$ by Jensen's inequality.

Chapter 30

Institutions for Long Lives

A society whose members lived for centuries would differ from ours in the structure of its institutions more than in the details of its technology, and the differences would not be anticipated by extrapolating from present practice. Nearly every institution that humans have devised presupposes a lifespan. Property passes by inheritance because owners die; offices rotate because officeholders retire or die; marriage, citizenship, apprenticeship, and the criminal law all assume stages of life that last decades. The assumption of mortality does silent work in preventing the accumulation of advantage, in refreshing the ranks of those who decide, and in allowing the world to be inherited by persons who did not choose it. Max Planck is reputed to have said that science advances one funeral at a time, and whether or not he said it, the remark identifies a mechanism by which societies renew themselves that a long-lived population would remove. The thesis of this chapter is that extended life, if it were achieved, would require deliberate substitutes for these mechanisms, that the substitutes can be designed, and that their design is an institutional problem whose features can be stated now with some precision.

Begin with demography, because it fixes the scale of the problem. In a population of constant size, births must balance deaths, and if the mortality rate is the reciprocal of a mean lifespan L , the population is the product of the annual number of births and the mean lifespan. A world of eight billion persons with a mean lifespan of eighty years has some hundred million births a year, as is roughly the case. If the mean lifespan were a thousand years and the population were to remain the same, births would have to fall to eight million a year, one-twelfth of the present number; at ten thousand years, to eight hundred thousand; and at a million years, to eight thousand. A population that wished to grow while living longer would have to accept still fewer births relative to its size, or would have to expand into new habitats, and a population that did not control births would grow at a rate that no environment could sustain. The consequences touch the most intimate decisions of family and the most contested questions of the right to reproduce. A program that extended life without addressing them would be proposing to resolve a constraint by default, and the default would be either an explosion of numbers or a rationing of births by whoever controlled access to the means of extension. It is

part of the argument of the book that neither is acceptable, and that the rate of births is a matter for the persons affected through procedures they can contest.

The age structure of a long-lived population has implications for representation. In a stationary population with a constant mortality rate, the proportion of persons older than a is $e^{-a/L}$. For a mean lifespan of a thousand years, 61 per cent of the population would be older than five hundred years and only 7.7 per cent younger than eighty. Most of the adults would be old in a sense that has no present counterpart, and in a society that gave votes and offices on the basis of age or of years of residence the young would be a permanent minority. The arrangements for voting, officeholding, and the allocation of public resources would have to be designed with this in view, and the principle that should govern them is that the participation of persons in decisions about their future should not depend on the accident of when they were born. This favors mechanisms that give weight to the interests of the young and to those not yet born, such as institutions with a mandate to represent them, and it counsels against rules that confer authority by age.

Wealth and power are the second concern. Capital, left to accumulate, grows at the rate of return, and an owner who does not die keeps all of the growth. If the rate of return exceeds the growth rate of the economy by $r - g$, the share of the total held by a perpetual owner of fixed initial wealth increases in proportion to $e^{(r-g)T}$. At a difference of two per cent, the multiple is 7.4 after a century, 22,000 after five hundred years, and nearly five hundred million after a thousand. These figures reflect the mathematics of compound growth, and not a forecast: no actual fortune has grown without interruption by taxation, expropriation, war, and misjudgment, and the fortunes of dynasties have regularly dissipated. But the dissipation has been largely a consequence of mortality, which subjects wealth to inheritance, division, and the unequal abilities of heirs. A population of immortals who made good investments would not be subject to these processes, and the equilibrium of wealth in such a society would be one of extreme concentration unless institutions prevented it. The instruments available are familiar. A periodic levy on wealth at a rate not less than $r - g$ prevents the share of any holder from growing without bound; limits on the duration of property rights, as in the law of patents and copyright, limit the accumulation of rent; and a requirement that wealth be returned to the common stock in part at stated intervals replicates for the living what inheritance taxation does for the dead. Equally important, the same reasoning applies to political power, which accumulates by the same mechanism of the compounding of advantage, and the terms of offices, the rotation of those who hold them, and the opening of positions to the young are the political counterpart of the levy on wealth.

The third concern is the durability of commitments. A government that binds itself for centuries is exerting a dead hand over those who come after it, and the dead-hand problem, familiar from the law of trusts and the debate over constitutional entrenchment, would be altered in a long-lived society, because the persons who made the commitments would be among those who lived under them. This has some advantages: the makers of a rule can be held to account for its consequences, and would have an incentive to revise a rule that did not work.

It also has dangers, since commitments made by powerful persons could be sustained against challenge by the same persons for as long as they lived. The analysis of Chapter 28 suggests a principle for the interval of renewal. A person's consent, and a body's mandate, decay with the connectedness of the stages that gave it, and the interval over which they can be presumed to hold is of the order of the window of strong connectedness, some decades. Commitments of greater duration should lapse unless renewed, with the renewal carried out by the persons then affected, who include persons not alive at the time of the original decision. A sunset provision of this kind is the mildest form of the principle, and it is the one most readily combined with the existing practices of legislatures.

The fourth concern, and the most difficult, is access. A technology of extended life would not at first be available to all, and the early distribution would be determined by cost, geography, and political influence. The history of medical advances is that those which are scarce are rationed by wealth, and that the gap between rich and poor in life expectancy, already large, would be widened. The program's principle of the admissible range requires that the distribution of such a technology be a matter for deliberate decision by procedures affected persons can contest, and the criteria of a just distribution are not for this book to determine. The book can say what is not acceptable: a distribution settled by the ability to pay without any public procedure, and a technology whose benefits are confined to the persons who are most able to influence the decisions about it. It can also say what must be provided in any scheme: a public record of who has access and on what terms, an independent audit of the effects on mortality and health, and the guarantee, already stated in Chapter 28, that the declining of the technology does not result in disadvantage.

The final concern is the continuing authority of the persons who hold the program together. A civilization that lasts for millions of years will need institutions that last as long, and the history of institutions is not encouraging: the longest-lived organizations in the human record, certain religious orders, a few commercial enterprises and ruling houses, have survived for centuries and, in a handful of cases, for more than a thousand years. Their persistence has depended on features that can be identified: a clear and limited purpose, a method for the training and selection of successors, a record that can be consulted, a capacity to change the rules while remaining the same body, and a culture of accountability. These are the features that the preceding chapters require of the institutions of verification and authority. What the programs of the book add is the requirement that the institutions be maintained for periods in which the survival of a single one of them is not to be assumed, which points again to redundancy, to diversity of design, and to the capacity to rebuild an institution from its records, and not to the hope that a single institution will persist unchanged.

Formalization

Three results are given, for population, for accumulation, and for renewal.

Consider a population of constant size N with birth rate B per year and a constant mortality hazard $\mu = 1/L$ at all ages (the extrinsic-hazard limit of Chapter 27), so that survival to age a is $e^{-\mu a}$.

Proposition 30.1. *The stationary population satisfies $N = B/\mu = BL$. The age distribution is exponential with density $\mu e^{-\mu a}$, so the proportion over age a is $e^{-a/L}$ and the proportion below age a is $1 - e^{-a/L}$. To maintain a fixed N as L rises the birth rate must fall as $B = N/L$.*

Proof. The balance $dN/dt = B - \mu N = 0$ gives $N = B/\mu$. The age density is proportional to the survival function $e^{-\mu a}$ and is normalized by $1/\int_0^\infty e^{-\mu a} da = \mu$. $\square$

For $N = 8 \times 10^9$: $B = 10^8, 8 \times 10^6, 8 \times 10^5, 8 \times 10^3$ per year for $L = 80, 10^3, 10^4, 10^6$ years. The proportion younger than 80 years is 0.632 at $L = 80$, 0.077 at $L = 10^3$, 0.0080 at $L = 10^4$, and 8×10^{-5} at $L = 10^6$. The proportion older than 500 years is 0.607 at $L = 10^3$ and 0.951 at $L = 10^4$. (The exponential law is the extrinsic-hazard limit and is used because it is the case in which senescence is removed; under it the mean age of the living is L .)

For accumulation, let an owner hold wealth $W(t)$ growing at rate of return r net of consumption, in an economy of total wealth growing at rate g , with a levy at rate λ per year on the owner's wealth.

Proposition 30.2. *The owner's share of total wealth is $s(T) = s(0)e^{(r-g-\lambda)T}$ while $s \ll 1$. The share is bounded for all T if and only if $\lambda \geq r - g$. Without a levy, the time for the share to multiply by a factor K is $T_K = \ln K/(r - g)$.*

Proof. $\dot{W} = (r - \lambda)W$ and the total grows as e^{gt} , so the ratio grows at $r - \lambda - g$. Boundedness requires a nonpositive exponent. The doubling-type time follows by solving $e^{(r-g)T_K} = K$. $\square$

For $r - g = 0.02$ the share multiplies by $K = 7.39, 2.2 \times 10^4, 4.85 \times 10^8$ in $T = 100, 500, 1000$ years; it doubles in 34.7 years; and a levy of 2 per cent holds the share constant. The result holds for the share while it remains small, and the qualification is relevant to the claim of extreme concentration: the model becomes inaccurate as $s \rightarrow 1$, at which point the owner's own return affects r .

For renewal, suppose a commitment made at time 0 by a body whose stages decay in connectedness as in Chapter 28, with mandate strength $c(t) = e^{-t/\tau}$. A commitment is renewed whenever $c(t)$ falls to the threshold θ .

Proposition 30.3. *The renewal interval is $\Delta_R = \tau \ln(1/\theta)$, and over a horizon T the number of renewals required is $\lfloor T/\Delta_R \rfloor$. For $\tau = 20$ years and $\theta = 1/2$, $\Delta_R = 13.9$ years, and a commitment intended to last 10^3 years requires 72 renewals.*

Proof. Solve $e^{-\Delta_R/\tau} = \theta$. $\square$

The requirement is unusually burdensome compared with the practice of legislation, which renews few things at such intervals, and it is intended as a limit case; a more tolerant threshold $\theta = 0.1$ gives $\Delta_R = 46$ years and 21 renewals for a thousand years. The choice of θ is a normative one, to be made by the society concerned, and the calculation shows how the burden of renewal depends on it.

Chapter 31

Present Obligations and Distant Futures

This chapter addresses the feature of the program that most closely resembles the ideologies the interview criticizes and that therefore requires the most careful treatment. Longtermism, as the interview presents it, is the view that the welfare of future persons deserves equal weight with the welfare of the living, and that if harms to present people are required in order to secure the welfare of a vastly larger number of future beings, those harms are acceptable. The interview presents this as a rationale for sacrifice in the present on behalf of speculative future populations and as a component of a larger set of commitments in which the construction of a successor intelligence is justified by the scale of its imagined benefits. This book need not endorse the interview's account of the history or the personnel of these movements to recognize the logical structure at issue. The target of the argument is a particular decision rule, namely unconstrained maximization of expected total welfare over all future persons without discount, and not longtermism as a family of positions, many of which do not endorse that rule. When the number of future beneficiaries is allowed to be astronomically large and their welfare is weighted without discount, the arithmetic of unconstrained maximization is dominated by the future, and any present cost, however severe, is justified if it increases the probability of the future benefit by even a tiny amount. The reasoning is vulnerable on several grounds: its conclusions depend on probability estimates that cannot be verified, it is sensitive to modeling assumptions that are not independently checkable, and it invites exactly the kind of justification that is difficult for those who bear the cost to contest.

The program responds not by assigning the future zero weight, which would be arbitrary, but by changing the structure of the decision. Obligations to existing persons are treated as constraints that define the set within which optimization over the future takes place, instead of as terms in a sum that can be outweighed. A person alive now is owed minimum provision of the conditions of a decent life, together with the procedural rights through which they can contest decisions affecting them, and these entitlements are not tradeable against benefits to

others, whether the others are alive or not yet born. Within the set of options that respect the floor, the long-term consequences of choices are given substantial weight, and the program regards the preservation of the conditions under which future persons can live well as a genuine obligation. The constraint is also a response to uncertainty: because the estimates on which speculative benefits rest are unverifiable, decisions that affect the present should be robust across the range of plausible models rather than dependent on any one. This is the same demand for independent verification applied in Chapter 9, now applied to the justification of sacrifice.

The distinction thereby drawn between this program and the ambitions the interview describes is institutional and does not rest on a claim about intentions. The program holds that capability be directed by procedures answerable to living persons, that the costs and benefits of projects be allocated by processes those persons can contest, that claims of benefit be verified by parties independent of those who make them, that the delegation of authority to machines be limited by the principle stated at the outset, and that present obligations constitute a floor and not a price. A project that meets these conditions can include astronomical engineering and extreme longevity without being a project whose legitimacy is established by the grandeur of its goal. A project that fails them does not become acceptable by being benevolently intended.

The argument from the weight of the future is usually made in the language of discounting, and the language is worth examining, because the disagreement it conceals is instructive. In the standard framework, associated with Ramsey's analysis of saving in 1928 [37], the value of a stream of future welfare is computed by weighting each period by a discount factor, composed of a pure rate of time preference, which gives less weight to the future simply because it is future, and a term reflecting the growth of consumption and the diminishing marginal value of additional consumption. The Stern Review of 2006 set the pure rate near one-tenth of one per cent a year, on the ground that to prefer the present to the future merely because of the date is without ethical justification, and obtained a high value for the prevention of climate harm; critics, among them Nordhaus, argued for rates an order of magnitude higher, partly on the ground that they reflect the observed behavior of savers and investors. The dispute does not turn on any fact that further research might discover, since it is a dispute about how to weigh persons by when they live, and the program's response is to avoid making the conclusion depend on it. A floor of provision for the living that is not subject to discounting, together with optimization over the remaining set, makes the choice of rate relevant only within the admissible set, where its influence is limited, and it is the use of the rate to override the floor that the program refuses.

A second difficulty is the uncertainty of the long future. The expected value of an action that affects the very distant future is the sum of a large number of terms, each of which is the product of a very small probability and a very large payoff, and the terms depend on estimates which cannot be checked. The result is a form of argument in which a speculative probability of benefit, multiplied by an astronomical number of beneficiaries, outweighs any certain cost to the living, and in which the conclusion can be driven in either direction by adjustments

of the probability that no available evidence constrains. This is the structure that has been called Pascal's mugging, after an example in which a stranger offers an enormous payoff for a small payment on a claim that cannot be verified, and it is the form in which the argument from the future becomes unfalsifiable. The remedy proposed in this book is the one that has been used throughout, the demand for independent verification, applied here to the claims on which a sacrifice would rest: a proposal to impose a cost on present persons for the sake of future ones must supply an estimate of the probabilities involved, the evidence for them, the sensitivity of the conclusion to the estimate, and a statement of what would count as evidence against it. A proposal that cannot do this has not stated a claim that can bind anyone.

A third consideration is that obligations to future persons are not obligations to a determinate set of individuals. Parfit's non-identity problem observes that the policies we adopt determine which persons will exist, so that a policy cannot be said to harm a particular future person who would not have existed had a different policy been adopted. The observation does not show that there are no duties to the future, since the existence of persons whose lives are bad is itself something that can be wrong, but it complicates the application of duties of harm to future persons and favors formulations in terms of the conditions that any future person should be able to expect. A principle of this kind, which Rawls put forward as a principle of just saving [38], holds that each generation should pass on to the next conditions at least as favorable as those it inherited, in respect of the institutions that sustain justice and the stock of capital, natural and made, on which a decent life depends. The program adopts this as a minimal intergenerational floor, complementary to the floor for the living, and it does not require that the future be treated as a source of claims that outweigh those of the present.

What does the floor for the living contain, and how is it connected to the readiness classification of Chapter 16? It contains clean water and sanitation, the energy needed for a decent life, health care, education, and the political and legal rights through which a person can contest what is done to them. These are all matters on which the technologies of the first readiness class are available or nearly so, and the programs of the megascale parts need not be completed, or even undertaken, in order to meet the floor. The rule that no measure that satisfies a present obligation may depend on a program of the second or third class is the means of keeping the two apart, and the audit of the plan is the means of enforcing it. The audit has a second function, which is to make the cost of the floor visible. The loss of potential value that the floor imposes, defined in the formalization, is a number that can be estimated, however roughly, and its publication allows the persons who bear the cost of the floor, and those who would benefit from its relaxation, to see what is being traded. A society that chooses to maintain the floor has made a choice, and a society that does not know the price of its choice has not made one.

The relation of this chapter to the preceding parts should be stated, because the chapters on space and on longevity are the places where the argument from the future has the greatest force. The reduction of the hazard of extinction by dispersal to other worlds, the preservation of options for the very long term, and the cultivation of knowledge that may be of value to

persons in the distant future are defensible aims, and the program does not oppose them. It opposes only their use as grounds for diminishing the provision made for living persons, and as grounds for removing the authority of living persons over the course of events. The distinction is between a program that seeks to leave the future more options and a program that seeks to choose the future's course. The first is compatible with the principle that capability is to be judged by the range of admissible futures it makes available; the second is the displacement of authority that the book was written to resist.

Formalization

Let time be indexed by $t \in \{0, 1, 2, \dots\}$, let N_t be the number of persons alive at t , and let $u_{i,t} \in \mathbb{R}$ be the welfare of person i at t . An action (policy) a in a set $\mathcal{A}$ induces welfare profiles $\{u_{i,t}(a)\}$. Unconstrained undiscounted utilitarian maximization selects $a \in \arg \max_a W(a)$ with

$$W(a) = \sum_{t \geq 0} \sum_{i=1}^{N_t} u_{i,t}(a).$$

Proposition 31.1 (Domination by the future). *Let a' be a policy that preserves present welfare and let a be a policy that costs present persons a finite total welfare $\ell > 0$ while raising by $\Delta p > 0$ the probability of attaining a future of total value V_f , other effects being ignored. Then the unconstrained expected-value criterion prefers a to a' if and only if $V_f > \ell/\Delta p$. Consequently, for every finite $\ell > 0$ and every $\Delta p > 0$ there exists a threshold $V_0 = \ell/\Delta p$ such that the criterion favors a whenever the claimed future value exceeds V_0 . Moreover, if $N_t \rightarrow \infty$ with mean welfare bounded below by some $\epsilon > 0$, then $W(a)$ diverges and the comparison of policies is undefined without a convention such as discounting or truncation.*

Proof. The expected change in W from adopting a is $\Delta p V_f - \ell$, which is positive exactly when $V_f > \ell/\Delta p$. For the final claim, $\sum_t N_t \bar{u}_t \geq \epsilon \sum_t N_t = \infty$ when $N_t \rightarrow \infty$, where $\bar{u}_t$ is the mean welfare at time t . $\square$

The proposition concerns a decision rule, namely unconstrained maximization of expected total welfare over all future persons, and not longtermism as a family of positions. Many positions described as longtermist do not endorse this rule, and the argument does not apply to them.

The constrained program replaces this with a feasible set. Let $w_{\min} \in \mathbb{R}$ denote the floor of welfare owed to each living person and let $\mathcal{R}$ be the set of procedural rights. Define

$$\mathcal{A}_{\text{adm}} = \left\{ a \in \mathcal{A} : u_{i,0}(a) \geq w_{\min} \text{ for all } i \leq N_0, \text{ and } a \text{ respects } \mathcal{R} \right\}.$$

Let $\mathcal{M}$ be a set of plausible models of the consequences of policies, each assigning to a a value

$W_m(a)$. The robust criterion selects

$$a^* \in \arg \max_{a \in \mathcal{A}_{\text{adm}}} \min_{m \in \mathcal{M}} W_m(a),$$

which is well-defined when W_m is finite for each m , for instance under a cutoff horizon T_{max} chosen by the legitimate procedure Σ .

Proposition 31.2. *For every $a \in \mathcal{A} \setminus \mathcal{A}_{\text{adm}}$ and every $a' \in \mathcal{A}_{\text{adm}}$, the constrained criterion prefers a' regardless of the magnitude of $W_m(a) - W_m(a')$. In particular no future benefit, however large, justifies a policy that violates the floor or the rights of persons alive at $t = 0$.*

Proof. The criterion maximizes over $\mathcal{A}_{\text{adm}}$ only, so policies outside the set are never selected, and the selection among members of the set does not depend on the values of policies outside it. $\square$

The proposition formalizes the statement that present obligations are a floor and not a price. Its price is a loss of potential value, namely $\sup_{a \in \mathcal{A}} \min_m W_m(a) - \sup_{a \in \mathcal{A}_{\text{adm}}} \min_m W_m(a) \geq 0$, which the program accepts and which should be reported to those who decide.

Chapter 32

Human Direction as a Program

The argument of this book is that the dominant opposition between salvation and extinction poses the wrong question, and that a better one is available. The better question asks what humanity should request of systems of greater capability if such systems exist, and it is answered by specifying the structure within which any request is made. The structure has been built in six parts. The first fixed the terms: the problems of the present are real whether or not any new capability arrives, they are coupled in ways that can make a collection of manageable crises unmanageable, the capacity to correct them can itself be eroded by them, and they present different kinds of obstacle, of which only the scientific and engineering kinds respond directly to capability. The second part examined the instruments, and found that the generation of proposals is cheap and its verification dear, that the acceleration of any field is limited by the slower of the two, that models fail in ways that optimization against them exploits, and that verification is worth only as much as the independence of the verifier from the generator and the margin by which a guarantee survives the gap between model and world. The third part considered institutions and held that what can be delegated in education, care, and public service is bounded below by a core of attention and judgment that cannot be, and that this core, because it does not shrink with automation, becomes the binding constraint and the proper object of public investment. The fourth and fifth parts applied a common standard of specification to the programs of very large scale, and in doing so separated the programs that survive their own arithmetic from those that do not. The sixth examined the prospect of extreme longevity as a problem of biology, of personal continuity, of reliability, and of institutions, and then fixed the constraint that prevents the distant future from being used to override the present.

Three findings of the middle chapters deserve to be recalled, because they show the method at work. The first is that energy is rarely the binding constraint on the megascale programs, and is more often a distraction. The energy to lift a cubic meter of water a kilometer, or to bring a kilogram to orbit, is a few kilowatt-hours, and the difficulties lie elsewhere: in the fifth power of the diameter of a pipe, in the dynamic pressure at the exit of an accelerator, in the

strength of a fiber that does not yet exist. The second is that several of the proposals fail on analysis, and that the book has said so. A compression tower to the edge of space is excluded by a scaling law that no choice of joint can alter, a rigid shell around a star by its instability and mass, a rigid ring near the ground by a demand for compressive strength greater than that of any known fiber in tension. Stating these failures is not an attack on the ambitions that prompted the proposals. It is a condition of taking them seriously, for the failures point to what must be different, in a material or in a sequence, for the ambition to be realized. The third is that the surviving programs share a feature: each depends, for its legitimacy as for its physics, on something outside the engineering. The freshwater corridors depend on the consent of the states they cross, the forest supports on the consent of the communities who live where they would stand, the mass accelerators on a regime of verification that does not exist, and the swarm of collectors on an authorization that no state can give by itself.

That last feature is the book's conclusion in small. The question which institutions may authorize a given program has no technical answer, and the book has not pretended otherwise. It has stated what an adequate answer must contain: an authorization that is multilateral where the effects are, that provides for the participation of those who are not parties, that allows a party to decline participation without loss of its existing rights, that is revocable by a procedure agreed in advance, and that is accompanied by a plan for termination that does not itself produce a harm worse than the one it ends. It has insisted that the accountability for every consequence rest on a person or an institution that can be named, questioned, and sanctioned, and not on an artifact, and in this it follows the interview that prompted the work. The interview's insight, that the attribution of agency to a machine relocates responsibility from those who can answer for it to an entity that cannot, is the premise from which the architecture of authority proceeds, and the book's contribution is to carry that premise through the engineering: to show where, in a program of robotic embodiment, orbital construction, or planetary restoration, the decisions are made that cannot be delegated, and to design the program so that they remain with persons.

Limitations should be stated plainly. The calculations of the book are of the first test only: they show that a concept is or is not consistent with its own premises, and they do not show that it is wise, safe, or affordable. Several of the concepts treated, the volsorial and intervolsorial pediments, the geothermal mass accelerators, the nested expandable structures, are described in the author's earlier notes in terms that leave their specifications incomplete, and the definitions adopted here are readings of those descriptions, to be superseded as the specifications are fixed. The numerical examples use stipulated parameters where parameters are not established, and each such place has been marked. The treatment of longevity assumes simple models of mortality and of memory that capture orders of magnitude and not the detail of any real population. The ordering of the readiness classes is a judgment, and a reasonable reader may reclassify a program on grounds that the book has not considered. The book has not attempted to settle the empirical questions on which the public debate turns, whether capable systems will arrive, how soon, and with what dispositions, and it does not need to, for

its argument is conditional. If they do arrive, the programs described here, with their gates, their verifiers, and their floors, are one account of what to ask of them and how to keep the asking in human hands. If they do not, the problems remain, the instruments of the second part are still those with which they must be addressed, and the institutions of the third part are still those that must be built.

Formal Summary

Collecting the constructions above, a *human-directed program* is a tuple

$$(\Omega, A, R_c, \Sigma, \kappa, \mathcal{D}, \{\varphi_j\}, r, w_{\min})$$

in which Ω is the space of trajectories, A the admissible set fixed by legitimate procedure and contained in A_{deleg} , R_c the realizable set at capability c , Σ a selection procedure, κ the classification of tasks into local autonomy, supervised execution, and nondelegable judgment, $\mathcal{D}$ the set of locally autonomous behaviors certified as robustly invariant, $\{\varphi_j\}$ a family of independent verification predicates, r a readiness function, and $w_{\min}$ the floor of present obligations. The program is *admissible* when (i) the realized trajectory is $\omega^* = \Sigma(R_c \cap A)$, where Σ exercises meaningful direction in the sense of Chapter 13 and every duty arising from its decisions is covered, that is, $\kappa_{\text{cov}}(h; \beta_h) = 1$ for every harm event h ; (ii) every deployed proposal lies in Π_c^* and has been checked by tests whose errors are not nested with those of the proposer; (iii) no judgment task is performed by a machine and every locally autonomous behavior lies in $\mathcal{D}$; (iv) the plan is readiness-respecting, which is a screening condition accompanied by a separate analysis of implementation risk; and (v) the selected policy belongs to $\mathcal{A}_{\text{adm}}$. The evaluative thesis is that the merit of capability is assessed first by strict enlargement of $R_c \cap A$ in the sense of inclusion, and by a measure μ only where its choice has been justified and its influence reported. The share $a(\omega)$ of tasks transferred to machines is not a proxy for admissibility, since admissibility depends on which tasks are transferred and not on how many, and the two comparisons are therefore not treated as measuring the same quantity.

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